



MÁSTER EN FINANZAS CUANTITATIVAS



GUÍA DIDÁCTICA Módulo 7

CURSO 2025/2026

Módulo 7: Valoración de Instrumentos Financieros

2025/2026

Coordinador:

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PRESENTACION

El Máster en Finanzas Cuantitativa se integra por 10 módulos que se agrupan en los siguientes tres cursos de experto.

Curso de Experto en Métodos Computacionales en Finanzas

Módulo 1: Programación en Python y en R

Módulo 2: Modelos de Simulación

Módulo 3: Datos Financieros y Aprendizaje Automático

Curso de Experto en Métodos Econométricos Aplicados en Finanzas

Módulo 4: Modelos de Series Temporales

Módulo 5: Modelos de Volatilidad y Correlación

Módulo 6: Modelos de No linealidad

Curso de Experto en Análisis Financiero y Gestión de Riesgos

Módulo 7: **Valoración de Instrumentos Financieros**

Módulo 8: Gestión de Carteras de Inversión

Módulo 9: Gestión de Riesgos Financieros

Módulo 10: Teoría del Valor Extremo

Atención directa al alumno:

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Quantitative Investment

Fundamentos Temas: 1 al 5 (m.d)

Introduction

This Chapter we are introduced to investment.

- Idea of investing, some idea of asset classes;
- Discussed why finance differs from other industries;
- Benefits of markets, idea of no “free lunch;”
- Talked about market participants; and,
- We played *Acquire!* (Yes, it is always competitive.)

Asset Classes

This Chapter we discuss asset classes.

- More than just “stocks and bonds;”
- Importance: differences in behavior, legal treatment;
- Briefly covered hard assets (real estate, commodities);
- Talked about wide range of fixed income investments;
- Explored equities and corporate actions; and,
- Covered derivatives and indices.

Markets in General

This Chapter we discuss markets.

- Not all markets are the same — for good reasons;
- Markets handle differences in flexibility, trust, liquidity;
- Discussed types of markets, benefits, drawbacks;
- Introduced orders, quotes, and trades;
- Mentioned financial intermediation; and,
- Considered short selling
- and regulation.

Modern Markets

This Chapter we discuss modern markets.

- Past collusion which led to reforms;
- Electronic vs algorithmic vs high-frequency trading;
- Intermediation without specialists (including theory);
- Recent competition-enhancing changes to markets;
- Effects of market structure on data, metrics;
- Problems and pseudo-problems with modern markets; and,
- Recent regulation.

Efficiency and the Macroeconomy

This Chapter we discuss efficiency and the macroeconomy.

- How do we think about incentives and production?
- What different types of efficiency are there?
- What do these imply for markets?
- How do we look at the macroeconomy?

Introduction

Chapter 1



Investment

- An *investment* commits resources now for future benefit.
- You commit time, cash, work, ideas, materials, space. . . now.
- Expect more (money? happiness?) than you put in. . . later.
- All of you, right now, are investing:
 - Spending time and work to get an education.
 - Have cash in a bank account to reduce theft, get interest.
- However, we should always consider *opportunity costs*:
 - How much benefit could you get doing something else?
 - Why not work straight out of high school? More seniority. . .
 - Why not study something easier? Less work. . .

Real Assets

- *Real assets* are goods which are used to produce other goods.
 - Land: home property, timberland, oil well;
 - Physical goods: buildings, manufacturing machinery;
 - Knowledge: training, experience, college, grad school;
 - Intellectual property: patents, copyrights, “secret sauce”;
 - Commodities: corn, coffee, copper, gold, oil, natgas; and,
 - Energy: electricity.
- Real assets can be harder to value, but we use them every day.
- Wealth is created from real assets; money alone isn't enough.
- Bitcoin is a real asset — lacking clear fundamental value.

Financial Assets

- *Financial assets* are claims on real assets or income.
 - Payments from letting land, oil well, buildings, equipment;
 - Royalties from use of intellectual property;
 - Interest on and return of monies lent to someone;
 - Foreign currencies;
 - Payments due to partial ownership of a firm; or,
 - Cashflows exchanged by prior agreement.
- We often further decompose financial assets:
 - Fixed-income: a promised stream of income (“bonds”);
 - Equity: ownership stake, but no income promise (“stocks”).
 - Derivatives: contracts for payments based on other prices.
- Our focus: investments in financial assets (and money).

Why Finance is Different

- Economists believe the finance sector is uniquely different:
 - Financing helps/encourages people to invest.
 - Bartering-based economies fail eventually. Why?
 - Cash: easily stored/transported; universally used.
- A damaged finance sector puts the whole economy at risk.
 - People cannot get loans for cars, homes, school.
 - Storeowners cannot use credit to buy goods to sell.
 - Firms cannot use loans to complete projects.
 - Firms cannot issue shares of ownership to expand.
- Why not just accrue cash? Inefficient; discipline of debt.

Benefits of Capital and Asset Markets I

Capital markets trade financial assets; asset markets trade real assets.

We believe capital and asset markets serve multiple purposes:

- Markets help individuals/companies *time their consumption*.
 - Earning more than you need now? Store wealth for later.
 - Earning less than you need now? Use wealth stored earlier.
 - Might also borrow from later wealth to get money now.
- Market prices *convey information* about expected value.
 - Decisions are made based on prices = signals to economy.
 - *Price Theory*: Prices thus most important role of markets.
 - Decentralizes economic planning; power to the people.

Benefits of Capital and Asset Markets II

Capital and asset markets also help us specialize:

- Allow us to *shift unwanted risk* to others.
 - Should farmers, bakers, brewers try to predict wheat prices?
 - Should Southwest, United be predicting jet fuel prices?
 - Or, should they lock in prices and focus on what they do best?
 - Do all Tesla investors want the same risk from battery factory?
- Facilitate *separation of ownership and management*.
 - Meet every day to manage firms you invest in?
 - Or, hire people to run the company for you.
 - Managers are charged with “maximizing shareholder value.”
 - *Agency problems*: possible conflicts of interest for managers.

Benefits of Capital and Asset Markets III

Capital and asset markets also serve an oversight role:

- Allow us to *compare firms and their performance*.
 - Investors can agitate to punish poor management.
 - Better management can be rewarded with stock/options.
 - Can see reaction to various announcements.
 - Reduce incentives to cheat (e.g. disciplining power of debt).
- Markets also *reward people who find problems*.
 - A key assumption of these markets is trust.
 - Financial statements, inventories, prices must be honest.
 - If not: investor can bet against firm, unveil its dishonesty.
 - Case in point: Enron scandal unveiled by Richard Grubman.

Investment Process

- Deciding where to invest involves a few concepts:
 - *Asset allocation*: choosing asset classes of investments.
 - *Security selection*: choosing instruments to invest in.
- Two common ways to determine where to invest:
 - *Top-down*: asset allocation, then security selection.
 - *Bottom-up*: Security selection ignoring asset allocation.
- We will discuss asset classes shortly, but in general:
 - Stocks vs bonds vs real estate vs commodities vs cash vs . . .

Competitive Markets

- Market participants are always looking for easy profits.
 - Electric car makers are all getting rich!
 - Bitcoin! COIN! COIN! HODL DAT COIN!
 - If true, you would all be investing in these areas. . .
 - Act I: easy \$; Act II: competitors; Act III: low prices, consolidation.
- This leads us to believe markets are *competitive*:
 - *i.e.* There is “No Free Lunch with Vanishing Risk” (NFLVR).
 - The economic equivalent of there being no perpetual motion.
- Thus we believe there is a *risk-return tradeoff*.
 - We expect more return from riskier investments.
 - If govt bonds pay 3%/year, why take risk for 3% or less?

Market Participants

Who are these “market participants?”

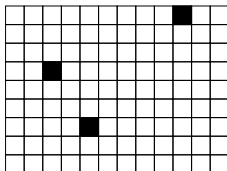
- Some make up what we call the *real economy*:
 - Individuals/households: save/invest cash; and,
 - Non-financial companies: raise capital for plants/equipment.
- Governments and agencies also participate in markets:
 - Borrow for capital projects: infrastructure, defense, etc.;
 - Trade in the bond market to set rates (aka monetary policy).
 - In some countries, governments invest to pay retirees.
- Finance firms are very active participants in markets.

Market Participants: Financial Firms

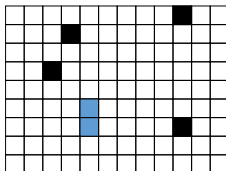
- Finance firms participate because they act as facilitators.
- *Investment bankers* assist with raising capital.
 - Help firms find investors and give feedback/advice.
 - Originate securities in the *primary market*; or,
 - Help buy/sell a firm or merge with another firm.
- Investments lead to *securities*: stocks or bonds.
- Those securities are traded in the *secondary market*.
- *Financial intermediaries* help buyers find sellers in market.
 - May match buyers with sellers (act as a *broker*);
 - May trade with buyer/seller (act as a *dealer*).
 - May also advise buyer on where to invest.
 - Banks take in money, pay interest, lend out money.

Acquire!

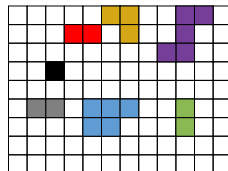
- To get at investment ideas, we are going to play *Acquire!*



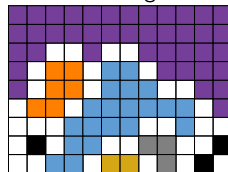
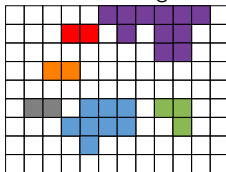
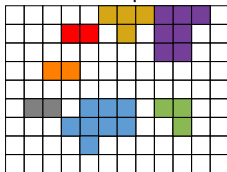
1. Place pieces



2. Hotel! Founder gets share.



3. Hotels grow



4. Merger! Cash payouts made. 5. Larger hotel takes over 6. End. Assess portfolio values.

- Then will talk about various asset classes.

Finance Culture & Fun

- May want to start reading one or more of these:
 - Markets pages of the *FT*;
 - Money and Investing section of the *WSJ*;
 - *The Economist* (best, funniest, but most reading).
- I *highly* recommend two books:
 - *Money Changes Everything* by Will Goetzmann.
 - *Why Nations Fail* by Daron Acemoglu and James Robinson.
- If your background is not strong in economics:
 - *Economics in One Lesson* by Henry Hazlitt.
 - Can also watch *Free to Choose* videos.
- For silly/fun books:
 - *Liar's Poker* and *The Big Short*: Probably Michael Lewis's only good books; however, they are amazing.
 - *King of Oil* and *Metal Men*: Oddly fascinating.

Asset Classes

Chapter 2

Introduction

This part discusses asset classes. Specifically:

- What is an asset class;
- Why we care about asset classes;
- An overview of various asset classes; and,
- Where derivatives fit.

What is an Asset Class?

- People talk about “asset classes;” what are they?
- *Asset class*: instruments w/similar characteristics.
- Thus we think members of an asset class have similar:
 - Types of risks that they are exposed to;
 - Cashflow structures and/or rules;
 - Laws and regulation; and,
 - Ways in which they behave, esp. in response to shocks.
- Why “instrument” and not “security?”
 - Money, real estate, and commodities are not securities.

Asset Classes

Can roughly break investments into ~~six~~ five asset classes:

- ① Commodities;
- ② Real Estate;
- ③ Money/Foreign Currency (FX);
- ④ Fixed Income; and,
- ⑤ Equities.
- ⑥ And sort of, not really: Derivatives (including indices).

This is not canonical; could draw lines differently.

Common claim: “**X** as an asset class.” (art/patents/??)

Whither Derivatives?

- Some texts claim derivatives are their own asset class.
- Derivatives get value from “underlying” instruments.
 - e.g. a call option may let you buy IBM at \$20/share.
- On the one hand: derivatives may not behave like underlying.
- Also: derivatives are regulated differently.
- On the other hand: they are clearly related to underlying.
- I know of no firm which treats them separately.
- Derivatives are a.e. a department in an asset class area.
- Less clear: where to put high-yield bonds vs preferred stock?

Why We Care About Asset Classes

- So would people care about a new asset class?
- Reason goes back to what defines an asset class.
 - Asset classes behave differently from one another.
 - Asset class A may do well when other asset classes stink.
 - Thus new asset classes may increase diversification.
- Also, undiscovered asset classes may hold bargains.
- However, you should also ask another key question:
 - How much money can be invested in this new asset class?
 - If not much (e.g. \$5 billion), not everyone can hold it.
 - Is this person biased, trying to sell me those assets?

Commodities

- *Commodities* are fungible goods which are inputs to industry.
 - *Fungible* means indistinguishable or agreed as equivalent.
- Some commodities rise and fall with the business cycle.
- As hard assets, commodities have value apart from money.
 - Thus they tend to be seen as inflation hedges.
 - Even in hyperinflation: gold, gasoline, ethanol retain value.
 - In fact, during hyperinflation these are main stores of value.
- Commodities offer good potential for diversification.
 - Correlation of -0.16 between WTI crude oil and S&P 500.
 - However, copper related to real estate; gold related to crises.

Real Estate

- *Real estate* is land, buildings, or portions of these.
- Real estate is the most widely held investment in the world.
- However, real estate is highly illiquid, unique.
 - Cannot easily bet on price declines.
 - No fungibility: each house/lot is unique.
 - No portability: Cannot transport real estate (usually).
- Most people invest in real estate via their home.
- Some hold other properties and rent them out. (Hassle.)
- Easy approach: managed real estate portfolios (REITs).
- Problem: some REITs are moderately correlated w/stocks.

Foreign Exchange (FX)

- Foreign exchange (FX) markets: most liquid in the world.
- About \$5 trillion traded in FX markets each day.
- Most currencies *float* against one another.
 - Exchange rate set by supply, demand. (trade, int. rates, risk)
- Example: If Swedish investments look bleak, money leaves Sweden.
 - People sell SEK, buy other currencies; SEK “falls.”
(i.e. 1 SEK gets you fewer and fewer USD.)
- “Big Five” currencies: USD, EUR, JPY, GBP, CHF.
- Why CHF? Seen as safe haven in crises. Low-tax on businesses.
- *Most USD cross rates* quoted as foreign per USD.

Money Markets

- *Money markets*: very short-term loans, very “liquid.”
- Due to short times to maturity, considered very low risk.
- Money markets may be called “cash” — but can lose money!
 - Foreign currencies are generally (foreign) money markets.
- We will mostly consider USD-denominated money markets.
- Money market funds: \$1/share; # shares increases over time.
- In industry: money markets are part of fixed income or FX.
- Sometimes referred to as STIRs: short-term interest rates.

Money Market Instruments: T-Bills and Interbank Rates

- *Treasury bills* (T-bills): short-term US government debt.
 - Issued at discount (ZCB); 28-, 91-, or 182-day maturities.
- *Federal funds*: overnight loans b/w financial institutions.
- *London Interbank Offered Rate*: short-term loans b/w large banks
 - Billions of \$ moves daily based on LIBOR.
 - Proper LIBOR rate needs currency, term (e.g. USD3MLIBOR).
- Lending rates often quoted as spread above these:
 - “LIBOR+30” (USD6MLIBOR+30bp^{*}); Fed funds + 50bp.

^{*} A *basis point* (bp) is 1/100th of 1%.

Money Market Instruments: Corporate Loans

- *Certificates of deposit* (CDs): time deposits at a bank.
- *Bankers' acceptances*: customer orders to pay funds later.
- *Eurodollars*: USD deposits at banks outside the US.
 - LIBOR-earning time deposits, <180-day maturities.
 - Futures traded on CME; among most liquid instruments on Earth.
- *Repurchase agreements* (repos): short-term security-backed loans.
- *Brokers' calls*: loans to broker making margin loans.
 - Investor borrows to buy security; must keep minimum LTV.
- *Commercial paper*: unsecured loans to corporations.
 - "CP" usu. backed by bank credit line, <2 months maturity.

Fixed Income

- *Fixed income* instruments are bonds, loans, and other debt.
- Term based on old idea: bonds pay stream of fixed amounts.
- Instruments are of longer terms than in money markets.
- Bond market in US is far larger than stock market.
- Debt spans a wide range of instruments:
 - Government/sovereign, agency, municipal;
 - Corporate (including high-yield); and,
 - Mortgage- and asset-backed (e.g. car loans).

Fixed Income Instruments: Government Debt

- The US Treasury issues debt paying semi-annual interest.
 - *Notes* issued maturity: 2–10 years. *Bonds*: >10 years.
 - Quoted as percent of par: “96.30” $\Rightarrow 96\frac{30}{32}\% = \969.375 .
- Govt/sovereign debt may also be *inflation indexed*:
 - Principal (par) changes w/an inflation index, e.g. TIPS use CPI.

Fixed Income Instruments: Agency and Municipal Debt

- Government agencies issue debt, to help subsidize desired behaviors.
 - FNMA, FHLMC, FAMC, TVA,... SLMA?
 - Sponsored agencies may have implicit government backing.
- *Municipal debt* issued by state+local governments/agencies.
 - “Muni” interest is exempt from federal and issuing state taxes.
 - General obligations (“GOs”) — backed by government credit;
 - Revenue bonds — backed by a revenue stream (more risky).
 - Equivalent taxable yield = $\text{yield}_{\text{muni}} / (1 - \text{tax rate})$.

Fixed Income Instruments: Corporate and Consumer Debt

- *Corporate debt* is issued by private corporations.
 - *Secured* bonds backed by collateral; *debentures* not.
 - Debt has an ordering of seniority in case of default.
 - May be issued in foreign currency; convertible to stock.
 - May be “puttable” if interest rates $\uparrow$; “callable” (refundable) if rates $\downarrow$
- 1980s: new structures eased investing in mortgages, consumer debt.
 - Usually, principal+interest payments are received monthly. May
 - have *prepayment risk*: full repayment if interest rates fall.
 - May have *default risk*: borrower may not repay debt.
- Investments often in loan pools or derivatives (CMOs, CDOs).

Equity

- *Equities* (stocks) are ownership stakes in a corporation.
- A cash dividend may be paid regularly.
- Have *limited liability* (lose original investment at most).
- Have *residual claim* on firm assets (last in line).
- *Common stock* typically allows voting on corporate issues.
- Companies may issue share classes with different voting rights.
- *Preferred stock* has no voting power, but:
 - Unpaid dividends cumulate; paid before common dividends.
 - May be *redeemable* (callable) by the issuer.
- *Depository receipts*^{*} represent shares of a foreign company.

^{*} Often American Depositary Receipts (ADRs) in the US.

Equity Corporate Actions

- Equities may pay cash dividend, offer capital appreciation.
 - Dividend and capital gains taxes vary by country.
- Equities may also pay a stock dividend.
 - “1.1” stock dividend grants 1 more share for every 10 held.
- Splits (“3 for 2”) and reverse splits are also possible.
- Price adjusts accordingly for stock dividends, splits.
- *Rights* to purchase new stock may be issued to shareholders.
 - Price typically at a discount; often, rights may be sold.

Derivatives

- *Derivatives* have values based on other instruments.
- *Options* allow purchaser to buy/sell underlying at a set price.
 - *Call* holders may buy (or not) underlying at set price.
 - *Put* holders may sell (or not) underlying at set price.
 - Right to buy/sell expires at a later date.
- *Futures* and *forwards* buy/sell underlying later at set price.
 - Purchase/sale occurs at contract maturity date.
- *Swaps* contract to exchange cashflows regularly until maturity.
- Portfolio-related derivatives are also very common.

Derivatives: Portfolio-Related Instruments

- Indices are technically portfolio derivatives.
 - Futures, options, swaps, ETFs, mutual funds trade on indices.
- *Basket swaps* exchange a portfolio's return for its financing.
- *Collateralized mortgage obligations* (CMOs) shift prepay risk.
- *Collateralized debt obligations* (CDOs) shift default risk.
- *Credit default swaps* (CDSs) swap bond for par upon default.

Derivatives: Financial Futures

- Futures allow for easy, safe hedging of business risks.
- Can lock in a price/level of interest.
- Why not apply this idea to financial risks?
- Chicago, 1972: creation of first FX futures.
- Chicago, 1975/6: first int. rate futures (GNMA, T-bills).
- KCBOT, 1982: first equity index futures (Value Line).
- Now, financial futures among most liquid global markets.
- Futures now dictate which indices professionals care about.

Indices

- *Indices* are aggregations of index member prices/returns.
 - Ideally, they provide a gauge of a particular market/sector.
- Most indices are arithmetic averages: $I = \sum_{i=1}^N w_i p_i$.
- *Price-weighted indices* average prices scaled by divisor.
 - Divisor adjusts to maintain continuity after splits, adds/deletes.
- *Capitalization-weighting* weighs prices by shares outstanding*.
- *Equally-weighted indices* average returns.
- *Total return indices* include dividends.
- *Geometric indices* compute geometric means of returns.

* May instead weight by *float* (shares freely trading).

Index Examples

- Indices are the most frequently quoted metrics for markets.
- US professionals mostly follow bolded indices; *not* the Dow.
 - **S&P 500**: cap-weighted, 500 large-cap stocks.
 - **Russell 2000**: cap-weighted, cap-rank 1001–3000 stocks.
 - ~~Dow Jones Industrial: price-weighted, 30 stocks.~~
 - **Barclays Capital Aggregate Bond**: 9000 bonds.
 - Dow Jones US Real Estate: 109 REITs/stocks.
 - S&P/Case-Schiller US Home Price: uses repeat sales.
- Goldman Sachs Commodity Index: 24 commodity futures.
- International stock indices:
 - TSX Composite/TSX 60; Bolsa IPC; Ibovespa;
 - EuroStoxx 50; FTSE 100; SMI; JSE All Shares; ASX 200;
 - Nikkei 225, Topix; KOSPI 200; CSI 300; Hang Seng; Taix.

The Road Ahead

We have covered asset classes; on to markets next time!

- Fundamentals: Markets, Efficiency;
- Measuring: Returns, Risk, and Diversification;
- Valuation: Fixed Income, Yield Curves, Equity Valuation;
- Valuation II: Factor Models, Microfoundations, Global Investing, FX;
- Risk Alleviation: Futures, Options, Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

How Markets Work

Chapter 3



Introduction

This part discusses how markets work. Specifically:

- Security issuance;
- Market structures;
- Orders and post-trade issues;
- Short sales; and,
- Regulation.

Market microstructure is the detailed study of how markets work.

The Primary Market



Primary Market

- The *primary market* is where securities are issued (sold).
 - Investors must choose (and be chosen) to participate.
 - The primary market exists to raise/borrow capital.
 - The *secondary market* is where issued securities are traded.
- New securities may be issued in a few ways:
 - *Private placement* (PP): issued only to “accredited investors.”*
 - *Initial public offering* (IPO): sells stock in firm to public.
 - A company may *spin-off* a part of itself to shareholders.
 - *Seasoned* issue: when a public company issues more securities.

* *Accredited investors*: \$1+MM net worth (exc. home), or \$200k (\$300k joint) income last 2 years.

Investment Banking

- *Venture capitalists* invest in young, pre-IPO firms.
- *Investment bankers* organize public offerings, PPs, M&A.
- Bankers responsible for an offering *underwrite* the deal.
 - Purchase offering, arrange firm commitments for resale.
 - Often form a *syndicate* to coordinate resale, share deal risk.
- Syndicate often places a *tombstone* to announce the offering.
- *Offering documents* describe security being issued, issuer.
 - Also list investment risks (mundane to outlandish/troubling).
- The preliminary documents are called a *red herring*.
- The approved documents are called the *prospectus*.

Offering Details

- Initial public offerings (IPOs) are sold to the public.
 - Underwriters typically organize a *road show*.
 - Road show promotes security, informs potential investors.
 - Road show also helps underwriters gauge interest, pricing.
 - *Bookbuilding* is the collecting of indications of interest.
- Private placements: sold directly to accredited investors.
 - Often called “144A” securities for the SEC rule allowing them.
 - Cheaper to issue: no road shows or extensive documents.
 - Cannot be traded openly like common stock.

Seasoned Offerings

- Seasoned offerings occur after an IPO.
- Companies might *shelf register* securities.
 - Securities are registered, gradually sold for up to 2 years.
 - Less expensive than a large public offering.
- Companies also may issue stock held by their treasury.
 - Often in response to exercise of a warrant or convertible bond.
- Some companies issue more shares via a *rights* issue.
 - Rights are given to current shareholders.
 - Allow purchase of additional (new) shares at a set price.
 - Thus rights are similar to warrants.

Secondary Markets

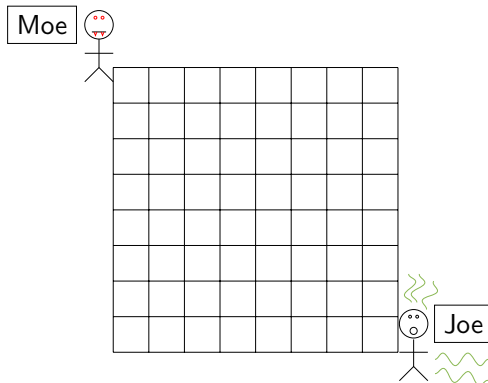


Market Types

- Markets are not all the same, nor should they be.
- Different market structures accomodate differences in:
 - flexibility;
 - fungibility; and,
 - liquidity.
- We can classify markets broadly (in increasing liquidity) as:
 - Direct Search
 - Over-the-Counter (OTC)
 - Auctions
 - Sporadic Call Markets
 - Continuous Trading

Direct Search Markets

- Direct search markets: highly illiquid; crudest of markets, e.g. :
 - Buying something via craigslist.com.
 - Asking friends who can loan you \$30 for beer.
 - Moe and Joe...seeking to trade “blow”...



OTC Markets

- OTC markets exist when liquidity is low/flexibility is crucial.
 - Swaps used to finance baskets of stocks;
 - Tranches of a CDO or CMO;
 - Mortgages on houses.
- Thus OTC markets often occur over the phone or face-to-face.
- OTC trades: between parties, may be custom contracts.
- Thus OTC markets are sources of *counterparty risk*.

Auctions and Sporadic Call Markets

- Auctions may be used in the primary market:
 - US Treasuries, WR Hambrecht's OpenIPO, emissions caps.
- Auctions are used in secondary markets also:
 - Day/session opening and closing equity prices.
 - Intraday if big news, high volume or volatility.
- Auctions are studied by economists using game theory (OR).
 - Typically classified as one- or two-sided, aka "fixings."
- Some markets conduct regular "call markets" (*i.e.* auctions).
 - Frequent auctions approximate continuous trading? Sort of.
 - Examples: LME, some illiquid Euronext stocks, *dark pools*.
 - Dark pools: no pre-auction data; *e.g.* ITG's POSIT (since 1987).

Continuous Trading

- *Market hours*: the time between open and close times.
- *Continuous trading*: during market hours; trading at any time.
 - Focus of most news, research; about 80% of volume.
- In the US, market hours vary for different markets:
 - Equities: 9:30–16:00 NY time (8:30–15:00 Chicago).
 - US Treasuries: 24 hrs/day (more liquid 8:00–16:00 NY time).
 - FX: Always open; most active currencies vary by time of day.
 - Non-Ag futures: Sun 17:00–Fri 16:00, exc. 16:00–17:00 Chicago.*
 - Ag futures: Sun 19:00–Fri 13:20, exc. 7:45–8:30, 13:20–19:00 Chicago.
- Futures are nearly 24 hours/day because they are used worldwide.

* Index futures also pause 15:15–15:30.

Specialists

- Many stock exchanges started as order-driven markets.
- One problem: Orders might not always match:
 - Market would then function only sporadically.
 - Lower volume would dissuade investors from trading there.
- Solution? “Hire” someone to watch market:
 - Ensure it is “fair and orderly.”
 - Constantly quote a bid and ask.
 - Also provide liquidity as needed.
- This person is called a *specialist*.

Genesis of Orders

- Expressing want: we can be attracted, repelled, or indifferent.
- Thus orders can be to buy or sell (or might not exist).
- Regulations in many countries constrain *short sales*.
- May even need to specify if a buy covers a short position.
- Thus there are more than just two types of orders.

Order Types

- In US equity markets, there are four basic order types:
 - Buy (B); Sell (S); Sell Short (SS); and,
 - Sell Short Exempt (SSE).
- SSE's are used by official market makers to sell short:
 - without borrowing the security; and,
 - (erstwhile) without being subject to a price test.
 - Thus they exist to increase liquidity.

Orders: Price Immediacy

Orders may trade immediacy for price sensitivity.

- *Market* orders execute immediately against “far side.”
- *Limit* orders specify worst acceptable price.
 - *Marketable limit* orders should execute immediately.
- *Market if touched*: activate at near price, e.g. exit at profit.
- *Stop (loss)*: activate at far price, e.g. exit at loss.

Quotes and Limit Order Books

- Market orders do not persist; they trade, then go away.
- *Quotes* are collections of active limit orders.
 - Buy orders are collected by price and become *bids*;
 - Sell orders are collected by price and become *offers/asks*.
- So quotes are prices at which people stand ready to buy/sell.
- A *limit order book* (“book”):
 - is the total amount at each price for standing limit orders.
 - changes if a limit order is received, corrected, canceled, filled.
 - often excludes non-“standard-way” orders.
 - may show individual order amounts at each price level.

Spread

- The highest bid amongst a set is the *best bid*.
- The lowest offer amongst a set is the *best offer*.
- Together these are the *best bid and offer* (BBO).
- BBO across all market centers: national BBO (NBBO).
- NBBO is better known as the *spread* or the *inside market*.
- If you buy/sell:
 - the best bid/ask is the *near side* of the spread; and,
 - the best ask/bid is the *far side* of the spread.
- In the UK, the spread is often called the *touch*.

Orders: Time in Force

Non-market orders may exist for a specified time period.

- *Good-til-canceled* (GTC) orders stay active until canceled.
- *Day* orders are canceled after the close/closing auction.
- *Immediate or cancel* (IOC), NOW orders.
 - Canceled after trading (or not) against book/CLOB_{fast}.
- Market auction orders may trade at any price:
 - *Market on open* (MOO/OPG), *Market on close* (MOC).
- Limit auction orders may trade at limit price or better.
 - *Limit on open* (LOO), *Limit on close* (LOC).
- Imbalance only (IO) orders trade against auction imbalances.

Market Intermediaries

- Some actors in secondary markets are intermediaries.
 - They may *broker* trades by matching buyers and sellers.
 - They may *deal* with just a buyer or seller.
- Brokering involves acting as an *agent*.
 - Must represent each customer's interests (fiduciary duty).
- Dealers act for their own (*principal, proprietary*) interest.
- There are also inter-broker dealers.
- Often, full-service investment firms are “broker-dealers.”

Intermediation and Order Responsibility

- Must mark orders as *agency* or *principal*.
- *Held* orders require “best execution.” (“agency” default)
- *Not held* orders give discretion to work the order.
- Order handling rules: fill agency before principal orders.
 - May not apply to odd-lot, block-size; across venues.
 - OTC bond markets: order handling is even more nebulous.
- *Front running*: agent favors principal over agency orders.
 - SEC, MiFiD* view this as a very serious offense.

* The EU Markets in Financial Instruments Directive.

After the Trade is Made: Settlement and Delivery

- Money and securities rarely transfer immediately after a trade.
- Money moves from buyer to seller at *settlement*.
 - Securities move from buyer to seller at *delivery*.
- Settlement and delivery usually occurs at the same time.
- Why not settle same-day (T+0)? Confirms need manual attention.
- Can specify non-standard settlement+delivery, but:
 -
 - Not “regular way” $\implies$ different, less-liquid market.
- Sometimes can specify different settlement, delivery dates.
 - Same as an unsecured, no-doc, no-questions-asked loan.

How Short Sales Work

- Selling short first requires borrowing securities from somebody.
- Most firms require that you get *locates* (find shares to borrow).
- You then sell those securities.
- Later, you buy securities back and return them to the lender.
- Short selling brouhaha a few years ago was about this.
 - Some traders shorted “naked” (without locates).
 - They then covered their shorts before delivery.
 - SEC required some shorts (banks) to have unique locates.

Selling Short: Foolish Controversy

Short selling is often controversial.

- Profit when a security/market declines in value (unpatriotic?).
- Has often been restricted, even in the US (via price tests).
- Even now: inconsistent attitude to shorts is stunning.
 - We blame shorts for bringing down bank stocks.
 - Who is thanking speculators for falling oil prices?
- Short selling rewards investors for exposing problems.
 - That oversight guides market prices toward fair value¹⁴.
- You cannot stop short selling:
 - Short exposure can also be had via swaps or options.
 - For indexed securities, you can synthetically short.

Regulation

Regulation I

- 1933 *Securities Act*: must register, disclose relevant info before issuing securities.
- 1933 *Banking (Glass-Steagall) Act* barred commercial banks from investment banking, brokerage; created *FDIC*.
- 1934 *Securities Exchange Act*: must register publicly-traded companies, report financials periodically; created *SEC*.
- 1936 *Commodities Exchange Act* regulates futures, commodities, commodities options trading.
- 1938 *Maloney Act* authorizes self-regulatory organizations (SROs).
- 1939 *Trust Indenture Act* requires debt issued w/indenture, regular financial reports, independent overseer.

Regulation II

- 1940 *Investment Company Act* requires registration, regulation by SEC; sets operating standards (incl. for short sales).
- 1940 *Investment Advisers Act*: advisers must register w/SEC, disclose conflicts of interest.
- 1956 *Bank Holding Company Act* required BHCs to register w/Fed; prohibits non-bank activities at subsidiaries.
- 1956 *Uniform Securities Act*: model for state anti-fraud (“blue sky”) laws.
- 1970 *Securities Investor Protection Act* protects investors vs misappropriation, brokerage failure; created *SIPC*.
- 1974 *Commodity Futures Trading Commission Act* created *CFTC*; allows creation of SRO.
- 1982 *Shad-Johnson Accord*: prohibited futures on stocks, narrow-based indices (SEC, CFTC “turf” detente).

The Road Ahead

We have covered market basics; on to modern markets next time!

- Fundamentals: Modern Markets, Efficiency;
- Measuring: Returns, Risk, and Diversification;
- Valuation: Fixed Income, Yield Curves, Equity Valuation;
- Valuation II: Factor Models, Microfoundations, Global Investing, FX;
- Risk Alleviation: Futures, Options, Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

Modern Markets

Chapter 4

Introduction

This part discusses modern markets. Specifically:

- Electronic vs algorithmic vs high-frequency trading;
- Intermediation without specialists;
- Recent pro-competitive changes;
- Market structure effects;
- Problems and pseudo-problems; and,
- Recent regulation.

Electronic Markets: How Soon Is Now?

- Almost all stock markets now allow electronic trading.
- Pre-1971, stocks listed on NYSE or AMEX — or were unlisted.
 - Listing: requirements on reporting, size, etc.
 - Unlisted stocks traded OTC by broker-dealers.
 - Natl Assn of Securities Dealers (NASD) oversaw them.
- 1969: Instinet; 1971: NASD automated quotation = NASDAQ.
- 1977: Toronto Stock Exchange Computer Assisted Trading System.
- Mid-1980s: Electronic derivatives trading introduced.
- Fully electronic stocks: Chile, 1989. (FI, SE, AU in 1990)

What Distinguishes Electronic Markets?

Jain (2005) compares electronic and floor-based markets:

Feature	Floor	Electronic
Counterparty ID	Known pre-trade	Rarely known pre-trade; often known post-trade
Order Book	Often none/fleeting	Constantly exists
Liquidity Seen	Bid-Ask	N best bid-asks or entire order book
Matching Speed	10s–minutes	Sub-second
Order/Oper. Costs	Higher	Lower
Settlement Speed	Slower	Faster
Settlement Cost	Higher	Lower

Advantages of Electronic Trading

Electronic trading has many benefits:

- Easily record history of interactions (*audit trail*).
 - Lets us assess orders and trades: Did we get good execution?
- Allows machines (*trading engines*) to do the work of trading.
 - Automation $\implies$ economies of scale, reduces explicit costs.
 - Engines are unemotional; behavior does not change/get “spooked.”
- Audit trail and automation can *reduce implicit order costs*.
- Easy data collection lets us conduct experiments, *tune behavior*.
 - Tunable/adaptive machines are the core of *algorithmic trading*.
- Forces market (and agent) logic to be documented (in code).
 - Disincentive to encode manipulative/unfair behavior.

Disadvantages of Electronic Trading

Electronic trading also has some drawbacks:

- Trading engines are only as good as we program them to be.
 - Recall: some concepts are not very well defined.
- Machines lack common sense; will do what you tell them.
- Startup (and incremental) infrastructure costs may be large.
 - This can be a barrier to entry for possible competitors.
 - May increase sophisticated–unsophisticated trader gap.
- Hard to gauge counterparty quality; important for some markets.

“Arms race” or constant evolution? (Sigh. Just. . . no.)

Market Makers (MMs)

- NASDAQ was run by competing broker-dealers; no specialists.
- These competing broker-dealers are called *market makers*.
- Market makers are different from specialists:
 - Compete w/each other; no monopolies on information or control.
 - Competition should yield fair and orderly market.
- Contrast the specialist with the market maker (MM):
 - Specialist chooses when to step in, can stop market.
 - MM has no advantage on discerning when to step in.
- Thus MMs constantly face a tough optimization: profits vs risk.

Electronic Communications Networks (ECNs)

- First electronic communications network (ECN): 1969.
 - Instinet created by institutional traders.
 - Way to trade large volume without MMs, paying full spread.
- An ECN is a limit order book that:
 - Displays orders; matches them in some priority; and,
 - Informs involved traders, outside world of execution.
 - Essentially, an ECN is an automated broker.
- Can see price you will get (low uncertainty).
- Superior software handles large volumes with low latency.
 - 2000: NYSE fill $\approx 3s$; Island fill $\approx 300ms$ round-trip.
 - 2003: Island fill $\approx 20-30ms$ round-trip.
 - 2008: BATS match $\approx 500\mu s$.
 - Now: BATS match $\approx 150\mu s$; most markets 2–20 ms round-trip.

Competition Lags: For a Few Dollars More

- Through mid-1990s, many investors insisted MMs overcharged.
- Christie and Schultz, "Why Do NASDAQ Market Makers Avoid Odd Eighth Quotes?" *Journal of Finance*, 1994.

- Is this pattern unusual? Yes, maybe 1-in-a-few-million.

Outcome: SEC charged >40 market makers with collusion.

1997: SEC changed rules to encourage hyper competition.

1998: settlements totalling more than \$1 bn.

From Rule Changes to Reg NMS

- To reduce collusion (and to punish), SEC adopted new rules:
 - 1997: 11Ac1-4, must reflect customer orders in bid+ask.
 - 1997: 11Ac1-1 changed, must quote if $>1\%$ of stock volume.
 - 2000: 11Ac1-5, public+standard execution quality reports.
 - 2000: 11Ac1-6, public+standard order routing reports.
 - Minimum tick size reduced to 1/16-ths (1997), \$0.01 (2001).
- 2005: Reg NMS repackaged these rules plus others.
 - Access Rule: intermarket linkages to access quotes.
 - Sub-penny Pricing Rule: No price increments $< \$0.01$.
 - Market Data Rules: fees re-allocated by execution quality.

Competition Arrives: Let a Hundred Flowers Bloom

- Changes came at a propitious time: internet bubble burst.
 - Good programming talent had just become very cheap.
- Also was a sharp rise in two ways to promote new venues:
 - *Rebates+Access Fees*: ECNs pay limit orders upon trading, and marketable order pays; aka “maker-taker fees”; and,
 - *Pay-for-flow*: MMs pay broker for marketable orders.
- These+execution quality reports= major shifts in market share.
- Theory and evidence so far are clear, though:
 - Maker-taker fees are OK, do not hurt investors; but,
 - Pay-for-flow raises cost of trading, is a *dominant strategy*.

Differences in Intermediation?

- Glosten (1989, 1994) studied different forms of intermediation.
- MMs: Price increases linearly with order size.
 - However: market can fail if too many informed customers.
- ECN: Price slightly convex function of order size.
 - Huge orders get worse prices; even tiny orders pay fixed cost.
- Specialist: large orders get better price than small orders.
(Loses on large orders to make money on small orders.)
- Are these necessarily good or bad? No.
- Rather, means we want a balance of all intermediary types.

Evidence of Success

- Stoll and Schenzler (2006): looked at spreads 1993–2001.
 - Quoted spreads dropped; mean: \$0.16 to \$0.01.
 - Effective spreads dropped; mean: \$0.12 to \$0.02.
- These changes are largely why you can now buy stock for \$10.
- Inspired EU Markets in Financial Instruments Directive (MiFID).
 - MiFID is like Reg NMS for EU markets; similarly successful.
- These changes have yielded decentralized “fragmented” markets.
 - But: huge cost savings; competition keeps markets together.

Decentralization Pros and Cons

- Main advantage, competition, brought many key changes:
 - Forced exchanges to execute orders faster and cheaper;
 - Popularized certain data (e.g. limit order book=LOB);
 - Punished cumbersome/anti-competitive market rules; and,
 - Made markets more robust to problems.
- However, decentralization fractures liquidity:
 - Liquidity tends to attract more liquidity;
 - Increases likelihood of being *traded through*;
 - Someone must connect to, monitor, route among N venues;
 - Unsophisticated customers likely ignorant of these issues.

Reg NMS and CLOB Issues

- Reg NMS stopped short of requiring a consolidated LOB.
- Requirements to achieve synchrony would have been onerous.
- For now, anti-competitive aspects of CLOB made clear.
- But requirements to route away have a cost (esp. speed). Some
- venues find this cost intolerable, anti-competitive. One way out:
- go “dark.”

Island Goes Dark (Hendershott and Jones, 2005)

- Early 2000's: Island took AMEX, NYSE market share by being *fast*.
- 2002: Island protested proposal to mandate intermarket linkage.
 - #1 venue for three most active ETFs; stopped publishing LOB.
- Effects were sudden and strong:
 - Market share: 58% ↓ 35%; price discovery moved to futures;
 - Trading costs increased overall; and,
 - Prices adjusted more slowly (quote changes became predictable).
- Island later re-illuminated; execution quality then increased.
- Government relented on proposal. . . but occasionally reconsiders.

Negotiated and Program Trades

- Some investors use negotiated trades for large orders.
 - Negotiated due to liquidity, adverse selection risks.
- Negotiated trades include:
 - block trades (10,000+ shares, direct search, declining);
 - arrival price (PRP) trades (priced wrt current bid/ask);
 - volume-weighted average price (VWAP) trades.
- Most negotiated trades are programs=large portfolio trades.

Program Trading and The Rise of the Machines

- Pre-1997, most trading automation was done for program trades:
 - *Program trade*: 15 or more names totalling >\$1 million.
 - Initially done w/runner+*program* (itinerary) of orders.
 - Program trading *does not* mean trading by computers.
 - However, program trading was an early adopter of computers.
- As trading got cheaper, more flow routed through program trading.
- True story bro: Systems trade cheaper, more reliably than humans.
 - Block desk VWAP: charge 8cps, miss by 12, s.d. of 150.
 - Algo VWAP: charge 2cps, miss by 1.5, s.d. of 0.5.
 - Guess which desk was empty six months later. . . .

Algorithmic Trading

- Thus *algorithmic trading* is the automation of trading.
- Uses execution, order management systems: OMS and EMS.
 - OMS: orders planned across time to hit a benchmark/trajectory.
 - EMS: order may be split across venues or updated strategically.
- This requires theory, careful thinking about how best to trade.
- Key OMS ideas: Bertsimas & Lo (1998), Almgren & Chriss (2000)
 - Idea: Split large (parent) order into smaller (child) orders.
 - Time orders to optimize mean vs variance of trading cost.
- EMS: key theory can be very complicated, may interact with OMS.

Index Arbitrage

- An early accomplishment of program trading: *index arbitrage*.
 - Trade index futures vs index stocks when implied prices differ.
 - Trader makes money at very low risk, low capital requirement.
 - Effect of trading: pushes prices closer together.
- Index arb is very beneficial to other investors:
 - Stocks, ETFs, index futures all kept at nearly same index level.
 - Thus I need not search for best deal to invest in index.
 - This also saves me money by reducing my IT spending.
- But... stocks traded in NYC; futures/options traded in Chicago.
- Hence high-speed (5ms vs 28ms) NY–Chicago microwave link.

Short Sales Redux

- 1994–2001: huge increase in market efficiency, trading speed.
 - Efficiency? Lower bid-ask spreads, quote changes less predictable.
 - If prices closer to true value, get better allocative efficiency.
- One major laggard: short sales; still required price tests.
 - Could not short stock if it just fell or at price below bid.
- 2005: SEC tries an experiment; Reg SHO eliminates price tests.
- Idea: Bring efficiency to pricing for overvalued stocks.
- Result: Positive... w/tweaks (must have capital+locates).

Pushing the Envelope: High-Frequency Trading

- At this time, simplest decisions became *highly* optimized.
 - Made by true/false choices; even multiplying is too slow!
- Idea: buy/sell instrument; exit position ASAP on another venue.
- True/false choices implied venue most likely to earn slight profit.
- Main speed advantage? From optimizing networks+connections.
- This is what constitutes *high frequency trading*.
- Make money on only 1 in 10000 (50000?) trades; else: break even.
- But they trade a lot, so money adds up. (LLN at work)
- How do HFTs add value? They keep prices the same across markets.
 - Use their speed to spread information across all venues.
 - Thus we need not fret so much about where we trade.

A Momentary Lapse of Reason

- Glitches with any automation can get bad quickly.
 - Trading engines may mishandle orders or break rules/laws.
 - However, code documents logic, eases finding+fixing errors.
 - 1 Aug 2012: Knight Capital bug costs \$440 MM in 45 minutes.
- However, human breaches of common sense can also be nasty.
- “Fat finger” errors are classic examples of this.
 - 9 Dec 2005 J-Com IPO; Mizuho employee fat fingers sell order.
 - Sells 610,000 at ¥1 (instead of 1 at ¥610,000).
 - TSE computers so busy, cancel takes 10 minutes. Cost: \$330 MM.

The Flash Crash

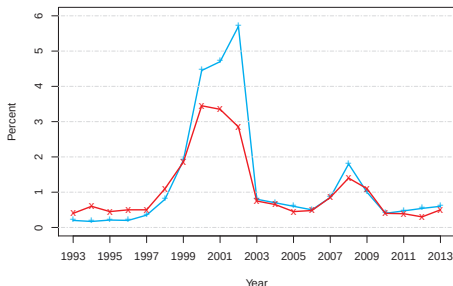
- 6 May 2010: US markets open down (concern about Greece).
 - 2:32 PM: Waddell & Reed sell, 75,000 ES via %-of-volume.
 - HFT firms provide liquidity, volume $\uparrow$ so W&R scale up too.
 - 2:41 PM: HFT firms hit risk limits, sell inventory.
 - 2:41–2:44 PM: S&P 500 futures (ES+SP), ETFs drop 3%.
 - Index arb engines trade heavily; effects bleed into equities.
 - Trading, MM engines exit as Δ volume triggers error flags.
 - 2:45:13–2:45:27 PM: Half the day's ES volume trades in 14s.
 - 2:45:28 PM: CME pauses ES, SP for 5s; sends reset signal.
 - 2:45:33 PM: Stocks and futures begin recovering.
 - 3:00 PM: Most price changes from 2:32 on have reverted.
 - Post-market-close: Some trades at price extremes canceled.
-

Flash Crash Questions

- Was it reckless to send a large %-of-volume order?
- Was it reckless to not supervise that order intensely?
- Should HFT firms have behaved differently? If so, how?
- How should trading engines handle unusual/erroneous data?
- Should trading+MM engines have stayed in the market?
- How did canceling trades affect traders the next day?
- Who was the main culprit? Who was hurt?
- Are there longer-term effects of canceling extreme trades?
- What are the longer term effects of such a crash?

Market Quality: Gao and Mizrach (2016)

- We may worry about how modern markets hold up to stress.
- Gao and Mizrach (2016) study prevalence of flash crashes.
- Turns out they are not new nor exploding in frequency.



Regulation III

1999 *Gramm-Leach-Bliley Act*

- repealed parts of Glass-Steagall, Bank Holding Co. Acts;
- allowed mixing banking, inv. banking/brokerage, insurance.

2000 *Commodity Futures Modernization Act*

- repealed Shad-Johnson Accord;
- exempted off-exchange energy trading.

2002 *Sarbanes-Oxley Act*

- CEOs, CFOs must certify reports;
- severe penalties for fraud, malfeasance; created PCAOB.

2010 *Dodd-Frank Act*

- demands central clearing, execution facilities, public quote+trade data for most derivatives;
- creates Financial Stability Oversight Council, OFR;
- Volcker Rule: restricts proprietary trading.

Some Thoughts on Regulation

- Often, talk of regulation is strongly polarized.
- Some insist gov't is the problem; deregulation yields growth.
- Others insist regulation is needed to make capitalism work.
- However, note the theme of the best regulation we've covered:
 - Encourage move from opacity to open markets; and,
 - Create standards so competition can flourish + self-police.
- Also, think carefully about what such changes look like:
 - A group which resists competition will accrue lots of money.
 - Smart move: use that money to depict change as bad.
 - Always ask: who says it's bad and how will this affect them?
- Theory and data — not claims — should guide policy.

The Road Ahead

We have covered modern markets; on to efficiency next time!

- Fundamentals: Efficiency;
- Measuring: Returns, Risk, and Diversification;
- Valuation: Fixed Income, Yield Curves, Equity Valuation;
- Valuation II: Factor Models, Microfoundations, Global Investing, FX;
- Risk Alleviation: Futures, Options, Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

Efficiency and the Macroeconomy

Chapter 5



Introduction

This part discusses efficiency and the macroeconomy. Specifically:

- How do we think about incentives and production?
- What different types of efficiency are there?
- What do these imply for markets?
- How do we look at the macroeconomy?

Incentives

- The overarching message of economics: incentives matter.
- Centuries of data show: changing incentives changes behavior.
- *Incentive compatibility*: when people's incentives are aligned.
 - Suppose I pay you \$5 for each object you pick up off the floor.
 - Is this incentive compatible? *cf.* Enron removing congestion.
- Goal: policies incentive compatible with growth, better outcomes.
- How do we measure growth? outcomes? trade-offs?

Production Functions

- Helps to use an economists' tool: *production functions*.
 - Economies create output Y from *factors*: capital K , labor L .
 - For example, economy i : $Y_i = A_i f(K_i, L_i)$.
 - Then we can talk about A_i : *total factor productivity* (TFP).
 - Why is TFP higher in Singapore vs Malaysia? Chile vs Argentina?
- Big question: is f same for all economies? Or different f_i ?
 - Increasing (IRS) vs decreasing returns to scale (DRS)?
 - IRS? More money you have, faster you grow; rich get much richer.
 - DRS? Eventually, growth tapers off. (Inefficiencies?)

Production Functions and Growth

- Certain policies seem to increase growth, relate to higher TFP or IRS.
- *Why Nations Fail*: inclusive economic+political systems.
 - What many economists refer to as “institution quality.”
 - TRADE, low corruption, representative democracy, freedoms.
 - Stable/fair laws (esp. property+contracts), competition.
 - Low taxes? Yes, but the others are more important (cf California).
- Will see these often affect efficiency.

Market Efficiency

- Often, “efficiency” in finance means *market efficiency*.
- Idea: Do prices incorporate new information quickly and fully?
- Common measure: can we easily predict quote changes?

$$\underbrace{\Delta q_t}_{\substack{\text{quote change} \\ = q_t - q_{t-1}}} = \alpha + \phi \underbrace{\Delta q_{t-1}}_{\substack{\text{previous} \\ \text{quote change}}} . \quad (1)$$

- Can also look at bid-ask spreads: wider $\implies$ more uncertainty.
- Or we can look at how much prices move apart from overall market.
- Why do we care? *Price theory*: prices are signals to the economy.
- Muffling, slowing, messing with prices $\implies$ lower growth, less jobs.
- cf. Transaction taxes in France, Italy vs UK, Germany.

Allocative Efficiency

- We often talk about market efficiency but prefer other efficiencies.
- Why? We assume — hope! — these efficiencies are correlated.
- *Allocative efficiency*: ideal capital allocation to economy's industries?
- If so, money goes where demand is; economy grows optimally.
- How do we know ideal industry mix? Two ideas:
 - ① Does increase in *value-added* attract increase in investment?
Strongest: Germany, Hong Kong, Denmark, New Zealand.
 - ② Does firm value respond quickly to capital shocks?
No $\implies$ inefficient firm; had “fat” to trim.

X-Efficiency

- Is industry's capital optimally allocated to firms/plants?
 - Do Ford, GM, Tesla, and Toyota all add equal value for \$1 spent?
 - Are all Ford, GM plants equally efficient?
- *X-efficiency*: efficiency of capital deployment within an industry.
- How to measure this? Tough; often measure firms' relative efficiency.
- X-efficiency: one of the most powerful and elusive of efficiencies.
 - Reallocating w/in industry: +30%–+60% to Chinese, Indian GDP. (!)
 - More developed economies have higher levels of X-efficiency.

Dynamic Efficiency: Creative Destruction

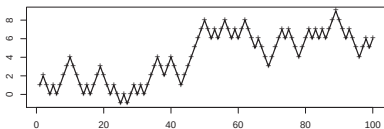
- *Dynamic efficiency* is the other strong force of growth.
- How easily better ideas/methods replace outdated ideas/methods.
- *Creative destruction* also known as *Schumpeter's Gale*.
- e.g. travel agents, manual metalworking, paper maps, snail mail.
- We see this efficiency impaired in many developing economies.
 - Healthy finance sector increases creative destruction.

How Efficiencies Relate

- It turns out that these efficiencies do all relate.
- More market efficiency $\implies$ more allocative efficiency, X-efficiency.
- Financial innovation increases creative destruction.
- Dynamic efficiency: mix of allocative, X-efficiency — across time.
- ! Creative destruction displaces workers = social upheaval.
- Should we “protect” jobs, stop/tax/regulate trade and innovation?
 - History is crystal clear: These lead to less growth, less jobs, brain drain.
- So what is the value of, e.g. high-frequency trading? Actually: a lot.
- If prices are not efficient, is price theory broken?
- No: Inefficiency likely dwarfed by uncertainty, optimally inefficient.

Random Walk

- Suppose markets in equilibrium, investments priced right.
- Might expect price changes to be *random walks*.



- Best guess at price next minute/day/month = price now.
- Equally likely to go above/below risk-free return.
- Proposed by Regnault (1863), Bachelier (1900, Brownian motion).

Forms of the EMH

- The EMH assumes that prices reflect past information.
- Thus three forms of EMH reflect increasing information:
 - *Weak form*: all past trade prices and volumes.
 - *Semistrong form*: all public information.
 - *Strong form*: all public, private information.
- Strong form: generally thought untrue, extreme, dogmatic.
- The semistrong form would imply any research is fruitless.
- Grossman and Stiglitz (1980): information cost matters.
 - Plainly: *somebody* has to make prices efficient; why not you?
 - Thus in equilibrium, research should be fruitful.

EMH Implications

- *Technical analysis* searches for patterns in prices.
 - Requires belief that prices adjust slowly to supply, demand.
 - Where we get terminology for “support,” “resistance.”
 - All EMH forms say technical analysis should not work.
- *Fundamental analysis* studies firm finances, cashflows.
 - Want to find firms better/worse off than most people think.
 - Semistrong, strong forms of EMH imply this is fruitless.
- *Active portfolio management* only strong form EMH opposes.
- Most people lack money, knowledge for intensive research.
 - EMH adherents suggest *passive investments*: index funds.
 - This is not bad advice since these funds have rock-bottom costs.

Noisy Market Hypothesis and Fundamental Indexing

- EMH implies holding “market.” (market-cap-weighted index)
- Arnott (2006) and others say this is suboptimal.
- *Noisy market hypothesis*: prices include random errors.
 - Some stocks over-/under-valued vs. intrinsic value.
- Cap-weighted indices over-invest in over-valued instruments.
- *Fundamental indexing*: weight by intrinsic value instead.
- How to determine intrinsic value? Tough.
- However, that does not mean that the criticism is invalid.

The Macroeconomy

- All of these thoughts on efficiencies play out on a larger stage.
- Thus we care about the *macroeconomy*: national/global economy.
- Predicting the macroeconomy is insanely hard; would be pointless. . .
- . . . except that even a so-so guess could be very valuable.

Global Economy

- Almost all firms affected by global economy.
 - May help/hinder competitors, foreign operations.
 - *Currency risk* may affect international revenues, costs.
 - Exchange rate governs relative *purchasing power*.
- *Political risk* also cannot be ignored.
- Currency risk may seem minor; beware *Peso Problems*:
 - Currency-adjusted interest rate may look attractive.
 - Except market may have priced in a possible devaluation.

Domestic Macroeconomy

Describing economy's health is complicated; use many metrics.

- *Gross domestic product* (GDP): all goods, services produced.
- *Unemployment rate*: % of workers, job seekers without a job.
- *Capacity utilization rate*: factory output $\div$ possible output.
 - Like an employment rate for factories.
- ~~*Participation rate*~~: distorted by retirees, gig economy.
- *Inflation*: aggregate rate of increase in nominal prices.
- *Interest rates*: the price of money at various times.
- *Budget deficit*: government spending less revenue*.
- *Sentiment*: confidence of people, businesses in economy.

* Zero deficit $\neq$ zero debt!

Sentiment Indicators

- *Breadth*: # risers – # fallers.
- Weight by volume to get *trin statistic*:

$$\text{trin} = \frac{\text{volume(fallers)}/\# \text{ fallers}}{\text{volume(risers)}/\# \text{ risers}} = \frac{\text{avg faller volume}}{\text{avg riser volume}}. \quad (2)$$

- *Confidence index* = $\frac{\text{Avg yield}(10 \text{ top-rated corporates})}{\text{Avg yield}(10 \text{ intermediate-rated corporates})}$.
 - Similar in spirit to a credit spread.
- *Put/Call Ratio*: = $\frac{\# \text{ outstanding put options}}{\# \text{ outstanding call options}}$ ^{usually} $\approx 65\%$.
- TED Spread: 3M T-bills–3MUSDLIBOR; TED > 50 bp $\implies$ crisis.
- VIX: CBOE S&P 500 volatility index
- Consumer confidence surveys

Demand Shocks

- *Demand shocks* affect consumption (spending).
- *Fiscal policy* deals with taxing and spending.
 - Spend to inflate demand for goods.
 - Raise taxes for income, to reduce consumption.
- *Monetary policy* deals with control of the money supply.
 - Fed may buy bonds: lowers yields, pays by “printing” money.
 - Fed may lower fed funds rate, increasing money circulating.
- In general, fiscal policy is more selective and targeted.
 - Unfortunately, setting fiscal policy is political, takes time.
 - Monetary policy: blunt, but less political, enacted quickly.

Supply Shocks

- *Supply shocks* affect producers (costs).
- Supply side policies seek to reduce frictions on business.
 - Burdensome regulation;
 - Laws/taxes which discourage innovation.
- Supply-siders also believe higher taxes may yield less revenue.
 - *Laffer curve* suggests this, implies revenue-optimal tax rate.
 - Idea: High taxes may remove rewards of taking risk.
 - Less risk-taking reduces innovation, revenue to tax.

Business Cycles

- Economy experiences periods of contraction and growth.
- *Business cycle*: the irregular pattern of recession, recovery.
- Often talk of *peaks* (local maxima), *troughs* (local minima).
- *Cyclical industries* are sensitive to business cycle.
 - High-cost items are generally considered cyclical goods.
- *Defensive industries* are often necessities, least cycle-sensitive.
- Much research on cycle prediction, typical cycle durations.
 - Typical peak–peak duration: 6–7 years.

Business Cycles: Prediction

- If business cycle prediction trivial, make easy money.
- Some indicators help predict or interpret cycle.
- *Leading indicators*: correlated with later cycle.
- *Coincident indicators*: correlated with simultaneous cycle.
- *Lagging indicators*: correlated with prior cycles.
- Indicators released regularly; published by financial press.
- Why do we care about lagging indicators?

The Road Ahead

We have covered efficiency; on to measuring and returns next time!

- Measuring: Returns, Risk, Diversification, and Modeling;
- Valuation: Fixed Income, Yield Curves, Equity Valuation;
- Valuation II: Factor Models, Microfoundations, Global Investing, FX;
- Risk Alleviation: Futures, Options, Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

Quantitative Investments

Medición Temas 6, 7 y 9

Returns

Chapter 6



Introduction

This part discusses returns:

- Time value of money;
- Compounding;
- Real and nominal interest rates;
- Comparing rates of return;

Time Value of Money

- #1 concept in finance: the *time value of money*.
- Simply: money now is worth more than money later.
- Many reasons for this:
 - Waiting = deferring gratification which deserves a reward;
 - \$1 now less risky: safe if you need it before “later;”
 - Opportunity cost: \$1 invested now likely worth more later.
- Do we need interest or a risk-free rate to believe this? No.
- However, to calculate equivalences we do need some rate.

Risk-Free Rates and Returns

- Often do not promise a *return*; if so, we call it a *rate*.
- Often calculate equivalences w/ “risk-free” interest rates.
- Can we avoid the concept of risk-free returns? Sure.
 - Use the same rates as those promised by lowest-risk investments?
 - Find risk-return relationship; then imply rate for no risk?
- Furthermore, are “risk-free” rates risk free? Surprisingly, no.
- But for convenience, clarity: we often use “risk-free” rates.
- Why? Financial analyses rest on a foundation of risk-free rates.
 - But politicians toy w/defaulting on US bonds! Yes, they do.
 - Default would blow up all financial models; like losing a compass.

Present and Future Values

- We then use a risk-free rate to find equivalences of money.
- Rates should reflect people's aggregate preferences.
- *Present value* of later money:
 - How much money market would accept now in lieu of \$1 later.
- This implies some risk-free *discount factor* d .
 - If \$1 in 6 months were worth \$0.97, then $d = 0.97$.
- To compute present values: $PV = \text{money} \times \text{discount factor}$.
- *Future value* is the opposite: what your money is worth later.
- Say we'd accept \$0.98 for \$1 in six months. Then $d = 0.98$, and:
 - $PV(\$1,6M) = \0.98 , and $FV(\$1,6M) = \$1/d = \$1.0204\dots$

Using Risk-Free Rates

- Typically, we calculate PV's and FV's using a risk-free rate.
- By convention, when we quote rates we quote them annualized.
- If $d = 0.98$ for 1 year, risk-free rate
$$1 + r = 1/d \implies r = 0.0204 = 2.04\%.$$
- But what if $d = 0.98$ for six months? Or two years?
- We use # of years T (0.5, 2) to get the annualized rate:
 - $\$1 \cdot d = \frac{\$1}{(1+r)^T} \implies \frac{1}{d} = (1+r)^T \implies r = \frac{1}{d^{1/T}} - 1$
 - 6M: $1/0.98 = (1+r)^{0.5} \implies r = 1/0.98^2 - 1 = 0.0412 = 4.12\%$
 - 2Y: $1/0.98 = (1+r)^2 \implies r = 1/\sqrt{0.98} - 1 = 0.01015 = 1.015\%$

Interest Payments

- If we realize gains occasionally, we get paid *interest*.
- Can reinvest interest; *compounding*: earning gains on prior gains.
- However, how we quote rates is still annualized; so we split gains.
- Say we get 3% interest on a \$100 investment; would be paid as:
 - Annually: \$3 once a year;
 - Semi-annually: \$1.50, twice a year;
 - Monthly: \$0.25, each month; or
 - Daily: \$0.0082, each day.

Compounding

- Reinvesting interest, however, changes how much we earn.
- If interest paid N times/year, our \$100 grows in 1 year to:

$$FV = \$100\left(1 + \frac{r}{N}\right)^{NT}. \quad (1)$$

- Annually: $FV = \$100 \cdot 1.03^{1 \cdot 1} = \103 ;
 - Semi-annually: $FV = \$100 \cdot 1.015^{2 \cdot 1} = \103.0225 ;
 - Monthly: $FV = \$100 \cdot (1 + 0.03/12)^{12} = \103.0416 ; or
 - Daily: $FV = \$100 \cdot (1 + 0.03/365)^{365} = \103.0453 .
- If we take the limit as the compounding frequency grows:

$$FV = \lim_{N \rightarrow \infty} \$100\left(1 + \frac{r}{N}\right)^{NT} = \$100e^{rT}. \quad (2)$$

- Continuous: $FV = \$100 \cdot e^{0.03 \cdot 1} = \103.0455 .

Compounding II

- What if we were only earning interest for 9 months?
- $FV = \$100(1 + \frac{r}{N})^{NT}$ still holds; in 9 months we'd have:
 - Annually: $FV = \$100 \cdot 1.03^{1 \cdot 0.75} = \102.2417 ;
 - Semi-annually: $FV = \$100 \cdot 1.015^{2 \cdot 0.75} = \102.2584 ;
 - Monthly: $FV = \$100 \cdot (1 + 0.03/12)^{12 \cdot 0.75} = \102.2726 ; or
 - Daily: $FV = \$100 \cdot (1 + 0.03/365)^{365 \cdot 0.75} = \102.2754 .
 - Continuous: $FV = \$100 \cdot e^{0.03 \cdot 0.75} = \102.2755 .

Comparing Rates of Return

- *Holding period return*: $R = \frac{\text{End Price} - \text{Start Price} + \text{Income}}{\text{Start Price}}$.
 - For zero-coupon bond bought at B_t , $R = \frac{100 - B_t}{B_t}$.
- How to compare instruments with different maturities?
 - *Effective Annual Return* (EAR): return over one year's time.
 - *Annual Percentage Rate* (APR): scaled period return.
Scaled by number of compounding periods/year.
- For a bond paying 10% every 6 months:
 - $\text{EAR} = (1 + 0.1)(1 + 0.1) - 1 = 0.21 \Rightarrow 21\%$
 - $\text{APR} = 0.1 \times 2 = 0.20 \Rightarrow 20\%$.
- Continuously compounded interest rate $r_{cc} = \ln_e(1 + \text{EAR})$.
 - $\text{Interest} = Pe^{r_{cc}T} - P$ (P : principal, T : time (years))

Interest Rates

- So we clearly need interest rates for many financial analyses.
- However, we can think of interest rates in different ways.
- *Nominal rates* give growth rate of invested money.
- *Real rates* give growth rate of investment purchasing power.
 - Thus real rates correct for inflation.
 - US inflation usually measured by Consumer Price Index (CPI).
- Under annual compounding (*i.e.* using YOY change in CPI):

$$1 + \underbrace{r}_{\text{real rate}} = \frac{1 + \overbrace{R}^{\text{nominal rate}}}{1 + \underbrace{i}_{\text{inflation}}} = 1 + \frac{R - i}{1 + i} \approx 1 + R - i \quad \text{for small } i. \quad (3)$$

Equilibrium Interest Rates

- Interest rates are, quite simply, the price to rent money.
- Supply, demand, government actions affect real interest rates.
 - Government borrowing shifts demand; Fed can shift supply.
 - With inflation, these determine nominal interest rates.
- Fisher (1930): Nominal rates anticipate inflation.

$$\underbrace{R}_{\text{nominal rate}} = \underbrace{r}_{\text{real rate}} + \underbrace{E(i)}_{\text{expected inflation}} . \quad (4)$$

- Problem #1: Data does not conclusively support (4).
 - Problem #2: r is also random and (perhaps) unobservable.
- Instead: $E(R) = E(r) + E(i) + \lambda \text{Var}(i)$

Inflation and Taxes

- Suppose nominal interest is taxed at a rate τ .
 - Thus we keep interest of $R(1 - \tau)$.
- Compare this to being taxed on real interest:

$$\underbrace{R(1 - \tau) - i}_{\substack{\text{after-tax} \\ \text{real rate}}} \approx (r + i)(1 - \tau) - i = \underbrace{r(1 - \tau)}_{\substack{\text{real rate} \\ \text{less taxes}}} - \underbrace{i\tau}_{\substack{\text{taxed} \\ \text{inflation}}} . \quad (5)$$

- Note that $R(1 - \tau) - i < r(1 - \tau)$.
- Thus real taxes increase under inflation.

What is Risk-Free?

- A few points to clarify about risk-free rates.
- “Risk-free” rates guarantee a promised rate of return on money.
 - What about risk that inflation erodes our money's value?
 - Often ignored; cashflows, valuations all computed in nominal \$.
- Some people say risk-free rate is shortest-term. Not so!
 - Has *reinvestment risk*: unknown future short-term rates.
 - Need to find risk-free rate for your investment horizon.
- Also, “risk-free” bond prices change; can lose money.
- Worst of all: What if government defaults?

Returns

- Many texts go to great pains in working with returns:
 - Introduce arithmetic and geometric averages; and,
 - Then work to reconcile them. Why?
- Arithmetic averages are easier to work with but inaccurate.
- Geometric averages are accurate measures of mean returns.
- Need to know how both of these work.
- However, capital gains/losses involve continuous reinvestment.
- So we work with returns like continuously-compounded rates.
- The way to do this: work with log-returns.
- Advantage: Math is simpler; log-returns add over time.

Log-Returns

- Log-returns^{*} make arithmetic averages accurate. The
- trick is to *define* returns slightly differently:

$$r_1 = \ell n(p_2/p_1) = \ell n(p_2) - \ell n(p_1). \quad (6)$$

- Then, arithmetic averaging works properly:

$$\frac{r_N + \cdots + r_1}{N} = \frac{\ell n\left(\frac{p_{N+1}}{p_N}\right) + \cdots + \ell n\left(\frac{p_2}{p_1}\right)}{N} \quad (7)$$

$$= \frac{\ell n\left(\frac{p_{N+1}}{p_1}\right)}{N} = \ell n((p_{N+1}/p_1)^{1/N}). \quad (8)$$

- To recover geometric mean, exponentiate:

$$1 + g = e^{\ell n((p_{N+1}/p_1)^{1/N})} = \sqrt[N]{p_{N+1}/p_1}. \quad (9)$$

^{*}Always use base-e logs, also known as $\log_e$ or ℓ

Risk versus Returns

Chapter 7



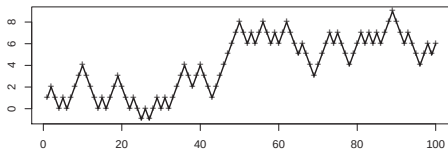
Introduction

This part discusses risky returns:

- Randomness;
- Statistical metrics for returns;
- How actual returns differ from normality; and,
- Risk premia.

Randomness

- When returns are uncertain, our perspective changes.
- Compounding is uncertain, may be bad: what if losses compound?
- Ideas on efficient markets guide how we model, assess risk.
- Recall Regnault and Bachelier modeling Paris stock prices.
- They, A.N.K., used a random walk (sum of disturbances):



$$X_0 = 0, \quad X_t = X_{t-1} + \epsilon_t \quad \epsilon_t \stackrel{iid}{\sim} \begin{cases} +1 & w.p. \, p \\ -1 & w.p. \, 1-p \end{cases} \quad \forall t = 1, 2, \dots$$

- If $p = 1/2$, X_t is a *martingale* ($E(X_{t+1}|X_t) = X_t$).

Honey, I Shrunk the Step Size

- This is not a bad model for prices, but it is a bit too “chunky.”
- Shrink time and space steps: process looks more like market prices.
 - Shrink time by factor of k , space by factor of $\sqrt{k}$.
 - Why $\sqrt{k}$ for space? Compounding, else growth differs.
- Get arithmetic or geometric Brownian motion (aka Wiener process):

$$dS_t = \mu dt + \sigma dW_t; \quad (\text{arithmetic}) \text{ or}, \quad (1)$$

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad (\text{geometric}). \quad (2)$$

- $E(W_t|W_0) = W_0$, $\text{Var}(W_t) = t$.
- Usually use geometric BM (stays >0); why we use log-returns.

Log>Returns Redux

- Derivative $d \log(S_t) = \frac{dS_t}{S_t}$ = continuous % change.
- To estimate μ and σ , sample market prices regularly.
- Then look at log-returns, differences of log(prices):

$$r_{t_i} = \log(S_{t_i}) - \log(S_{t_{i-1}}) = \mu \Delta t_i + \sigma(W_{t_i} - W_{t_{i-1}}). \quad (3)$$

- Breaking News! Beautiful mathematical model slain by ugly facts!
- Geometric Brownian motion $\implies$ log-returns normally-distributed.
- Problem: They are certainly not normally-distributed.
- Thus we often look at *moments*: expected powers, e.g. $E(r_{t_i}^2)$
- Especially *central moments*: remove mean, scale to unit variance.

Statistics Review: Mean, Variance, Volatility

- In general: we look at these metrics scaled to be annual.
- *Expected return*: probability-weighted average possible return.

$$\mu_r = E(r) = \int_{-\infty}^{\infty} r \underbrace{f(r)}_{\substack{\text{pdf} \\ \text{of } r}} dr \stackrel{\text{or}}{=} \sum_{i=1}^N r(i) \underbrace{p(i)}_{\substack{\text{pmf} \\ \text{of } r}} \quad (4)$$

- *Variance*: central second moment (aka $\text{var}(r)$).

$$\sigma_r^2 = \int_{-\infty}^{\infty} (r - \mu_r)^2 f(r) dr = \sum_{i=1}^N (r(i) - \mu_r)^2 p(i). \quad (5)$$

- *Volatility*^{*} is square root of variance: $\sigma_r = \sqrt{\text{Var}(r)}$.

^{*}Volatility is the standard deviation of annual log-returns.

Statistics Review: Higher Moments

- Higher-order moments are of concern for non-normal data.
- *Skewness*: standardized third moment.

$$\gamma_r = \int_{-\infty}^{\infty} \left(\frac{r - \mu_r}{\sigma_r} \right)^3 f(r) dr = \sum_{i=1}^N \left(\frac{r(i) - \mu_r}{\sigma_r} \right)^3 p(i). \quad (6)$$

- *Kurtosis*: standardized fourth moment.

$$\kappa_r = \int_{-\infty}^{\infty} \left(\frac{r - \mu_r}{\sigma_r} \right)^4 f(r) dr = \sum_{i=1}^N \left(\frac{r(i) - \mu_r}{\sigma_r} \right)^4 p(i). \quad (7)$$

- Normal distribution: $\gamma = 0$, $\kappa = 3$.

Sample Distribution Metrics

- For regularly-spaced returns, can easily estimate metrics.^{3*}
- Assume each observation is equally likely: $p(i) = \frac{1}{N}$.
 - Put “hats” on quantities we estimate.

$$\text{mean} = \hat{\mu}_r = \frac{1}{N} \sum_{i=1}^N r_i \quad (8)$$

$$\text{variance} = \hat{\sigma}_r^2 = \frac{1}{N-1} \sum_{i=1}^N (r_i - \hat{\mu}_r)^2 \quad (9)$$

$$\text{skewness} = \hat{\gamma}_r = \frac{1}{N-2} \sum_{i=1}^N \left(\frac{r_i - \hat{\mu}_r}{\hat{\sigma}_r} \right)^3 \quad (10)$$

$$\text{kurtosis} = \hat{\kappa}_r = \frac{1}{N-2} \sum_{i=1}^N \left(\frac{r_i - \hat{\mu}_r}{\hat{\sigma}_r} \right)^4 \quad (11)$$

*This estimation gets much more complicated with intraday returns.

Dependence

- Need dependence measures to find portfolio variance.
- *Covariance* of X , Y : $\Sigma_{XY} = E(XY) - E(X)E(Y)$.
- *Correlation* is covariance over $\sigma_X\sigma_Y$: $\rho_{XY} = \frac{\Sigma_{XY}}{\sigma_X\sigma_Y}$.
 - Correlation is bounded: $-1 \leq \rho_{XY} \leq +1$.
 - Thus covariance is bounded: $-\sigma_X\sigma_Y \leq \Sigma_{XY} \leq \sigma_X\sigma_Y$.
- Often write variance, covariances in covariance matrix Σ :

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \Sigma_{XY} & \cdots & \Sigma_{XZ} \\ \Sigma_{XY} & \sigma_Y^2 & \cdots & \Sigma_{YZ} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{ZX} & \Sigma_{ZY} & \cdots & \sigma_Z^2 \end{pmatrix}. \quad (12)$$

Portfolio Variance

Example portfolio variance, $\sigma_P^2 = w^T \Sigma w$, for three-asset portfolio:

- ① Write the weights along the top and left sides:

$$\begin{array}{c} w_X \quad w_Y \quad w_Z \\ w_X \left(\begin{array}{ccc} \sigma_X^2 & \Sigma_{XY} & \Sigma_{XZ} \\ \Sigma_{XY} & \sigma_Y^2 & \Sigma_{YZ} \\ \Sigma_{XZ} & \Sigma_{YZ} & \sigma_Z^2 \end{array} \right) \end{array} \quad (13)$$

- ② Then cross-multiply headings times cells:

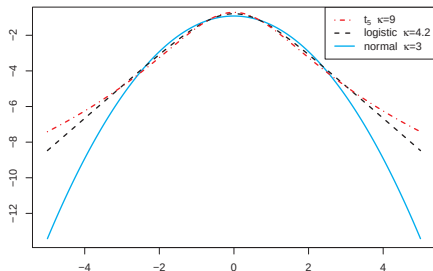
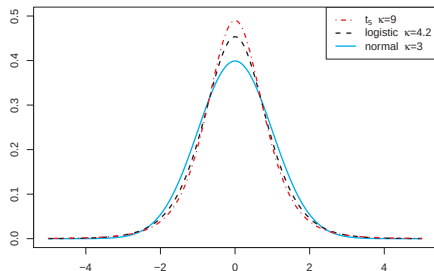
$$\begin{array}{c} w_X \quad w_Y \quad w_Z \\ w_X \left(\begin{array}{ccc} w_X^2 \sigma_X^2 & w_Y w_X \Sigma_{XY} & w_Z w_X \Sigma_{XZ} \\ w_X w_Y \Sigma_{XY} & w_Y^2 \sigma_Y^2 & w_Z w_Y \Sigma_{YZ} \\ w_X w_Z \Sigma_{XZ} & w_Y w_Z \Sigma_{YZ} & w_Z^2 \sigma_Z^2 \end{array} \right) \end{array} \quad (14)$$

- ③ Adding gives $\sigma_P^2 = w_X^2 \sigma_X^2 + w_Y^2 \sigma_Y^2 + w_Z^2 \sigma_Z^2 + 2w_X w_Y \Sigma_{XY} + 2w_X w_Z \Sigma_{XZ} + 2w_Y w_Z \Sigma_{YZ}$.

Comparing Distribution Metrics

- For comparison, we generally scale metrics to be annualized.
- Mean and variance (μ_r, σ_r^2) scale linearly with time.
 - For US stocks, $\mu_{\text{annual}} = 250\mu_{\text{daily}}, \sigma_{\text{annual}}^2 = 250\sigma_{\text{daily}}^2$.
 - A 0.04 daily volatility annualizes to $0.04 \times \sqrt{250} = 0.63$ ("63%").
- Skewness, kurtosis (γ_r, κ_r): No scaling needed.
- A symmetric distribution has zero skewness.
- Skewness < 0 : more negative than positive surprises.
- The normal distribution has a kurtosis of $\kappa = 3$.
 - If $\kappa > 3$, we say data have *fat tails*.
 - Fat tails: More data near mean *and* at extremes.

Fat Tails and Pointy Heads



- The problem with high kurtosis distributions is not just fat tails.
- It's the pointy heads! Returns show deceptively little variation.
- This creates a false sense of security... so you take more risk.
- Then a surprise hits — a big surprise. You lose extra big.

Risk and Speculation

- Let's consider a risky game/investment:
 - Coin flip: Heads wins \$1 BB + \$2, tails loses \$1 BB.
 - We expect to earn \$1; how much does it cost to play?
- If playing costs \$1, we say this is a *fair game*.
- What if playing costs less than \$1?
 - For some *risk-averse* investors, game will be worthwhile.
 - Risk premium (\$1–cost) must compensate for variability.
 - This is *speculation*: taking a worthwhile risk.
- If risk premium ≤ 0 , only *risk-loving* (*gamblers*) will play.
- Speculator or gambler? Depends on investor's expectations.

Excess Returns and Risk Premia

- Government bonds determine a currency's *risk-free rate* r_f .
- Governments need not default; can print money if needed.
 - And yet they do default... WHY?
- *Excess return*: actual risky return – risk-free return.
- Investing in something riskier should earn more, on average.
 - Average compensation for that risk is the *risk premium*.
 - Risk premium = average excess return = $E(r_i - r_f)$.
- *N.B.* Some managers use a target rate instead of risk-free rate.
- *Risk aversion*: how unwilling people are to take risk (λ).
- We can then measure our outcomes in terms of *utility*:

$$U = E(r) - \lambda \text{Risk}^\omega \quad \omega > 1. \quad (15)$$

Historical Risk Premia

- Mean excess returns suggest US stock risk premium of 6–8%.
- US risk premia similar to other industrialized nations.
- Do risk premia change over time?
 - Academic research, common sense both suggest they do.
- US stock risk premium ranges over 3–15%.
 - Varies across time, subset (“universe”) of US stocks.
 - Probably persistent/exhibits serial correlations.
- Should consider various risk premia in a thorough analysis.

Historical Record of Returns

Return Distribution	Volatility σ	Skewness γ	Kurtosis κ
Normal	N/A	0	3
Stocks	0.2–0.8	-0.5	5–7
Bonds	0.1–0.2	0.5	5–7
Currencies	0.1	-0.1	5–10
WTI Crude	0.4	0.1	6
Copper	0.2	-0.4	7.5
Electricity	0.5–1.5	0–1	7–25
Natural Gas	0.7	-1.0	31
Wheat	0.3	-0.8	60

- Skewness is negative, not “effectively” 0.
- $\gamma < 0$ is bad for longs.
- Kurtosis of large indices is less than for single stocks.

Whence Normality?

- Averaging real-life random numbers gives normal dist'n.*
- But data show mild to severe departures from normality. Why?
- Not all random: competition; firm's troubles may help competitors.
- Skewness < 0 for stocks, copper, natgas, wheat.
 - Stocks: surprises are often bad; firms rarely hide good news.**
- Skewness > 0 for govt bonds and maybe crude, power.
 - Govt bonds: Occasional jolts of fear cause "flight to quality."
- Kurtoses > 3 for everything.
 - Sporadic info releases mean sometimes there are big shocks.
 - In between info releases: price adjustments are small.
 - Also, market has different states ( , ), volatile periods.

*Assuming that by "real-life" we mean having a finite volatility.

**Stocks are also final claimants on a firm in bankruptcy.

Diversification

Chapter 9



Introduction

This part discusses diversification, specifically:

- Risk aversion;
- Utility and indifference;
- Allocation between risk-free and risky assets;
- Finding the optimal risky portfolio; and,
- Finding the optimal complete portfolio.

Risk Aversion and Utility

- Can think of risk-averse investor's *utility* function.
 - Values expected return less some multiple of risk. What
 - captures risk? If returns log-normal, variance σ^2 .

$$U_{\lambda}(E(r_P), \sigma_P) = E(r_P) - \lambda \text{Risk}^{\omega} \quad (1)$$

$$(\text{usually proxied as:}) = E(r_P) - \frac{\lambda}{2} \sigma_P^2. \quad (2)$$

- To invest: Maximize utility when choosing a portfolio.
 - λ expresses the investor's risk aversion.
 - How to know your risk aversion λ ? HARD.
 - How to simulate the fear of losing your life's savings?
 - For same variance, do you prefer more/less skewness, kurtosis?
 - Ignoring these concerns: common λ 's in 2–4 range.
-

Risk Aversion and Utility: Example

- Consider 3 portfolios:

Risk	Risk Prem.	$E(r_P)$	σ_P
None	0%	4%	0%
Low	2%	6%	8%
High	8%	12%	20%

- The utilities of these for various risk aversions λ :

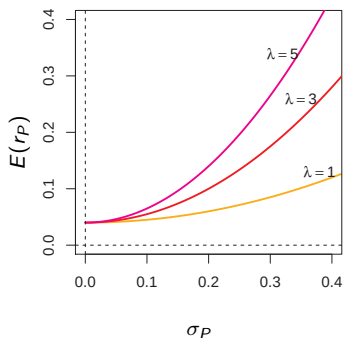
λ	U(no-risk)	U(low-risk)	U(high-risk)
0	0.04 = 4%	6.00%	12.00%
2	4%	5.36%	8.00%
4	4%	4.72%	4.00%
6	4%	4.08%	0.00%
7	4%	3.76%	-2.00%

Indifference

- We are *indifferent* between outcomes of equal utility.
 - In other words, we would be equally happy with any of them.
- Rearrange (2) to get *indifference curve*.

$$E(r_P) = U + \frac{\lambda}{2}\sigma_P^2. \quad (3)$$

- *N.B.* Different λ 's yield different indifference curves.



If $\lambda = 3$, we're equally happy with:

- $E(r_P) = 0.04, \sigma_P^2 = 0$;
- $E(r_P) = 0.10, \sigma_P^2 = 0.04, \sigma_P = 0.20$;
- $E(r_P) = 0.20, \sigma_P^2 = 0.10\bar{6}, \sigma_P = 0.327$.

Allocating Among Risky, Risk-Free Portfolios

- How do we create different portfolios on indifference curve?
 - Idea: Put some money in risk-free investment F .
 - *Complete portfolio* includes risk-free, risky portfolio (F, P).
- Say P is 40% corp bonds, 60% equities; investing \$300k:
 - 1/3 in F , 2/3 in P : \$100k T-bills, \$80k corp bonds, \$120k equities.
 - 1/2 in F , 1/2 in P : \$150k T-bills, \$60k corp bonds, \$90k equities.
- As we change allocation between F and P :
 - *We do not change what is in F and P .*

One Risky and One Risk-Free Asset

- Allocate fraction y of investment to P , $1 - y$ to F .
- Let r_f , r_P and 0 , σ_P be rates of return, volatilities of F , P .*
- Then the rate of return for complete portfolio C is:

$$E(r_C) = yE(r_P) + (1 - y)r_f = r_f + y \underbrace{(E(r_P) - r_f)}_{\text{risk premium}} \quad (4)$$

- The volatility of the complete portfolio C is:

$$\sigma_C = y\sigma_P \Rightarrow y = \sigma_C / \sigma_P. \quad (5)$$

- Thus we can restate the expected return on C as:

$$E(r_C) = r_f + \sigma_C \frac{E(r_P) - r_f}{\sigma_P}. \quad (6)$$

*Why 0 for F volatility?

Capital Allocation Line (CAL)

- Look more closely at what we have:

$$E(r_C) = r_f + yE(r_P - r_f) = r_f + \frac{E(r_P - r_f)}{\sigma_P} \sigma_C = r_f + S_P \sigma_C. \quad (7)$$

- Defines a line relating $E(r_C)$ and σ_C .
- We call this line the *capital allocation line* (CAL).
 - Slope: Sharpe ratio of portfolio P .
 - Gives transition from all F to all P : F at $y = 0$, P at $y = 1$.

Utility Along the CAL

- Where to be along the CAL? What y makes us happiest?
- To find out, maximize utility along the CAL.

$$U^* = \max_y U_\lambda(y) = \max_y E(r_C) - \frac{\lambda}{2} \sigma_C^2 \quad (8)$$

$$= \max_y r_f + y(E(r_P) - r_f) - \frac{\lambda}{2} y^2 \sigma_P^2. \quad (9)$$

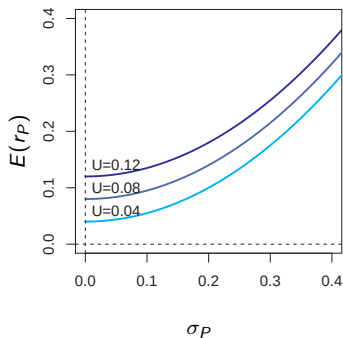
- We could also constrain $0 \leq y \leq 1$ (i.e. no SS).
- Since U concave, set $U' = 0$ and solve:

$$U' = E(r_P) - r_f - y^* \lambda \sigma_P^2 = 0 \quad \implies \quad y^* = \frac{E(r_P) - r_f}{\lambda \sigma_P^2} \quad (10)$$

- *N.B.* Must be very risk averse to invest only risk-free.

Indifference Curves Redux

- Recall the indifference curve definition (for portfolio C):
 - Equally-pleasing $(E(r_C), \sigma_C)$ pairs given our λ .
 - Thus any pair of $(\lambda, \text{utility})$ defines a curve.
- Utility gives indifference curves $E(r_C) = U + \lambda/2\sigma_C^2$:



- Curves for $\lambda = 3$.
- Ceteris paribus*...
- ...prefer a higher curve for given λ .
- But higher curves may not be possible.

Optimal Risky Portfolios



Diversification and Risk

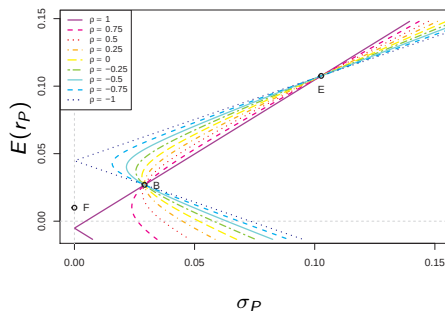
- *Diversification* is splitting capital among many instruments.
 - Idea: Reduces instrument-specific risk.
 - How? *Insurance principle**: many small bets safer than one big bet.
- Some risk eliminated (or greatly reduced) by diversification.
 - Call this *idiosyncratic risk* or *specific risk*.
- However, some risk may be common to all instruments.
 - Call this *systematic risk*. (“Market risk” is inaccurate.)
 - This risk cannot be made to vanish under diversification.

*This is merely an expression of the Gauss-Markov Theorem.

Portfolios of Two Risky Assets

- Explore diversification further; consider two-asset portfolio:
 - Invest in risky bond portfolio B or equity portfolio E .
 - Shares invested: w_B and w_E ; $w_B + w_E = 1$.
- Portfolio return: $r_P = w_B r_B + w_E r_E$.
- Portfolio variance: $\sigma_P^2 = w_B^2 \sigma_B^2 + w_E^2 \sigma_E^2 + 2w_B w_E \text{Cov}(r_B, r_E)$.
- Don't forget: $\text{Cov}(r_B, r_E) = \rho_{BE} \sigma_B \sigma_E$.

Two Risky Assets: Varying Weights



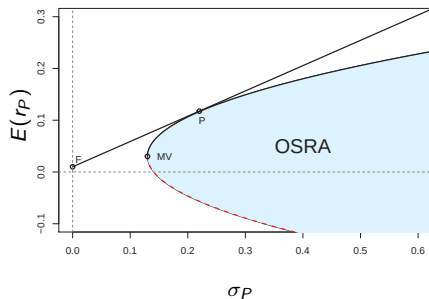
- Can vary w_B ($w_E = 1 - w_B$).
- Frontiers: $B - E$ curves.
- $|\rho_{BE}| = 1$? Can make $\sigma_P = 0$.

- *Minimum variance portfolio* may have $\sigma_P < \sigma_i$ for all stocks i .
 - *N.B.* green and blue curves have points with lower volatility.
- Points right of curve define *opportunity set of risky assets* (OSRA).

Two Risky and One Risk-Free Asset

- What if we add money market to “bond and equity” universe?
 - CAL gives possible allocations to risky, risk-free portfolios.
 - OSRA gives possible risky portfolios.
 - Thus we must invest on the CAL and OSRA.
- Also means the all-risk endpoint of CAL must be on OSRA.
 - This gives many possible CALs; which to use?
 - Want CAL with greatest slope (*i.e.* Sharpe ratio).

Optimal Risky Portfolio



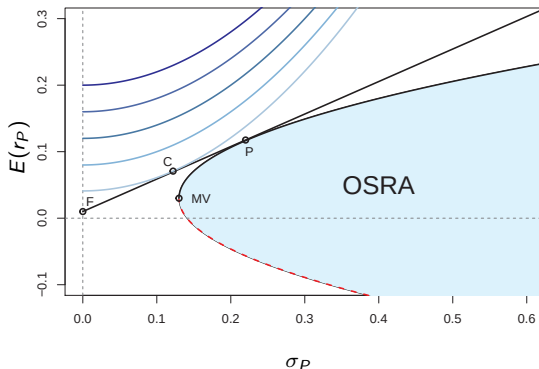
- Find the CAL
- Want to maximize Sharpe ratio.
- CAL for P has slope $\frac{E(r_P) - r_f}{\sigma_P}$.

$$\max_{P \in \text{OSRA}} S_P = \max_{P \in \text{OSRA}} \frac{E(r_P) - r_f}{\sigma_P}. \quad (11)$$

- That gives the line from F tangent to the top of the OSRA.
- Point of tangency yields the *optimal risky portfolio*.
 - One Risky Portfolio to Rule Them All! (... Sort of.)
- Why not pick a point on red dashed frontier?

Optimal Complete Portfolio

- Maximizing utility on CAL gives *optimal complete portfolio*.
- Thus utility guides allocation between F and optimal P .
- Find tangency indifference curve to get tangency point C .
- C is optimal complete portfolio, mix of F and P .



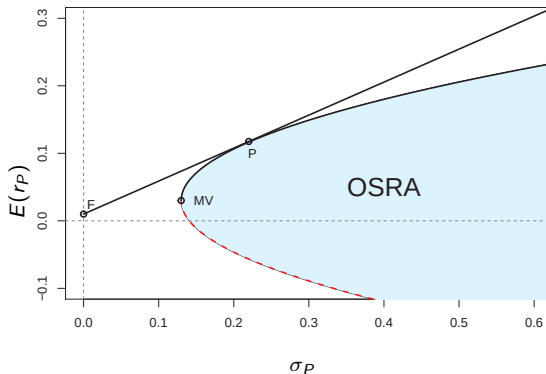
Markowitz-Roy Portfolio Selection

- We can now outline the process to find C :
 - Get $r_f \Rightarrow F$;
 - Get expected returns $E(r_i)$, variances σ_i^2 , correlations $\rho_{ij} \Rightarrow$ OSRA;
 - Find optimal risky portfolio P and resulting CAL;
 - Use risk aversion λ to get indifference curves; and,
 - Find optimal complete portfolio C .
- This is the idea of Markowitz (1952) and Roy (1952).*
- Usually, we skip finding the OSRA and just solve for P .

*Harry Markowitz received the Nobel Prize for this. He has said that Arthur Roy should have also won.

Markowitz-Roy Terminology

- Markowitz-Roy approach has yielded useful terminology.
- OSRA includes minimum variance portfolio *MV*;
- Minimal variance for any target return: *minimum variance frontier*.
- Top-half of OSRA is the *efficient frontier* (of risky assets).
- Bottom-half of OSRA is the *inefficient frontier* (of risky assets).



One Optimal Risky Portfolio to Rule Them All?

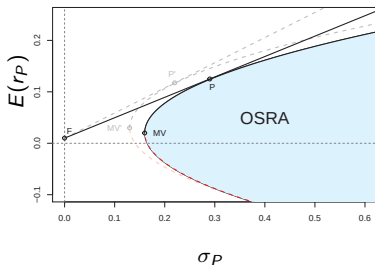
- Why doesn't everyone use the same optimal risky portfolio?
- People estimate inputs with different data;
- Taxes differ based on holding period, etc.
- Self-imposed constraints (risk, ethical/religious issues).
- Life exposures: Do not forget exposure to your employer!
- Home bias: You don't own Canadian, Mexican stocks??
- Different levels of financial education.

Other Problems with Markowitz-Roy

- Even with same inputs, taxes, Markowitz-Roy is not perfect.
- Must estimate $> n^2/2$ inputs; quickly exhausts info in recent data.
- Consider randomness! $\max E(\text{utility}) \gg \max$ based on averages.
 - However, stochastic optimization is not easy.
- Why not use higher moments, better risk measures?

$$U = E(r) - \frac{\lambda}{2}\sigma^2 + \frac{\lambda^2}{6}\gamma - \frac{\lambda^3}{24}\kappa + \dots \quad (12)$$

- Should consider transactions costs: lower return, add risk.



Diversification in the Limit

- How well does diversification work? Consider an example:
 - Suppose we invest equal amounts in n risky assets*.
 - All risky assets: same $E(r)$, σ , pairwise correlation ρ .
- Portfolio variance is then:

$$\sigma_P^2 = \sum_{i=1}^n \frac{\sigma^2}{n^2} + \sum_{i=1}^n \sum_{j \neq i}^n \rho \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n} + \frac{n-1}{n} \rho \sigma^2. \quad (13)$$

- As $n \rightarrow \infty$, $\frac{\sigma^2}{n} \downarrow 0$, $\frac{n-1}{n} \rho \sigma^2 \rightarrow \rho \sigma^2$.
- Thus $\rho \sigma^2$ is the common (systematic) variance.
- Portfolio variance mostly due to correlation among assets.
- Why we like assets that are uncorrelated with other assets.

*The well-known “1/ n portfolio” which sometimes also beats MPT.

Quantitative Investment

VALORACIÓN I

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Fixed Income

Chapter 11

Introduction

This part discusses fixed income. Specifically:

- Bond features and innovations;
- Accrued interest;
- Pricing and yields;
- Interest rate sensitivity;
- Risks to bondholders; and,
- Repos.

Basic Terminology

- A *bond* is a security relating to a loan.
 - Typically, the bond is the rights to loan repayment.
- The *par* (*face*) value is the amount borrowed.
 - Often, par is repaid in whole at *maturity date*.
- Bonds may receive regular *coupon* (interest) payments.
 - Annual interest = *coupon rate*; $\times$ par.
- *Zero coupon bonds* pay only par at maturity.
- Par, coupon, maturity, other details listed in *bond indenture*.

Bond Features: Sweeteners

- Some features entice investors.
- *Floating rate bonds* pay interest based on current rates.
 - Bond prices less affected by interest rate movements.
- *Indexed bonds*: par values change with inflation.
- *Interest rate puts*: lenders force repurchase if rates $\uparrow$.
 - Reduces interest rate risk of investors.
- *Convertible bonds*: firm repays in equity if investor chooses.

Bond Features: Risk Reducers

- Other features help issuers reduce risk.
- *Interest rate calls* let firm buy bond if rates ↓.
- *Sinking fund provisions*: retire bonds over time.
 - Smooths cashflow over time vs one big outflow.
- *Inverse floaters* pay less when rates rise, more when rates drop.
- *Asset-backed bonds* often backed by mortgages, other loans.
 - Let lenders shift loans/receivables to other investors. (even Bowie!)
- *Catastrophe bonds* pay back less if certain rare events occur.
 - e.g. severe hurricane/earthquake; a method of reinsurance.

Basic Bond Pricing

- Pricing a bond is theoretically simple:

$$\text{bond value} = E(\text{PV of par}) + E(\text{PV of coupons}). \quad (1)$$

- If bond matures in T years, constant rates, just after coupon:

$$\text{bond value} = \frac{\text{par}}{(1+r)^T} + \sum_{t=1}^T \frac{\text{coupon}}{(1+r)^t} \quad (2)$$

$$= \frac{\text{par}}{(1+r)^T} + \frac{\text{coupon}}{r} \left(1 - \frac{1}{(1+r)^T} \right). \quad (3)$$

- $PV(\text{perpetuity}) = \frac{\text{notional}}{r} \Rightarrow PV(\text{coupons}) = \frac{\text{coupon}}{r} - \frac{\text{coupon}}{r(1+r)^T}.$
- Between coupons, also add accrued interest.

Accrued Interest

- Bonds in most countries quoted *clean* (without interest).
 - Buyers of bonds between coupons are owed a partial coupon.
- Thus the actual price paid includes accrued interest.
 - *Invoice price* (aka *dirty price*) quote includes accrued interest.
- Why we, most countries quote prices clean? Comparability.
- How to calculate accrued interest (AI)?

$$\tau = \frac{\text{days from last coupon}}{\text{days, last to next coupon}} \quad (4)$$

$$AI_{\tau} = \frac{\text{coupon rate}}{\text{coupons/year}} \times \tau. \quad (5)$$

- Daycount for most government bonds is 30/360.
 - Many corporates, some countries use Actual/365.

Slightly Better Bond Pricing

- These formulas work if the bond pays interest annually.
- For a bond which pays interest n times/year, we can do better.
- If bond matures in T years, just after a coupon payment:

$$\text{price} = \frac{\text{par}}{(1 + \frac{r}{n})^{nT}} + \sum_{t=1}^{nT} \frac{\text{coupon}}{(1 + \frac{r}{n})^t} \quad (6)$$

- Between coupons: fractional discounting – accrued interest.

$$\text{dirty price} = \frac{\text{par}}{(1 + \frac{r}{n})^{\lceil nT \rceil - \tau}} + \sum_{t=1}^{\lceil nT \rceil} \frac{\text{coupon}}{(1 + \frac{r}{n})^{t-\tau}}, \quad (7)$$

$$\text{clean price} = \text{dirty price} - Al_{\tau}. \quad (8)$$

Bond Yields

- *Yield to maturity*: annual return if bond held to maturity.
 - Just invert the pricing formula (*i.e.* solve for r).

$$\text{YTM} = \{r : \text{bond market price} = \text{bond value (6) or (7)}\}. \quad (9)$$

- *Current yield* = annual coupons/current price.
- May also consider other payoff scenarios: *yield to call*, *yield to put*.

Bond Returns

- Yields based on current prices; what if rates change?
 - We would like to know *realized compound return* (RCR).
- Problem: We don't know where rates are going.
- *Horizon analysis* considers RCR for various scenarios, holding periods.
- How to compute log-returns for bonds?

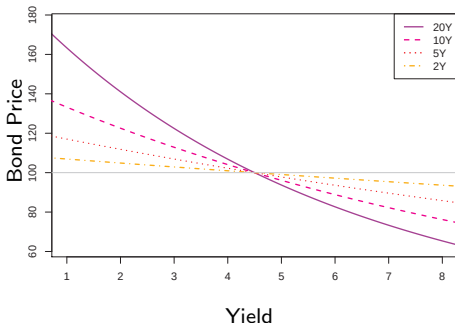
$$r_{t \rightarrow \tau} = \log \left(\frac{P_{\tau} + AI_{\tau} + C_{\tau}}{P_t + AI_t} \right) \quad (10)$$

where C_{τ} captures coupons paid from time t to time τ .

Bond Prices Over Time

- Even with constant rates, bond prices change over time.
- *Interest rate risk*: when interest rates rise, bond prices fall.
- This is important to remember; bond prices affect gains:
 - You can lose money if you sell risk-free bonds at a low price.
 - Furthermore: longer-term bonds = more sensitive to interest rates.
 - Long-term bonds (e.g. 30Y UST) can be as volatile as the S&P 500.
 - Bonds tend to be least volatile when issued and near maturity.
- If YTM $\uparrow$, holding period return $\downarrow$

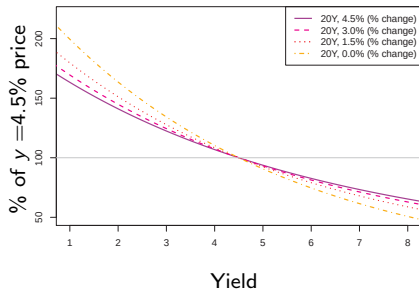
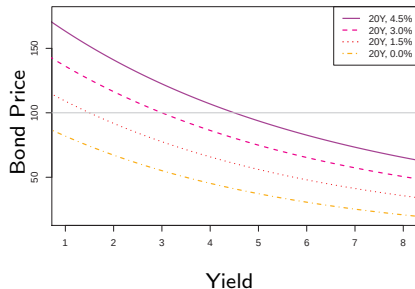
Bond Prices versus Yields by Maturity



Look at price vs yield for varying maturities of 4.5% coupon bonds:

- Price of 4.5% bond yielding 4.5%: 100.
- Bond prices change inversely with interest rates.
- Longer-dated bonds more sensitive to interest rates.
- Interest rate sensitivity higher at lower interest rates.

Bond Prices versus Yields by Coupon



Look at price vs yield for varying coupons of 20Y bonds:

- Zero-coupon bond always below 100 if rates >0 : *discount bond*.
- Yield $>$, $<$ coupon? Bond trades at a *discount*, *premium*.
- Bond prices decrease as coupon decreases.
- Lower-coupon bonds are more %-sensitive to rates.

Interest Rate Sensitivity: DV01

- We can directly analyze interest rate sensitivity:

$$P_0 = \sum_{t=1}^T \frac{CF_t}{(1+y)^t} \quad \text{or} \quad \sum_{t=1}^{nT} \frac{CF_t}{(1+\frac{y}{n})^t}, \quad (11)$$

$$\frac{dP_0}{dy} = \sum_{t=1}^T -t \frac{CF_t}{(1+y)^{t+1}} \quad \text{or} \quad \sum_{t=1}^T -\frac{t}{n} \frac{CF_t}{(1+\frac{y}{n})^{t+1}}. \quad (12)$$

- *DV01*: dollar value of a 1 basis point yield change.²

$$DV01 = \frac{dP_0}{dy} \cdot 0.0001. \quad (13)$$

- One basis point scale is much more sensible.

²One basis point is $\frac{1}{100}$ -th of 1% = 0.01%.

Interest Rate Sensitivity: Duration

- Can formalize idea that maturity/cashflow timing relate to sensitivity.
- *Duration*: PV-weighted average time of cashflows.

$$D = \sum_{t=1}^T t \underbrace{\frac{CF_t/(1+y)^t}{P_0}}_{PV(CF_t) \text{ weight}} = \frac{-(1+y)}{P_0} \frac{dP_0}{dy}. \quad (14)$$

- Modified duration D^* : gives us % bond price change.

$$D^* = \frac{-1}{P_0} \frac{dP}{dy} \Big|_{y_0} \quad (15)$$

- Sensitivity, cashflow timing, maturity... all wrapped up together.
- *N.B.* $DV01 = -D^* P_0 \cdot 0.0001$.

Interest Rate Sensitivity: Insights

- Interest rates $\uparrow$? Bond prices $\downarrow$.
- Bonds trade at premium/discount based on coupon vs interest rates.
- Duration of a T -year zero-coupon bond? T .
- Lower-coupon bonds more sensitive to interest rates.
- Interest rate sensitivity is not linear: more sensitive at low rates.

Duration Intuition

From earlier observations come simpler rules:

- Duration of zero coupon bond = maturity (in years).
- For same maturity, lower coupon $\Rightarrow$ higher duration.
- For same coupon, duration usually $\uparrow$ as maturity $\uparrow$.
 - If bond is not trading at a discount, this is always true.
- All else equal, duration of coupon bond $\uparrow$ as YTM $\downarrow$.
- Duration of a perpetuity: $D_{\text{perp}} = \frac{1+y}{y}$.

Convexity

- Note that we mentioned how duration changes.
- Q: Are these changes linear?
- Q_A: Would $\Delta\text{YTM} = 2\epsilon$ affect prices twice as much as $\Delta\text{YTM} = \epsilon$?
- A: Usually, no. This is due to *convexity*.
- Recall plots of prices vs. ΔYTM were curved.
 - Modified duration related to current derivative wrt y .
 - Convexity related to second derivative wrt y :

$$\text{Convexity} = \frac{1}{P_0} \frac{d^2 P}{dy^2} = \frac{1}{P_0(1+y)^2} \sum_{t=1}^T (t^2 + t) \frac{CF_t}{(1+y)^t}. \quad (16)$$

Convexity: A Complication

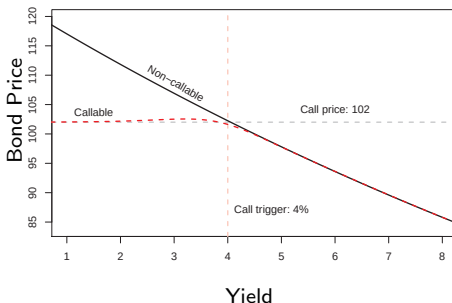
- Convexity seems like trouble:
 - Durations must be recalculated after every yield change.
 - Formula seems complicated; isn't duration alone sufficient?
- But convexity is helpful:
 - Interest rate cuts benefit bondholders more than hikes hurt.
 - Allows us to better measure interest rate sensitivity:

$$\frac{\Delta P}{P_0} = -D^* \Delta y + \frac{\text{convexity}}{2} (\Delta y)^2. \quad (17)$$

- And convexity is a reality; you should understand it.

Convexity of Callable Bonds

- Recall that some bonds may be called by issuer if rates drop.
- Thus callable bonds rarely trade much above the call price.
 - Creates *negative convexity*: rate drops affect prices less.
- Problem: Called bond is extinguished before maturity.
- Solution: Compute *effective duration* $D_{\text{eff}} = -\frac{\Delta P/P_0}{\Delta r}$.
- Mortgages are similar: if rates drop, holders refinance loans.
 - But people often wait to refinance (minimize fees; “dumbos”).
 - Thus mortgages more likely than corps to trade above principal balance.



Discount Bonds

- *Original-issue discount bonds* issued with low coupons.
 - OID bonds are less common than *par bonds*.
 - Includes *zero-coupon bonds* (“zeros”) such as T-bills.
 - Zero-coupon bonds only pay at maturity.
- Longer term zeros are often created by *stripping* a bond.
 - Each payment becomes a unique zero-coupon bond.
 - These bonds are called *strips*.

Default

- A borrower who misses bond payment is in *default* = serious.
- Default may happen for a number of reasons:
 - *Balance sheet insolvency*: liabilities $>$ assets.
 - *Cashflow insolvency*: cannot make a payment.
 - *Strategic default*: purposeful default, may be negotiating tactic.
- Corporations, municipalities, even federal governments may default.
- Creditors may seize specific assets if pledged as bond *collateral*.
- Otherwise, creditors may force sale of general firm assets.
- Often cheaper, better: *debt workout* (a compromise).

Protective Covenants

- Bond indentures have *protective covenants* to assure lenders.
- *Subordination clauses*: “No cutting in line.”
 - Later borrowing is subordinated = paid later if bankruptcy.
- *Cross-default clauses*: “All for one, one for all.”
 - Cannot default on some bonds and pay others to play games.
- *Dividend restrictions*: “Don’t spend more than you make.”
 - Typical: Cannot pay more in dividends than earnings.
- *Asset withdrawal restrictions*: no selling assets to pay shareholders.
- *Asset substitution restrictions*: do not change to make firm riskier.
- *Underinvestment clauses*: must also invest in lower-risk projects.
- *Force majeure clause*: (rarer) excuses default in war, disasters, etc.

Credit Spreads

- One of the most informative measures: *credit spread*.
- Basically, yield difference from similar-maturity government bond.
- Short-term: interbank — govt bills, eg TED spread.
 - Very liquid; top credit vs “risk-free”.
 - Can indicate when economy nearing crisis.³
- Intermediate term: Corporate bonds — notes/bonds.
 - Less liquid; sign of spreading crisis.
 - Exemplar: 10-year BBB/Baa-rated corporates — UST 10Y.
- High-yield rates vs investment-grade rates.

³cf Boudt, Paulus, and Rosenthal (2017): TED spread > 50 bp indicates crisis.

High Yield

- Unrated/sub-investment-grade bonds are *high yield bonds*.
 - Also known as “junk” bonds.
- Pre-1977: All junk was “fallen angels” (downgraded issues).
- 1977: Drexel and Milken premier first original-issue junk.
 - Investors found diversified high-yield portfolios attractive.
- High yield often trades somewhat like equity.
 - Equity receives residual cash after paying bondholders.
 - High yield should trade similarly if assets < liabilities.

Liquidity Effects

- New bonds like puppies: most quickly find a “forever home.”
- Newly-issued bonds are liquid for a short period.
- After that, they become illiquid; few circulate and little trading.
- Insurers, endowments, pensions hold for long-term to hit targets.
- Government bonds: on-the-run vs off-the-run.
- Yield difference can be tens–hundreds of basis points.
- Not free money! Liquidity risk is vicious type of risk.
- May become more liquid near certain remaining maturities.

- *Repurchase agreement* (repo): sale+later buy of assets.
- Pledge asset as collateral in exchange for loan of cash.
- Lender holds back some of the proceeds (the *haircut*)
- Why repo assets?
 - Frees up cash in case needed for day-to-day operations.
 - A form of leverage; allows making other investments.⁴
- Special rates quoted if many people want to short asset
- Repos are an important part of liquidity management.
- Repos have declined; Basel III/Dodd-Frank made them costly.

⁴This does increase your risk, however.

Yield Curves

Chapter 12

Introduction

This part discusses yield curves. Specifically:

- Yield curves;
- Monetary policy; and,
- Interest rate dynamics.

Term Structure of Interest Rates



Curves and Intertemporal Smoothing

- For many liquid goods, we can lock in prices at a later time:
 - Power; jet; natgas; USD/HKD: trade futures or forwards.
 - Interest rates: trade futures, bond, or enter a CD.
- Later: get agreed price — or P&L compensates for difference.
- Often look at *curves*: Price or yield vs later dates.
- Sometimes call these curves the *term structure*.
- If one date's price (or yield) is far from nearby dates:
 - Trade adjacent dates;
 - Store goods/money for less than difference;
 - PROFIT! (Maybe?)
- This *intertemporal smoothing* is what market makers, HFT do.

The Yield Curve

- We have assumed constant interest rates. Unrealistic.
- Supply and demand affect prices at all maturities.
- Thus different maturities have different yields.
- A *yield curve* plots yields versus times to maturity.
 - Usually use this terminology for government bond yields.
 - However, very related is the LIBOR-based *swap curve*.
 - Also look at *credit spreads*: yield increase for risky debt.

Yield Curve versus Zero Curve

- Suppose we had zero-coupon bond yields (aka *spot rates*).
- Curve of discount rates often called a *zero curve*.
- Should think about bonds used to make yield curve.
 - On-the-runs (recently-issued) are most liquid; most accurate?
- The yield curve relates to the zero curve like so:

Maturity (years)	YTM (%)	Price	Discount Factor
3M	0.5	\$998.75	0.9988
6M	1.1	994.53	0.9945
1	3.0	970.87	0.9709
2	3.5	966.18	0.9662
3	4.0	961.54	0.9615
4	5.0	952.38	0.9524

- If 6M yield calculated as EAR, price = \$994.54.

How to Plot a Yield Curve

Plotting a yield curve is not done like other curves.

- Plot yields of non-indexed govt bonds: most liquid, most *tenors*.^{*}
- Consistently plot the same tenors so you can compare across time.
- Abbreviate tenors: O/N (overnight), 3M (3-month), 10Y (10-year).
- Plot yields vs time: time is ordered but NOT to scale.

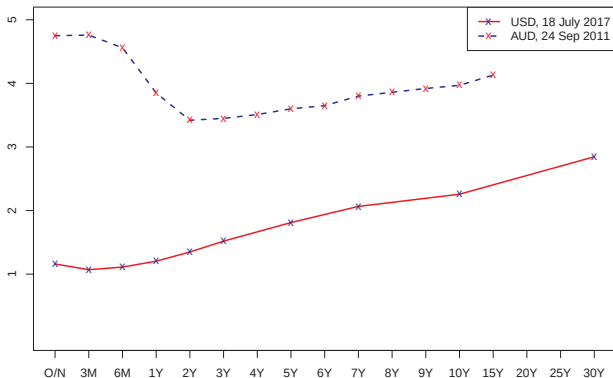
^{*} In bonds, “tenors” refers to maturities and not opera stars.

What Yield Curves Tell Us

- Some yield curve shapes are associated with macroeconomy.
- *Inverted* curve shows investors value long-term bonds highly.
 - Investors may be anticipating an economic slow-down.
- *Humped* curve is typical of a currency defense.
 - Government props up mid-term rates to entice investors.
- (Gently) *upward-sloping* curve often implies stable economy.
- High *front end* of the curve slows short-term growth.

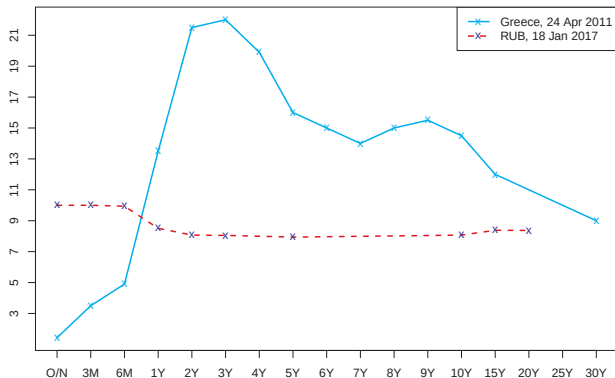
What Yield Curves Tell Us: The Good,...

- AU, US: (Gently) upward-sloping curves; stable economic growth.
- AU: Elevated front end of curve; govt trying to cool rapid growth.



What Yield Curves Tell Us: ... The Bad, and the Ugly

- RU: Inverted curve; investors fear slow growth/contraction.
- GR: Humped curve; currency defense, begging for money to stay.



Types of Interest Rates

- Zero-coupon bond yields imply price of money at one time; clean.
- We call such yields *spot rates*, e.g. 6M spot rate $r_{z,6M}$.
- Interest rates are uncertain; helpful to lock-in later rates.
- *Forward rates*: interest rates starting in the future.
 - Future rates can lock in now, e.g. 2Y bond in 1 year $f_{1 \rightarrow 3}$.
 - Often implied by differences in other bond yields.
- *Short rates*: rates of shorter-term bonds starting now.
 - 6M spot rate above is a short rate.

Yield Curves in a Certain World

- If rates certain, all bonds to same horizon have same return.
- This implies particular certain forward rates.
- Thus compounded rate from a 3-year bond should equal:
 - 1-year bond reinvested in 2-year bond 1 year forward;
 - 2-year bond reinvested in 1-year bond 2 years forward;
 - 1-year bonds reinvested in 1-year bonds 1&2 years forward.
- In our example zero/yield curve, rates seem to be increasing:
 - $2Y \text{ YTM} = 1Y \text{ YTM} \times 1Y \text{ 1-yr-fwd YTM}$
 $1.035^2 = 1.03(1 + r_{1,1}) \implies f_{1 \rightarrow 2} = 4.00\%$
 - Quick-n-dirty approximation:
 $3.5\% \times 2 = 3\% + f_{1 \rightarrow 2} \implies f_{1 \rightarrow 2} = 4\%.$

How to Determine the Yield Curve

- We want spot rates for yield curves.
- Coupon bond yields mix in prices of money at multiple times.
- How to find pure yield curve: *bootstrap* the curve.
- Use spot rates we know to infer those we do not.
- Typically work from front end of curve toward back.
- Example: 1.25% 1.5Y bond yields 1% $\implies P_{1.5Y} = 100.3713$.

$$100.3713 = \frac{1.25/2}{1 + y_{z,6M}/2} + \frac{1.25/2}{(1 + y_{z,1Y})^2} + \frac{100 + 1.25/2}{(1 + y_{z,1.5Y}/2)^3} \quad (1)$$

$$y_{z,6M} = 0.5\%, \quad y_{z,1Y} = 0.75\% \implies y_{z,1.5Y} = 1.021\%. \quad (2)$$

- Then we would find 2Y, etc.

About Forward Rates

- In our example, 1-year rate 1 year forward: $f_{1 \rightarrow 2} = 4\%$
- 3Y, 4yrs fwd: rate for what *will be* 3Y bond in 4 years, $f_{4 \rightarrow 7}$.
- If rates were certain, forward rates might be future interest rates.
- With uncertainty, forward rates often above E(interest rates).
- Must entice people to commit money in an unknown future state.
- This gets us to theories for the term structure. . .

Monetary Policy and Yield Curve Dynamics



Term Structure Theories

- A few major theories are used to explain yield curve shapes.
- *Expectations hypothesis*: Forwards predict later short rates.
- *Liquidity preference*: must pay people to lock up cash.
 - Short-term lenders value liquidity, dominate market; so,
 - Forward rates above later rates: $f_2 > E(r_2)$.
- *Market segmentation*: independent investors w/differing horizons.
- *Preferred habitat*: investors diverge from horizon... for a price.
- *Interest rate volatility* also seems to affect curve shape.
- Theories form scale of lender willingness to alter horizon.
- Also know central banks target yields to implement monetary policy.

The Fed's Dual (Triple?) Mandate

- Why do we let central banks control monetary policy?
- So unemployment and/or inflation do not get out of hand.
- Thus the Fed's *dual mandate*:
 - Maintain price stability (keep inflation low); and,
 - Maximize employment.
- Fed watchers literally plot these on a bulls-eye chart.
- Post-2008: Also tasked w/mitigating financial instability.
 - Is preventing bank panics/market crashes a third mandate? Seems so.

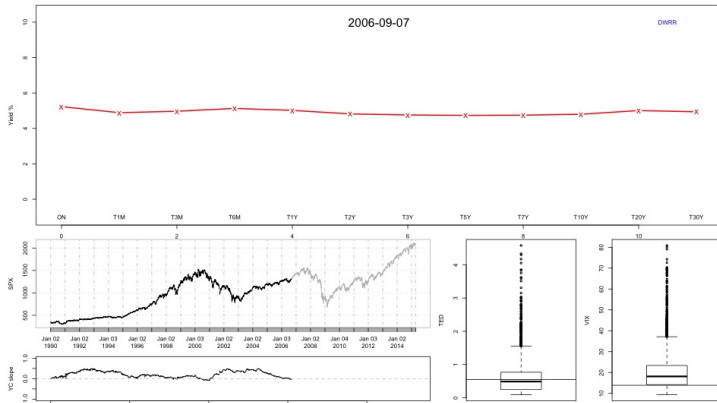
Open Market Operations

- How central bankers change rates: *open market operations*.
- In other words: they trade bonds to move yields.
- Open market operations allow central banks to act quickly.
- Most central banks target front end of the curve: overnight rate.
- Since 2008: Fed, others have used *non-standard monetary policy*.
 - Unleashed open market operations on other parts of yield curve.
 - Have also discussed targeting risky bonds/repo markets.

Yield Curve Dynamics

- Putting ideas together gives factors which drive yield curve:
 - expectations of short rates;
 - government targeting of rates;
 - mandated investment horizons;
 - equilibrium forward rates; and,
 - liquidity premia.
- Inflation also matters; but, its effect may not be clear.
- Curves in the text show some of these ideas.

Yield Curve Movie



- Really. Never-before seen. (Well, not in this way.)
- Curve; S&P 500; curve slope; TED; VIX.

Yield Curve: The Movie!

- Look for: crises (1998, 2008); dot-com burst (2000).
- Yield curve goes red when inverted.
- TED spread (credit), VIX (equity vol.) indicate crisis.
- Declining yield curve slope often presages SPX declines.
- Litterman-Scheinkman: common movements = shift, twist, bow.

<https://www.youtube.com/watch?v=6z8g9xaj3g8>

Modeling Yield Curve

- Yield curve models help us assess over-/under-valuation of bonds.
- Lots of models; could take a course on just yield curve modeling.
- General ideas: yields tend to revert to targeted levels (front end).
 - Vašíček model has Ornstein-Uhlenbeck short rate:

$$dr_t = \lambda(\bar{r} - r_t)dt + \sigma dW_t. \quad (3)$$

- Nelson-Siegel: Short-/medium-/long-term effects.
 - Like slope/hump/level; Svensson adds a second hump.
- May assume some bonds more liquid; their prices more reliable.
- Many models use Litterman-Schenkman factors: level, slope, curve.

Example: Yield Curve Arbitrage

- Suppose the following set of UST yields:
 - 1-year: 2.0% 1-year 1 year forward: 2.2%
 - 2-year: 2.1% 1-year 2 years forward: 2.2%
 - 3-year: 2.3%.
- Remember: Draw the paths w/yields as slopes.
- Does everything look correct?
- Can make money: Buy 3Y; pay by selling 2Y & 1Y 2yrs fwd.
- What are Sharpe, Sortino ratios of this investment?
- How will this affect prices of these bonds?
- What if we instead did this trade using GM bonds?

Equity Valuation

Chapter 13

Introduction

This part discusses equity valuation. Specifically:

- Sector and industry analysis;
- Fundamental analysis and comparables;
- Issues with discounted cashflow (DCF) models;
- Dividend discount models (DDMs);
- Free cashflow methods;
- The value of growth; and,
- Thoughts on the pricing kernel.

Sector and Industry Analysis

- Below economy, must consider sectors and industries.
 - Investing well is easier in well-performing industries.
 - Sectors/industries have different sensitivities to economy.
 - Firms within an industry are exposed to similar risk factors.
- Economic sensitivity metric: degree of operating leverage.

$$\text{DOL} = \frac{\% \text{ change in profits}}{\% \text{ change in sales}} = 1 + \frac{\text{fixed costs}}{\text{profits}}. \quad (1)$$

- Pure borrowing (leverage) also increases cycle sensitivity.
- Thus we expect certain ranges of DOL for different industries, e.g. .
 - Software: high fixed cost, tiny variable costs = high DOL; vs
 - Construction: medium fixed cost, large variable costs = low DOL.

Sector and Industry Analysis: Rotation

- Sectors often respond coherently to the business cycle.
- Suggests choosing sector weights in light of business cycle.
- *Rotation* reweights sectors optimally given business cycle.
- Thus if we are entering a contraction, we might:
 - overweight defensive stocks; and,
 - underweight cyclical stocks.

Sector and Industry Analysis: Industry Definition

- How to define sectors and industries?
 - Barra has defined national/world sectors, industries.
 - S&P also has their own definitions.
 - Nat'l/Int'l industry codings, e.g. NAICS, UKSIC, NACE, ISIC.
- National/international codings use a hierarchy: SSIIi...i
 - SS = sector (1–2 characters); II = industry;
 - i...i = characters/symbols for sub-industry.
 - e.g. NAICS: 20 sectors; 92 industries (sub-sectors).
 - Often use code substrings SS or SSII as categorical variable.

Fundamental Analysis

⚠ Accounting measures may be “smoothed” or “managed.”
This make them less reliable.

- Fundamental analysts compare companies using:
 - financial filings (in US: from SEC's EDGAR); and,
 - derived measures, e.g. from Compustat.
- We often compare firms by *comparables*:
 - Price ratios: earnings (P/E), book (P/B), even website clicks;
 - Per-share ratios: earnings, book, sales, cashflow.
- Idea: Similar companies should have similar measures.
- All imply prices the analyst takes into consideration.

Price-to-Earnings: Criticism

Criticisms of P/E analysis abound, among them:

- Earnings is an accounting construct.
 - Economic earnings may diverge (e.g. depreciation).
- Companies manage (inflate, smooth) earnings.
- Assumes earnings rise smoothly; ignores lumpiness.
- P/E for cyclicals is less stable.
- P/E may be meaningless for start-ups.

Price-to-Earnings: Alternatives

- Can use P/E and forecasted dividends to predict target price.
 - Feed this into DCF models to get intrinsic value.
- Can look at other ratios:
 - *Price/book*: how aggressively market values assets.
 - *Price/cashflow*: avoids earnings management.
 - *Price/sales*: useful for valuing firms without earnings.

Limitations of Book Value

- Equity holders are “residual claimants.”
 - But, what are assets (and hence residual) worth?
- Problem: book value = original cost less depreciation.
- We might instead ask: What would assets sell for?
 - This is critical for debt analysts.
 - If *liquidation value*^{*} exceeds market cap, expect raiders.
- Consider *replacement cost* of assets less liabilities.
- Leads to *Tobin's q*: $q = \frac{\text{market price of firm}}{\text{replacement cost of assets}}$.
 - q should tend to 1 due to competition.
 - If $q > 1$, competitors can replicate firm for cheaper.

^{*} Money left after selling assets and repaying debt.

Expected Holding Period Return

- Basic approach for a certain time horizon:
 - For now, assume we have prediction of price at year-end.

$$\text{Expected HPR} = \frac{E(\text{annual div.})}{P_0} + \frac{E(\text{annual } \Delta \text{price})}{P_0} \quad (2)$$

$$= E(\text{div. yield}) + E(\text{cap gain yield}) \quad (3)$$

- Is expected HPR attractive for risk? Think about expected return.
- Think of *stochastic discount factor* d for return factor R
 - We then expect that competitive pricing yields $E(dR) = 1$.
- We often think in terms of *pricing kernel* $k = 1/d - 1$.

Jensen's Inequality and Discounted Cashflows

- When discounting cashflows, we use the pricing kernel k .
- However, if k is uncertain, that biases the valuation.
- For a convex f , Jensen's Inequality says:

$$E(f(k)) > f(E(k)) \quad \forall f : f'' > 0 \quad (4)$$

- This is trouble; cannot just plug in averages. Must correct:

$$E(f(k)) \approx f(E(k)) + \frac{1}{2} f''(E(k)) \sigma_k^2. \quad (5)$$

- Thus if we discount cashflows:

$$\tilde{V} = \sum_t \frac{C_t}{(1+k)^t} + \frac{1}{2} \sum_t \frac{t(t+1)C_t}{(1+k)^{t+2}} \sigma_k^2. \quad (6)$$

- We will come back to this shortly.

Intrinsic Value

- *Intrinsic value* uses price target, pricing kernel k .
 - Computes what would be a fair price for stock.
- One-year horizon, annual dividend D_1 , EOY target price P_1 :

$$V_0 = \frac{E(D_1) + E(P_1)}{1 + k} \quad (7)$$

- For a two-year investment horizon:

$$V_0 = \frac{E(D_1)}{1 + k} + \frac{E(D_2) + E(P_2)}{(1 + k)^2} \quad (8)$$

- This prepares us for the world of DDMs.

Dividend Discount Model (DDM)

- Intrinsic value hints at valuing stock at any horizon.
- Basic *dividend discount model* (DDM) for H -year horizon³:

$$V_0 = \frac{D_1}{1+k} + \frac{D_2}{(1+k)^2} + \cdots + \frac{D_H + P_H}{(1+k)^H} \quad (9)$$

- If dividends constant D , annuity price yields P_H , V_0 :

$$P_H = \frac{D}{k} \Rightarrow V_0 = \sum_{t=1}^{\infty} \frac{D}{(1+k)^t} = \frac{D}{k} \quad (10)$$

- Thus we value stock as the sum of all discounted dividends.
- But for estimated $\hat{k}$, we adjust for Jensen's Inequality:

$$V_0 = \frac{D}{k} + \frac{D}{k^3} \sigma_k^2. \quad (11)$$

³We suppress most of the expectations from here forward.

Gordon-Shapiro Constant-Growth DDM

- Reasonable extension: dividends which grow at rate g .⁴
- This yields the *constant-growth dividend discount model*:

$$V_0 = \frac{D_0(1+g)}{1+k} + \frac{D_0(1+g)^2}{(1+k)^2} + \dots = \frac{D_0(1+g)}{k-g} = \frac{D_1}{k-g} \quad (12)$$

- Instant implication: dividend growth at/above k is unsustainable.
- We can also invert this formula to estimate k :

$$k = \frac{D_1}{P_0} + g = \text{div. yield} + \text{capital gains rate.} \quad (13)$$

- For estimated $\hat{k}$ and $\hat{g}$, we Jensen correct to get:

$$\hat{V}_{0,G-S} = \frac{\hat{D}}{\hat{k} - \hat{g}} + \frac{\hat{D}}{(\hat{k} - \hat{g})^3} (\hat{\sigma}_k^2 + \hat{\sigma}_g^2 - 2 \text{Cov}(\hat{k}, \hat{g})). \quad (14)$$

⁴Discovered by Gordon and Shapiro (1956).

Reinvestment, Gordon-Shapiro, and DuPont

- Does Gordon-Shapiro suggest companies pay out all dividends?
 - No; Gordon-Shapiro shows this may be unwise.
- Suppose we lower our dividend payout ratio.
 - Reinvest fraction b of dividends in company.⁵
 - If company's $ROE > k$, this seems sensible.
- Then, dividends grow at rate $g = b \cdot ROE$, implying

$$V_0 = \frac{E_1(1 - b)}{k - g} = \frac{E_1(1 - b)}{k - b \cdot ROE}. \quad (15)$$

- The DuPont model decomposes ROE into separate tasks:

$$ROE = \underbrace{\frac{\text{net profits}}{\text{sales}}}_{\substack{\text{profit margin} \\ \text{(sales efficacy)}}} \times \underbrace{\frac{\text{sales}}{\text{average assets}}}_{\substack{\text{asset turnover} \\ \text{(production efficiency)}}} \times \underbrace{\frac{\text{average assets}}{\text{average equity}}}_{\substack{\text{equity multiplier} \\ \text{(funding efficiency)}}}. \quad (16)$$

⁵Unpaid dividends are also called retained earnings.

Multistage Growth DDMs

- Constant ROE sounds great; contradicts observed life cycles.
- Instead, use a life cycle model and basic DDM to value firm.
- Basic idea: value modeled dividends via discounting.
 - Near dividends are discounted individually.
 - Assume decreasing dividend growth during maturity stage.
 - Then use Gordon model for dividends thereafter.
 - This also handles one-time special dividends.

Multistage Growth DDM Example

Insert example here valuing an actual company.

Except that multi-stage modeling was an exercise in frustration.

- The multistage growth DDM is flexible — maybe *too* flexible.
- You can work hours trying to make a nice example of a real firm.
- But when the model works, it gives highly varying prices.
- If you torture the model, it will confess.

Free Cashflow Models and WACC

- Instead of dividends, can value free cashflow to all/part of firm.
- Discount by firm's *weighted average cost of capital* (WACC).
- Funding: consider sources of funding for cashflow destination.
 - Firm? Use equity, bond, bank loans in fractions $w_e + w_b + w_\ell = 1$.
 - Equity? $w_e = 1$. Bondholders? $w_b = 1$.
- For funding sources having rates of return r_e, r_b, r_ℓ :

$$WACC = w_e r_e + (w_b r_b + w_\ell r_\ell)(1 - \tau). \quad (17)$$

- Why multiply bonds, loans by $1 - \tau$? Tax advantage of debt.

Free Cashflow to the Firm

- Find cost of replicating *free cashflow to the firm* (FCFF):

$$FCFF = EBIT(1 - \tau) + \text{depreciation} - \text{capex} - \Delta NWC \quad (18)$$

where τ is tax rate, NWC is net working capital.

- Then discount FCFF by WACC:

$$\text{Firm value} = F_0 = \sum_{t=1}^T \frac{FCFF_t}{(1 + WACC_t)^t} + \frac{F_T}{(1 + WACC_T)^T}, \quad (19)$$

$$\text{or } F_0 = \frac{FCFF_1}{WACC - g}. \quad (20)$$

- Subtract debt, compute per-share value: $V_0 = \frac{\text{firm value} - \text{debt}}{\text{shares outstanding}}.$

Free Cashflow to the Firm: Estimated FCFF, WACC

- If we estimate FCFF and WACC (i.e. $r_e = \hat{k}$), firm value is:

$$F_0 = \sum_{t=1}^T \left(\frac{\widehat{FCFF}_0(1 + \hat{g})^t}{(1 + \widehat{WACC}_t)^t} + \frac{t(t+1)\widehat{FCFF}_0(1 + \hat{g})^t}{2(1 + \widehat{WACC}_t)^{t+2}} \hat{\sigma}_{WACC-g}^2 \right) + \frac{\hat{V}_T}{(1 + \widehat{WACC}_T)^T}, \quad (21)$$

$$\hat{V}_T = \frac{\widehat{FCFF}_0(1 + \hat{g})^T}{\widehat{WACC} - \hat{g}} + \frac{\widehat{FCFF}_0(1 + \hat{g})^T}{(\widehat{WACC} - \hat{g})^3} \hat{\sigma}_{WACC-g}^2, \text{ where} \quad (22)$$

$$\hat{\sigma}_{WACC-g}^2 = \hat{\sigma}_{WACC}^2 + \hat{\sigma}_g^2 - 2 \text{Cov}(\widehat{WACC}, \hat{g}). \quad (23)$$

Free Cashflow to Equity

- Find cost of replicating *free cashflow to equity* (FCFE).

$$\text{FCFE} = \text{FCFF} - \text{interest}(1 - \tau) + \Delta \text{net debt}. \quad (24)$$

- Find change in firm leverage from time $0 \rightarrow t$: $\ell_t = \frac{1 + \frac{\text{debt}_t(1-\tau)}{\text{equity}_t}}{1 + \frac{\text{debt}_0(1-\tau)}{\text{equity}_0}}$.
- Adjust k for change in leverage: $k_{E,t} = r_f + \ell_t(k - r_f)$.
- Discount cashflows at $k_{E,t}$ and apportion per share:

$$V_t = \frac{\text{FCFE}_t}{k_{E,t} - g}, \text{ or } \hat{V}_t = \frac{\widehat{\text{FCFE}}_t}{\hat{k}_{E,t} - \hat{g}} + \frac{\widehat{\text{FCFE}}_t}{(\hat{k}_{E,t} - \hat{g})^3} \hat{\sigma}_{k_{E,t}-g}^2, \quad (25)$$

$$V_0 = \frac{\sum_{t=1}^T \frac{\text{FCFE}_t}{(1+k_{E,t})^t} + \frac{V_T}{(1+k_{E,T})^T}}{\text{shares outstanding}} \text{ or estimated version. } \dots \quad (26)$$

Valuation Models: Comparisons

- In theory, these valuation models should all be equivalent.
 - *Modigliani-Miller Theorem* says dividends, debt irrelevant.
 - Thus DDM, G-S, FCFF, FCFE, P/E should all yield same values.
- Why do these valuations differ then?
 - Uncertainty in assumptions. (e.g. When is firm “mature?”)
 - Uncertainty in direct inputs.
 - Non-constancy of inputs (r_f varies).
 - M&M ignores taxes, creditor rights, costs, other frictions.
 - $WACC_T - g$ and $k_{E,T} - g$ near 0 $\implies$ fat-tailed valuations.
- A good analyst would consider all these models.

Growth Opportunities and P/E

- If $ROE > k$, re-investing is better than value without: $\frac{E_1}{k}$.
- Difference = *present value of growth opportunities* (PVGO):

$$PVGO = P_0 - \frac{E_1}{k} \quad (27)$$

- Can rearrange PVGO formula to see P/E relationship:

$$\frac{P_0}{E_1} = \frac{1}{k} \left(1 + \frac{PVGO}{E_1/k} \right). \quad (28)$$

- Thus if PVGO is 0, stock trades like an annuity.
- As growth opportunities become more valuable, P/E rises.
- What is $PVGO/(E_1/k)$? Ratio of growth value to asset value.
 - If growth value high relative to assets, P/E will be high.

Price-to-Earnings and the Gordon-Shapiro Model

- Recall Gordon-Shapiro model: $P_0 = \frac{D_1}{k-g}$.
- Substituting in $D_1 = E_1(1 - b)$ and $g = b \cdot \text{ROE}$:

$$P_0 = \frac{E_1(1 - b)}{k - b \cdot \text{ROE}} \Rightarrow \frac{P_0}{E_1} = \frac{1 - b}{k - b \cdot \text{ROE}}. \quad (29)$$

- Note effect of b , ROE on P/E for $k = 10\%$:

	Retained earnings b			
ROE	0%	25%	50%	75%
7%	10.0	9.1	7.7	5.3
10%	10.0	10.0	10.0	10.0
13%	10.0	11.1	14.3	100.0

- Rough guide: $P/E \approx g \Rightarrow \frac{P/E}{g} < 1$ attractive.

The Discount Rate

- Big question: what *really* is the pricing kernel k ?
 - Efficient markets: $k = r_f +$ some multiples of risk premia.
 - But then we estimate k , yielding $\hat{k}$.
 - Why not use average excess returns of a stock? Could, but. . .
 - ① We think that model reduces noise (and thus valuation variation);
 - ② We can correct for time trends; and,
 - ③ We can exclude alpha — which allows for mispricings.
 - ④ Also nice: it helps reveal relevant economic forces.
 - Use risk-free rate based on our investment/decision horizon.
 - Probably should allow for multiple risk factors.
-

The Capital Asset Pricing Model

Chapter 14

Introduction

This part discusses the capital asset pricing model. Specifically:

- Single-index model;
- Markowitz curse of dimensionality;
- Equilibrium and capital asset pricing model (CAPM);
- Problems with the CAPM;
- Improving the CAPM;
- Hedging; and,
- The Zero Beta Portfolio.

Notation

There are some bits of notation to get straight first.

r_i = return on asset i ;

r_M = return on portfolio M ; often M means “market;”

r_f = risk-free rate for period returns r_i are over;

R_i = excess return of asset i ; $R_i = r_i - r_f$;

ϵ_i, η_i = new randomness from new info, supply/demand;

β_i = asset i 's sensitivity to a risk factor; and,

α_i = asset i 's abnormal/unexplained return.

Index Models



General Model

- In general, our models split expectations and risk:

$$r_i = \underbrace{E(r_i)}_{\text{mean}} + \underbrace{\epsilon_i}_{\text{risk}} \quad \epsilon_i \sim (0, \sigma_i^2). \quad (1)$$

- This often takes a form like so:

$$\underbrace{y_{it}}_{\text{response}} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\delta_j}_{\text{group mean}} + \underbrace{\beta x_t}_{\text{risk factor}} + \underbrace{m_t}_{\text{overall randomness}} + \underbrace{g_j}_{\text{group randomness}} + \underbrace{\eta_{it}}_{\text{unexplained randomness}}, \quad (2)$$

$$m_t \stackrel{\perp}{\sim} (0, \sigma_m^2) \perp g_j \stackrel{\perp}{\sim} (0, \sigma_g^2) \perp \eta_{it} \stackrel{\perp}{\sim} (0, \sigma_{\eta_i}^2) \perp m_t. \quad (3)$$

- Let's walk through this; it is nowhere near as scary as it looks.
- Then you will not be scared by subsequent models.
- Remember: this just expresses what we think/know.

Markowitz Model Input List

- Recall n -instrument-universe portfolio construction used:
 - estimates of n expected (log-)returns r_i ; and,
 - estimates of n variances σ_i^2 and $n(n-1)/2$ covariances Σ_{ij} .
- To get around this, split log-returns into expectation+surprise:

$$r_i = \text{expectation} + \text{surprise} = E(r_i) + \epsilon_i \quad \epsilon_i \sim (0, \sigma_i^2). \quad (4)$$

- With a common random surprise we get a random-effects model:

$$r_i = E(r_i) + m + \eta_i \quad m \sim (0, \sigma_m^2) \perp \eta_i \stackrel{\text{ind}}{\sim} (0, \sigma_{\eta_i}^2). \quad (5)$$

- Thus m, η_i are systematic, idiosyncratic parts of ϵ_i :

$$\sigma_i^2 = \sigma_m^2 + \sigma_{\eta_i}^2, \quad \Sigma_{ij} = \sigma_m^2. \quad (6)$$

Single Factor Model

- Our life is easier if we use a proxy for m .
 - Want: factor common to all firms; differing sensitivities to factor.
- This gives us the *single factor model*:

$$r_i = E(r_i) + \beta_i m + \eta_i \quad (7)$$

$$\Rightarrow \sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\eta_i}^2, \Sigma_{ij} = \beta_i \beta_j \sigma_m^2. \quad (8)$$

- ! None of this requires normality.
- In fact, m and η_i are not likely to be normal.
- So what is m ? That is a question you should keep asking.

Single-Index Model

- Want an observed common factor m ; why not impose one?
 - Use **broad-based** index log-returns; average should exceed r_f .
- Excess returns $R_i = r_i - r_f$ give *single-index model*:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \eta_{i,t}. \quad (9)$$

- Taking expectations, we get expected excess returns:

$$\underbrace{E(R_i)}_{\text{risk premium}} = \underbrace{\alpha_i}_{\text{idiosyncratic premium}} + \underbrace{\beta_i E(R_m)}_{\text{systematic premium}}. \quad (10)$$

- Taking variances: $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\eta_i}^2$, $\Sigma_{ij} = \beta_i \beta_j \sigma_m^2 = \rho_{im} \rho_{jm} \sigma_i \sigma_j$.
- Note flip from returns r_i 's to excess returns R_i 's!

Single-Index Model: Implications

- We greatly decreased the input set for portfolio optimization:
 - $n \hat{\alpha}_i$'s + $n \hat{\beta}_i$'s + $n \hat{\sigma}_{\eta_i}$'s + $\hat{R}_m + \hat{\sigma}_m + \hat{r}_f = 3n + 3$ estimates.
 - For 50 instruments, 153 versus 1,325 estimates.
- So we get more precise estimations of α 's, β 's, σ_{η_i} 's.
- Unfortunately, we trade efficiency for flexibility:
 - Cannot see unusual pairwise correlations (e.g. UN-UL).
 - Miss correlations b/w instruments w/ $\beta = 0$. (oil-diesel!)

The Capital Asset Pricing Model



Capital Asset Pricing Model (CAPM)

- *Capital asset pricing model* (CAPM) related to index model.
 - Sharpe (1964), Lintner (1965), Mossin (1966)... and Treynor (1961).
- Makes a number of assumptions:
 - ① Many small, price-taking, impactless investors.
 - ② No taxes, no transactions costs.
 - ③ Invest in public universe (only) without capital limits.
 - ④ All investors plan for same single period (myopic).
 - ⑤ Investors maximize mean-variance (Markowitz) utility.
 - ⑥ All investors have same knowledge, *homogeneous expectations*.
- If these hold, they imply an economic *equilibrium*:
 - All investors inherently hold “the market.”

CAPM: A Trip to *Equitopia*

Let's take a magic trip. . . to *equitopia* where all is in balance.

- Investors all see the return/risk ratio: excess return/variance.
- There is a typical ratio of excess return/variance.
- Investments with a higher/lower ratio look more/less attractive.
- Ah! But investors would have already bought/shorted those.
- Thus ALL investments have the same ratio of excess return/variance.
- So the “market” = everything in proportion to current size.
- Investors all hold the same risky assets, weights as market ($P = M$).
- M is also optimal risky portfolio (CAL=CML).

CAPM: Living in *Equitopia*

What are the implications of life in *equitopia*?

- All investments have the same ratio of excess return/variance.
- In well-diversified portfolio, instrument i risk contribution is equal:

$$\frac{R_i}{\Sigma_{iM}} = \frac{R_M}{\sigma_M^2} \implies R_i = \frac{\Sigma_{iM}}{\sigma_M^2} R_M. \quad (11)$$

- Sharpe-Lintner CAPM:

$$E(r_i) - r_f = R_i = \frac{\Sigma_{iM}}{\sigma_M^2} R_M = \beta_i R_M \implies \alpha_i = 0. \quad (12)$$

- We do not live in equitopia, so this is wrong.
- Even w/o equilibrium, CAPM is good first-guess linear model.
 - Except no statistician would just assume $\alpha_i = 0$.
- But, hold on! CAPM is *useful*.

CAPM: We Are All Indexers Now?

- One optimal risky MARKET portfolio to rule them all!
 - If CAPM assumptions true and...
 - ...we all have the same inputs, constraints, risks.
- Invest in risk-free bonds, market-representative portfolio.
- But those assumptions do not hold! Well, no; but:
 - Can view M as good first guess at optimal risky portfolio.
 - CAPM still has good advice: cheap index fund is good starting point.
- Since we all hold market and net borrowing+lending=0:
 - Average weight of M must be $\bar{y} = 1$.
 - Since $y = \frac{E(r_M) - r_f}{\bar{\lambda} \sigma_M^2}$, $E(R_M) = \bar{\lambda} \sigma_M^2$.
- Market risk premium based on variance, average risk aversion.

Security Market Line

- Plot expected return-beta relationship ($E(r_i)$ vs. β).
 - Get the *security market line* (SML).
- Fairly-priced assets lie on the SML.
- Instruments with expected return $>$ CAPM lie above SML.
- Instruments with expected return $<$ CAPM lie below SML.
- Alpha is distance above SML.
- For firm reinvestment, SML gives required rate of return:
 - Use required rate of return to value projects.

CAPM versus the Index Model

- CAPM is related, but different, from single-index model.
 - β_i is same in both model; but,
 - CAPM assumes single-index model $\alpha_i = 0$.
 - Extreme: CAPM says we cannot usefully forecast α_i 's.
- Thus we should not expect to “beat the market.”
- Can also consider realized – expected returns (*market model*):

$$r_i - E(r_i) = \beta_i(r_M - E(r_M)) + \eta_i. \quad (13)$$

- Return surprises tend to be noisy beta-scaled market surprises.

CAPM and Single-Index Model: Theoretical Problems

Both CAPM and single-index model have theoretical issues.

- There is probably not just one risk factor, the market.
- All correlations based on correlation to market proxy? No.
- Investors are not all mean-variance optimizers. (mean-risk?)
- Not clear that an equilibrium exists.
- Single-index model: Is α_i “money for nothing?”

CAPM and Single-Index Model: Theoretical Problems

CAPM and single-index model also have empirical issues.

- Universe of all assets is unknowable, so cannot test models.
- Using proxies for the market portfolio is flawed approach.
- Heteroskedasticity! Volatility is *not* constant.
- Inefficient markets mean data have serial correlations.
- Past betas may not predict future betas well (stats, econ reasons).

(Flawed) CAPM Tests

Issues do not stop people from (flawed but inevitable) testing of CAPM.

- Many “tests” of alphas do not clarify source of problem.
 - Is model flawed, missing risk factors? trend in data? bad proxy?
- Basic randomness: some tests will fail even for true models.
- Many tests introduce noise into test; biases test outcomes.
- Some tests use sorted data; that also induces a bias.

Ultimately, we cannot “prove” a theory with data.

Improving the CAPM

We can improve the CAPM to get a better model, better estimated.

- Find better beta estimates.
- Consider the macroeconomic state.
- Add other investments to our universe.
- Include other risk factors.
- Maybe look at a precursor model.

Better Betas

- Estimated past betas may not best predict future betas.
- We can *squeeze* betas toward the overall mean:

$$\tilde{\beta}_i = \frac{1}{3} + \frac{2}{3}\hat{\beta}_i \quad (\text{Blume}) \quad (14)$$

$$\equiv \frac{\hat{\sigma}_{\bar{\beta}}^2}{\hat{\sigma}_{\hat{\beta}_i}^2 + \hat{\sigma}_{\bar{\beta}}^2} \hat{\beta}_i + \frac{\hat{\sigma}_{\hat{\beta}_i}^2}{\hat{\sigma}_{\hat{\beta}_i}^2 + \hat{\sigma}_{\bar{\beta}}^2} \bar{\beta} \quad (\text{Vašíček}). \quad (15)$$

- We can try to predict future betas (Rosenberg and Guy):

$$E(\beta_{i,t+1}) = a + b_1\beta_{i,t} + b_2\text{mktcap}_{i,t} + b_3\frac{\text{debt}_{i,t}}{\text{assets}_{i,t}}. \quad (16)$$

- Industry also matters; betas vary over business cycle.

A Better Model

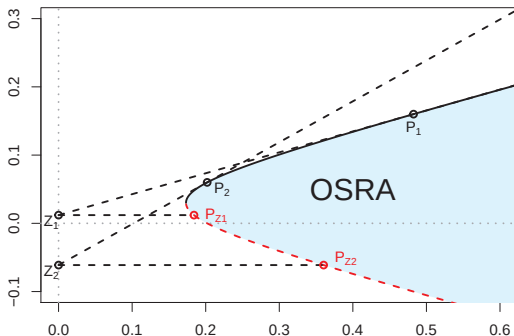
- Betas vary over business cycle; so condition on macroeconomy.
 - Idea of the conditional CAPM (Rothschild/Harvey).
 - Similar: intertemporal CAPM adds economic state factor (Merton).
 - Thus risk-shifting (and speculators) are beneficial.
- We should also incorporate other investments into the model.
 - Heaton and Lucas added private firms; that performs better.
 - Adding human capital investment also helps (Jagannathan+Wang).
- Finally, we should allow for a more complete “market” index.
 - Add more equity indices, other asset-class indices?
- Back to economic basics; look at consumption? Not so helpful.

Hedging

- One of the great ideas of the CAPM: hedging market exposure.
- Say we find portfolio P : $R_P = 0.02 + 1.3R_M + \eta_P$.
- Suppose M is Russell 2000; can find *tracking portfolio* T .
 - T has same $\beta = 1.3$, minimal idiosyncratic variance.
 - T designed to match systematic return of P .
 - T is just 1.3 Russell 2000 -0.3 T-bills.
- Can then eliminate market risk by shorting T .
 - Long P , short T is *market neutral*.
- Could go further: separate alpha and market exposure.
- This is idea behind *alpha transport* or *portable alpha*.
 - Much money lost learning how this relies on assumptions.

The Zero Beta Portfolio

- What if we do not have a risk-free rate? Fisher Black to the rescue!
- Black's idea: market-neutral portfolio implies risk-free-ish rate.
 - Let market M = optimal risky portfolio;
 - Tangent line to frontier at M intersects y-axis at r_z .
 - Frontier portfolio P_{ZM} with $E(r_{P_{ZM}}) = r_z$ has zero beta.
- Yields Black form of the CAPM: $r_{it} = r_z + \beta_i(r_{Mt} - r_z) + \epsilon_{it}$.



Nous sommes tous CAPMeurs?

- Clearly, the CAPM has been used for many models.
- Why is CAPM so insanely great?
- Excellence is not due to CAPM *per se*.
- Concept is great because it joins useful ideas:
 - Linear modeling (aka regression) of benefits;
 - Risk/return tradeoffs should tend toward average; and,
 - Tradeoffs between desirable features of inputs/a model.
- These key ideas will serve you well throughout your life.
- Are they perfect? No; but, they are useful.
- The trick is finding trade-offs you can model.

Factor Models/ The Arbitrage Pricing Theory

Chapter 15



Introduction

This part discusses factor models and the arbitrage pricing theory. Specifically:

- Arbitrage pricing theory;
- Trading mispricings;
- Macro risk factors;
- Micro risk factors; and,
- Style factors.

Risk Factors: Betcha Can't Use Just One

- Conditional CAPM, ICAPM show us: one risk factor is not enough.
- When we use multiple factors, we get models like:

$$R_i = \alpha_i + \beta_{i,GDP}(\Delta GDP - r_f) + \beta_{i,r_f}\Delta r_f + \eta_i. \quad (1)$$

- Call β_i 's *factor sensitivities/loadings*. . . or just betas.
- We are now more likely to see negative β 's:
 - Maybe instrument is really a hedge for that factor;
 - Maybe the model considers a spread between factors; or,
 - The risk factors may be multicollinear.

Arbitrage Pricing Theory (APT)

- Arbitrage pricing theory (APT) developed by Ross (1976).
- Three assumptions of APT:
 - ① Returns may be described by a factor model;

$$r_i = \nu_i + \beta_{i1}f_1 + \dots + \beta_{ik}f_k + \epsilon_i. \quad (2)$$

- ② Can diversify away idiosyncratic variance (ϵ_i effect); and,
 - ③ Markets do not allow persistence of arbitrage.
- If these hold, APT makes a bold assertion:

$$R_i \approx \beta_{i1}F_1 + \dots + \beta_{ik}F_k, \quad F_j = f_j - r_f. \quad (3)$$

- *Arbitrage*: riskless profits not requiring capital.
- Often trade factor *tracking portfolios*, e.g. $T_1 \sim F_1$.

So... We're All Done Here?

- APT seems to say: find a factor model and risk goes away.
- Ah, but what is it they say about beautiful theories?
- Lehmann and Modest: APT does not hold without noise.
 - So that approximation and dropping the error term? Never mind.
- We still need risk management. . . and risk managers.
- However, the APT offers a very useful perspective.
- Asset management conferences: many talks on risk factors.

Arbitrage vs. Risk-Return Dominance

What happens when we are in *disequilibrium*?

- *Dominance argument*: many change holdings slightly.
 - Changes due to differing risk aversion (CAPM).
- *Arbitrage argument*: few take large positions; smash prices together.
 - This implies any disequilibrium closes quickly (APT).
- APT-based arguments stronger than CAPM-based arguments.
- *Law of One Price* (LOOP): economically same = priced same.
- What is not arbitrage? Stat arb, risk arb, ... most “arb.”
- We will use *well-diversified portfolios*: $\eta_P \approx 0$.

$$r_P \approx E(r_P) + \beta_{P,F}F. \quad (4)$$

Theory for Factor Models

A *lot* of theory for factor models and their validity. Briefly;

- Absence of arbitrage $\iff$ can price all sorts of options.
- Factor model return Sharpe ratio $\leq$ optimal risky portfolio S_P .
- Factor model explain less than R^2 of true model variation.
- What if a factor model breaks one of these rules?
 - Then we know it is not an accurate model of risk factors.

Rules for Arbitrage

- If two instruments should be equally priced:
 - Cheaper/higher-yielding = *cheap*; expensive/lower-yielding = *rich*.
- General rule for arbitrage situations.

Buy the cheap; sell the rich.

- Amounts must cancel risk factors; use T-bills to balance money.
- This captures difference between their prices/returns.
- Mispricings may be absolute or relative.

Example Relative Mispricing

- Suppose we have two well-diversified portfolios A , B .

$$E(r_A) = 0.1 + 1.2F \quad (F \text{ is a risk factor}); \text{ and,} \quad (5)$$

$$E(r_B) = 0.09 + 1.2F \quad \text{where } E(F) = 0. \quad (6)$$

- APT/LOOP says these should be priced the same.
 - Thus $E(r_A)$ should equal $E(r_B)$. . . but it does not.
 - Thus we have a relative mispricing of A versus B .
 - Buy cheap (A), sell rich (B) to cancel out risk.
 - Total return on \$1 is then $\$1(r_A - r_B) = \0.01 .

Example Absolute Mispricing

- Assume $r_f = 0.02$; factors $E(f_1) = 0.05$, $E(f_2) = 0.08$:

$$r_A = r_f + 1F_1 + 0.5F_2. \quad (7)$$

- Arb-free $E(r_A)^* = 0.02 + 1(0.05 - 0.02) + 0.5(0.08 - 0.02) = 0.08$.
- If observed $E(r_A) = 0.10$, A return $>$ expected: absolute mispricing.
- A yielding more than expected = cheap.
- Arb portfolio G : Buy \$1 A , \$0.5 T-bills; short \$1 F_1 , \$0.5 F_2 .

Example Absolute Mispricing: The Bookkeeping

- Arb portfolio G : Buy \$1 A , \$0.5 T-bills; short \$1 F_1 , \$0.5 F_2 .
- Return on G is expectation plus surprises (differences from expected):

$$r_G = E(r_A) + 1(f_1 - E(f_1)) + 0.5(f_2 - E(f_2)) - 1T_1 - 0.5T_2 + 0.5r_f \quad (8)$$

$$= E(r_A) - E(f_1) - 0.5E(f_2) + 0.5r_f \quad (9)$$

$$= 0.10 - 0.05 - 0.5 \cdot 0.08 + 0.01 = 0.02. \quad (10)$$

- *N.B.* captured $|\text{arb-free} - \text{observed}|$ ($|0.08 - 0.10|$).

Example Multi-factor Relative Mispricing

- Suppose the prior setup but we also look at portfolio B :

$$r_A = r_f + 1F_1 + 0.5F_2, \quad (11)$$

$$r_B = r_f + 0.8F_1 + 2F_2. \quad (12)$$

- Arb-free $E(r_B)^* = 0.02 + 0.8(0.05 - 0.02) + 2(0.08 - 0.02) = 0.164$.
- If observed $E(r_B) = 0.14 < \text{expected}$: A is cheap, B is rich.
- Arb portfolio G : buy \$1 A , \$1.5 T_2 ; short \$1 B , \$0.2 T_1 , \$1.3 r_f .

Example Multi-factor Relative Mispricing: Bookkeeping

- Arb portfolio G : buy \$1 A , \$1.5 T_2 ; short \$1 B , \$0.2 T_1 , \$1.3 r_f .

$$\begin{aligned} r_G = & E(r_A) + 1(f_1 - E(f_1)) + 0.5(f_2 - E(f_2)) \\ & - E(r_B) - 0.8(f_1 - E(f_1)) - 2(f_2 - E(f_2)) \\ & - 0.2T_1 + 1.5T_2 - 1.3r_f \end{aligned} \quad (13)$$

$$= E(r_A) - E(r_B) + 0.2E(f_1) - 1.5E(f_2) - 1.3r_f \quad (14)$$

$$= 0.10 - 0.14 - 0.02 \cdot 0.05 + 1.5 \cdot 0.08 - 0.026 = 0.044. \quad (15)$$

- N.B.* captured mispricing in A (0.02) and B (0.024).

Risk Factors

Useful Risk Factors?

- The one issue we have yet to address: what are useful risk factors?
- This gets tricky: factors need to have long-run importance.
- In general, we can break factors into three groups:
 - Macroeconomic factors;
 - Microeconomic factors; and,
 - Style factors (which maybe are not real factors).
- Many academics made an industry of finding factors.

Macro Factors

- Already saw a fixed-income macro model (Litterman-Scheinkman):

$$r_{jt} = \alpha_j + \beta_{j,IV} \Delta \text{Level}_t + \beta_{j,slope} \Delta \text{Slope}_t + \beta_{j,curv} \Delta \text{Curv}_t + \eta_{jt}. \quad (16)$$

- Five-factor macro model (Chen-Roll-Ross):

$$r_{jt} = \alpha_j + \beta_{j,IP} \overbrace{\Delta \% IP_t}^{\text{industrial production growth}} + \beta_{j,Ei} \overbrace{\Delta E(i_t)}^{\text{expected inflation change}} + \beta_{j,\epsilon i} \overbrace{\epsilon_{it}}^{\text{inflation surprise}} + \beta_{j,CMG} \overbrace{\Delta CMG_t}^{\text{credit spread change}} + \beta_{j,LMS} \overbrace{\Delta LMS_t}^{\text{yield curve slope change}} + \eta_{jt}. \quad (17)$$

- ϵ = surprise in inflation vs expected; $i_t - E(i_t)$,
- CMG = corporate – govt bond credit spread,
- LMS = long–short-term govt yields $\approx$ YC slope.

Macro Factors: Hedge Fund

- There are three macro factor models often used for hedge funds.
- Hasanhodzic-Lo six-factor model:
 - excess return on market, AA corporate bond index;
 - return on currency, commodity indices; and,
 - corp–govt bond credit spread, implied volatility index (e.g. VIX).
- Fung-Hsieh seven-factor model:
 - excess return on market; small-minus-large-cap return;
 - change in UST 10Y yield; corp–govt bond credit spread; and,
 - trend factors for bonds, FX, and commodities.
- Fung-Hsieh eight-factor model:
 - excess return on US, non-US, emerging market equity indices;
 - return on US, non-US govt bonds and 1M Eurodollars; and,
 - return on gold and trade-weighted USD index.
- Commodity index returns often related to unexpected inflation.

Micro Factors: Dynamic

- Micro factors relate to firm-specific behavior.
- Factors are often dynamic or liquidity-based.
- Engle-Lilien-Robins ARCH-in-mean:
 - Captures spikes in market volatility w/time series model;
 - Current volatility affects returns: clean risk-return tradeoff.
- Bollerslev-Engle-Wooldridge GARCH-in-mean:
 - Like ARCH-in-mean model but allows for cross-correlations.
 - Stock market volatility can affect bond market returns.

Micro Factors: Liquidity, Firm Risk

- Micro liquidity models try to capture effect of market liquidity.
- Liquidity is a very powerful factor; driving force of many crises.
- Amihud-Mendelson model just includes %bid-ask spread surprise:

$$r_{jt} = \alpha_j + \beta_{M,j} R_{Mt} + \beta_{S_j} \underbrace{(S_{jt} - E(S_j))}_{\substack{\text{unexpected} \\ \% \text{spread}}} + \eta_{jt}. \quad (18)$$

- Acharya-Pedersen Liquidity CAPM relates spread-excess returns:
 - $R_{jt} - S_{jt} = \beta_{all}(R_{Mt} - S_{Mt})$.
 - β_{all} : components from market, stock returns, spreads.
- Pastor-Stambaugh looks at stock-vs-market outperformance:
 - Looks at Fama-French factors, surprise in dollar volume effect.*
- Could also look at firm-specific risk factors:
 - Idiosyncratic volatility, assets vs liabilities, cashflows in vs out.

* We will see the Fama-French model shortly.

Style Factors

- Style factors relate to firm aspect/cross-sectional differences.
- Fama-French three-factor model uses small—big, high—low B/P:*

$$r_{it} = \alpha_i + \beta_{i,M}R_{Mt} + \beta_{i,SMB}SMB_t + \beta_{i,HML}HML_t + \eta_{it}, \quad (19)$$

- Carhart model adds “momentum” winners—losers (WML_t) factor.
- Fama-French five-factor model adds to three factors:
 - RMW_t : robust—weak profitability, and
 - CMA_t : conservative—aggressive portfolios.

* Related: The Morningstar style box.

Will the Real Risk Factors Please Stand Up?

- Are these really risk factors? The right factors?
- Research suggests: No. Not risk, not the driving forces.
- Palacios-Huerta: adding human capital is better than FF3.
- Chan-Chen-Hsieh: SMB related to credit spreads.
 - May also add small-stock effects = more complete market.
- Das-King-Sinha: HML only active around earnings announcements!
- Sadka: Momentum related to post-earnings announcement drift?
- Sinha: News sentiment subsumes momentum.
- Wang: initial jobless claims (IJC) surprises explain momentum.
- *SMB*=credit/small-caps, *HML*=earnings risk, *WML*=news/macro?

Modeling Issues

- Factor models are useful; but, modeling questions remain.
- Should we look at levels of variables or changes?
 - Levels if level relates to risk or prices take time to adjust.
 - Changes/surprise if markets efficient, respond to info.
- Excess returns, returns, or surprises?
 - Excess returns for traded instruments w/risk premium;
 - Returns for stationary/mean-zero factors; and,
 - Surprises for macro, strongly-recurring factors.
- Finally, what do we want from factor models?
 - Pricing kernel? Meaning/insight? Benchmark? Replication?

Coda: Real Risk Factors

- The proliferation of claimed risk factors raises skepticism.
- Harvey-Liu-Zhu note this is like testing multiple hypotheses.*
- Must hold models to a stricter standard when many tests.
- They find most-likely factors: market, HML, momentum, and . . .
 - . . . durables consumption, short-term volatility, liquidity.
- Factor actively traded? Also suggests it is a risk factor.

* If you roll 1000 dice pairs, are the double-sixes really lucky dice?

The Failure of Arbitrage Pricing

- Huberman-Stanzl: Markets do not allow quasi-arbitrage.
 - Cannot build a sequence of trades that generates money.
 - Cost of exiting APT trades = cost of entry; makes no money.
- “Siamese Twin” shares: strong evidence that APT is not exact.
- Model risk: what if your factor model is wrong?
- Fundamental risk: no idea when you make money, so you sit out.
- Credit, trust: venues check credit; counterparties must trust you.
- Market segmentation: different horizons? different risk pricing.

Quantitative Investment

VALORACIÓN II

Temas 18 al 21

Foreign Exchange

Chapter 18



Introduction

- This part discusses foreign exchange.
- In particular, we will discuss:
 - Global Currencies;
 - Currency Risk;
 - Interest Rate Parity;
 - Purchasing Power Parity and Other Models;
 - Growth Implications; and,
 - Current Issues.

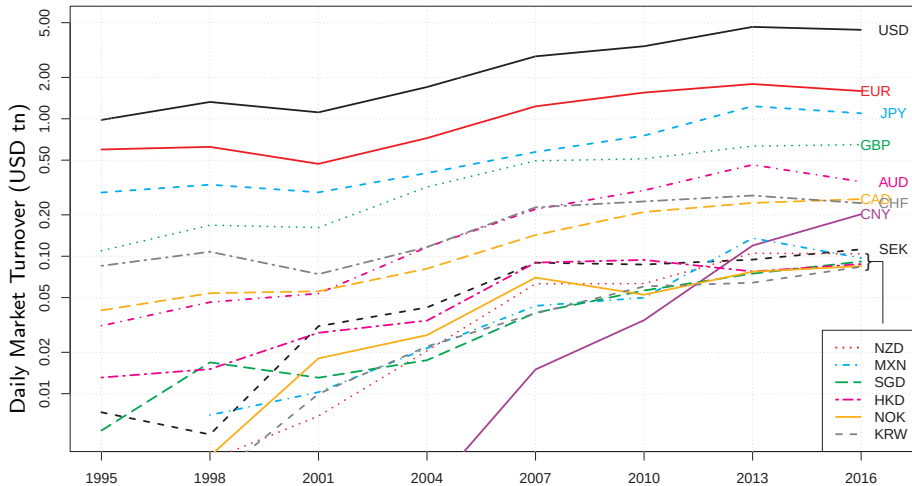
Currency Codes

- Even more than countries, we need a shorthand for currencies.
 - \$? USD, CAD, AUD, NZD, HKD, SGD, TWD? MXN, ARS? \$? BRL?¥?
 - JPY? CNY?
- ISO code = country code + currency character. . . mostly:
 - EUR, CAD, AUD, NZD, HKD, MOP, CNY, JPY, INR, IDR.

Global Currencies

- A few issues, conventions we need to know for FX.
- Most currencies *freely convertible*; some subject to *capital controls*.
- Some currencies are also *pegged* to others.
- Quoting convention of Base/Quoting: counterintuitive.
 - $ABC/DEF = x$: 1 unit of ABC is x units of DEF; $1 : x$.
 - $ABC/DEF \approx \frac{1}{DEF/ABC}$ (due to transaction costs)
- Preferred base: EUR, then GBP, AUD, NZD, USD, CAD, CHF....
 - Quote JPY as USD/JPY; GBP, EUR as GBP/USD, EUR/USD.
- “Long” = long base currency (and so short quoting currency).

Market Size



FX: biggest market in the world; never sleeps.

Handling Risks

- How do we handle these risks? Ideally, we hedge:
 - Find instruments matching those risks; trade to counter risk.
 - Trade FX futures to (roughly) lock in exchange rates.
 - Buy volatility, out-of-the-money puts in case of political issues.
- Should also hire analysts who understand local language/culture.
- Analysts can also help parse differences in financial reporting.
- Diversify: don't put too much of your portfolio in one country.

Handling Risks: Example

- Suppose I am going to invest USD 30 mn in the Euro Stoxx 50.
- To hedge my Euro exposure, I can lock in the reconversion to USD.
- When I sell stocks, will sell EUR and buy USD; so: short EUR.
- Sell USD 30 mn = EUR 25.5 mn $\implies$ 204 CME 6E futures.
- If EUR $\uparrow$, less USD per EUR:
 - Unhedged: I might lose even though index is up.
 - Hedged: futures decline; short position accrues a profit.
- If I sell at a value $>$ \$30 mn, I am unhedged on the gains.
- Could also be over-hedged (equally bad) if we have losses.
- Thus we adjust the FX hedge day-by-day.

What Drives FX Rates

- Investment opportunities in a country drive FX rates.
- In particular, interest rates are a strong driver of FX rates.
- Will see two such models: uncovered, covered interest rate parity.
- Will also look at purchasing power parity.
- And then consider some other drivers.

Interest Rate Parity: Uncovered

- If we know interest, FX rates: should not be easy money.
- This is the idea behind *uncovered interest rate parity*.
 - Interest rate differences imply expected FX rate change.
- If $r_{\tau,USD}$ = USD risk-free rate over t to $t + \tau$:

$$1 + r_{\tau,USD} \stackrel{\text{should}}{=} \frac{E(X_{t+\tau,EUR/USD})}{X_{t,EUR/USD}} (1 + r_{\tau,EUR}) \quad (1)$$

$$\Rightarrow \tilde{X}_{t+\tau,EUR/USD} = \frac{1 + r_{\tau,USD}}{1 + r_{\tau,EUR}} X_{t,EUR/USD}, \text{ or} \quad (2)$$

$$\% \Delta \tilde{X}_{t+\tau,EUR/USD} \approx r_{\tau,USD} - r_{\tau,EUR}. \quad (3)$$

- If $r_{\tau,USD} = 4\%$, $r_{\tau,EUR} = 3\%$: expect EUR/USD $\uparrow 1\%$.

Carry Trades and a Puzzle

- Beautiful theory! There's just one problem. . .
- Yet again: it's those ugly facts.
- *Forward premium puzzle*:
 - Currencies of high-interest rate economies should depreciate.
 - However, they often appreciate. Why? Not clear.
- This has led to *carry trades*:
 - Borrow in low-interest rate economies. . .
 - . . . invest in high-interest rate economies.
-  *Aviso!* Is this a *Peso Problem*? YES.

Interest Rate Parity: Covered

- What if we could lock in future FX rates?
- There should not be easy money; else, we can arbitrage.
- This is the idea behind *covered interest rate parity*:
- If we know the later FX rate $X_{t+\tau, USD/EUR}$:

$$\underbrace{X_{t, EUR/USD}(1 + r_{\tau, USD})}_{\text{EUR } 1 \rightarrow \text{USD, earn USD interest}} \stackrel{\text{should}}{=} \underbrace{(1 + r_{\tau, EUR})X_{t+\tau, EUR/USD}}_{\text{EUR 1 earns EUR interest, } \rightarrow \text{USD}} \cdot \quad (4)$$

- FX rates should compensate for different interest rates.
- If relationship doesn't hold, *carry trade*: lock in profit.
 - If $r_{\tau, USD} = 4\%$, $r_{\tau, EUR} = 3\%$: $X_t = X_{t+\tau} = 1$.
 - Borrow EUR @ 3%, EUR $\rightarrow$ USD, lock in USD $\rightarrow$ EUR at time τ .
 - Invest USD @ 4%. End: USD $\rightarrow$ EUR, repay loan.
 - PROFIT!

Interest Rate Parity: Covered?

- Suppose we can lock in future FX rates and see an arbitrage.
- This should be easy money; fire up the money machine!
- Makin' cash! Amirite?!? U jelly bro? Well... hold on.
- The truth is that even with hedging we could end up unhappy.
- What if the bank we invest in fails?
- What if the money is stolen through fraud?
- What if the country limits conversions out of their currency?
- Why would banks offer an arbitrage if they were healthy?

Interest Rate Parity and Icelandic Banks: Act I

- Consider people who put money in Icelandic banks:
 - 2003 Iceland: 290k people, \$9.7 bn GDP (=\$39k).
 - Pre-2008: banks expanded w/easy loans from Eurobonds.
 - High interest rates: 8% (2005), 10% (2006), 13% (2007).
 - Meanwhile, rates in Europe in 4%–6% range, US in crisis.
- Icelandic banks court investors through online banking.

Interest Rate Parity and Icelandic Banks: Acts II & III

- Money flows into ISK; huge bank deposits.
- Banks: more money than borrowers need. . . so lend foolishly.
 - 2007 Iceland: 312k people, \$21 bn GDP (=\$69k).
 - 2007–2008 Icelandic inflation of 14% $\gg$ target of 2.5%.
- Sep 2008: Central bank raises rates to 15+ % to stem inflation.
- Banks are unable to replace old debt w/new debt; begin failing.
- Depositors lose money; Iceland limits conversions out of ISK.
- Government fell; bank executives imprisoned.
- Capital controls not removed until 2017!

Purchasing Power Parity

- Money is not useful *per se*; useful for what it can buy.
- Often measure purchasing power w/*basket* of goods.
- Adjusting for costs of goods: *purchasing power parity* (PPP).
- In the US, we usually adjust based on CPI.
- One of the more effective, unusual indices: *Big Mac Index*.
 - Reason: Big Mac is common, unchanging basket of goods+services.

Purchasing Power Parity and FX Rates

- Two ideas about FX rates and PPP: absolute and relative.
- Absolute: Differences in PPP imply FX rate.*

$$\tilde{X}_{t,EUR/USD} \stackrel{\text{should}}{=} \frac{CPI_{t,USD}}{CPI_{t,EUR}}. \quad (5)$$

- Relative: Change in PPP implies change in FX rate.

$$\tilde{X}_{t+\tau,EUR/USD} \stackrel{\text{should}}{=} \frac{\Delta CPI_{t \rightarrow t+\tau,USD}}{\Delta CPI_{t \rightarrow t+\tau,EUR}} X_{t,EUR/USD}. \quad (6)$$

- Problem: Prices usually higher in more productive economies.
 - Could we adjust for productivity? Sure!**
 - Also consider anti-distortionary FX policies, trade.

* *N.B.* Both CPIs must have same basket.

** If you thought that, welcome to being an analyst.

Current Issues

- Big issues in the land of foreign currencies now?
- The Eurozone: can it stick together?
 - How to handle the next Greece?
 - What will Brexit do to EUR? European financing markets?
- Cryptocurrencies
 - What happens when govts with capital controls crack down?
 - Already happening. Results? Often hurts cryptocurrencies.
 - What about counterparty risk? Operational risk? KYC?

The Road Ahead

We covered Foreign Exchange; on to Risk Alleviation and Futures next!

- Risk Alleviation: Futures, Options, Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

Forwards, Futures, and Swaps

Chapter 19

Introduction

- Here we will discuss forwards, futures and swaps. In
- particular, we will discuss:
 - Cash/Spot;
 - Forwards;
 - Futures and the effects;
 - Swaps; and,
 - Futures curves.

Spot and Cash Prices

- *Spot price*: paid for a good delivered now “on the spot.”
 - Thus location, quality, and immediacy all affect price.
- Disseminating all possible spot prices would be onerous.
- Markets center on locations (Chicago, NYC, London, Tokyo).
 - Economies of scale, deep labor pools, ports, rail termini.
 - These spot prices more important (early standardization).
 - Leads to quoting prices “basis*” key spot prices.
- *Cash price*: paid for a good now/ASAP in standard location.
 - People often use “spot” and “cash” interchangeably.
- *Location differential* = Spot price – Cash price.
 - ① Q: What if demand for coal to make steel in Gary is high?
 - ② A: Gary location differential for coal will be high.

* “Basis” refers to a price differential.

Speculators and Risk Pricing

- Standardized cash prices allow comparison, liquid markets.
- Liquid markets tend to attract speculators.
- Speculators seek profit by betting on price direction, costs.
- Successful speculators attract more speculators.
- Competition helps establish the cost of assuming borne risks.
- Thus speculators offer three major benefits:
 - *Risk Transfer*: assume risk others do not want.
 - *Risk Pricing*: help determine costs of those risks.
 - *Liquidity*: trade when others will not; attract others.
- These three benefits are central to all derivatives markets.

Derivatives

- Derivatives: contracts traded on *underlying* risk.
- Thus they derive their value from the price of the underlier.
- Contracts are negotiated b/w two individuals (*counterparties*).
- May be traded on-exchange or *over-the-counter* (OTC).
- OTC trading typically done over the phone.
 - Allows for flexible terms, customized contracts.
 - Must often call for quotes; trade rarely reported publicly.
 - Thus OTC markets are less price-competitive.

Looking Forward

- Some people cannot tolerate risking where prices will be.
 - They must make expensive business decisions in advance.
 - They base these decisions on prices.
- Thus producers who act on prices often want to lock in prices.
- Consumers act likewise, also want to lock in prices.
- These needs led to first *forward contract*:
 - Agreement between two people to trade later at a set price.
 - Specifies amount, quality, delivery location/method, and date.
 - Money changes hands at later specified date.
- First forwards traded in ancient Babylonia (1750 BCE).

Making Forwards Better

- Obvious steps to increase liquidity, convenience:
 - Standardize contracts (quality, size, expiry, delivery location).
 - Trade contracts in one location (an exchange).
 - Protect against counterparty impairment.
- With standardization, prices become comparable.
- Can require traders to deposit money in an account.
- Mark to market: regularly credit/debit account P&L.
- Net positions daily for determining proper margin.
- Comparable prices let us determine unrealized P&L.
- Structure all contracts to involve well-capitalized CCP.

Marking to Market

- Central clearing involves *marking to market*:
 - Paying/receiving payment for daily P&L change.
- What does marking to market (MTM) look like?

Day 1 Buy 10 Mar CME corn (ZCH9) @ \$4.2700.
No money paid; post margin ($\approx 5\%$).
Mar corn (ZCH9) closes @ \$4.2625.
Pay MTM debit of $10 \times 5000 \times \$0.0075 = \375 .

Day 2 Mar corn (ZCH9) closes @ \$4.3550.
Receive MTM payment of \$4,625.

Day 3 Mar corn (ZCH9) expires @ \$4.3500.
Pay MTM debit of \$250.

- At expiry, most of P&L has been *previously* paid/debited.
- At closeout, we get our margin balance back — $\pm$ P&L
- Without MTM: realize \$4,000 of P&L on day 3.

Margin

- Traders on a derivatives exchange must post margin:
 - Cash, letter of credit, or govt bonds in segregated account.
 - Margin (collateral) ensures trader can pay mark-to-market.
- Four important margin terms:
 - *Initial margin*: initial deposit (max expected daily change).
 - *Maintenance margin*: account level requiring more margin.
 - *Margin call*: the request to post more margin.
 - *Variation margin*: amount needed to satisfy (meet) margin call.

Margin Example

Day 1 Short 10 contracts of Mar CME corn (ZCH9) @ \$4.27.

Contract size = 5000.

Notional = $10 \times 5000 \times \$4.27 = \$213,500$.

Initial margin = $10 \times \$1,000 = \$10,000$. ($\approx 5\%$)

Mar corn closes at \$4.2625

Mark-to-market credit = \$375; margin balance = \$10,375.

Day 2 Mar corn closes at \$4.3550

Mark-to-market debit = \$4,625; margin balance = \$5,750.

Below maintenance margin (75% of \$10,000 = \$7,500).

Margin call: variation margin of \$4,250.

After meeting margin call, margin balance = \$10,000.

Derivatives Trading and Positions

- B sells A 10 contracts of Mar CME corn^{*} at \$4.27.
 - A's position = +10; B's position = -10.
- ZCH9 goes up \$0.10: A makes $\$0.10 \times 10 \times 5000 = \5000 .
 - B loses \$5000.
 - Derivatives markets are *zero-sum games*.
- *Open interest* = 10 contracts; *volume* = 10 contracts.
- A closes out position by selling 10 contracts back to B.
 - Open interest = 0; volume = 20.
 - Had A sold to C instead, open interest would be 10.
 - When B made money, C would lose money (still zero-sum).

^{*} Mar CME corn contract (ZCH9) is for 5000 bushels.

Forward to Futures

- These features change a forward into a *futures contract*.
 - Standardized contracts (quality, size, expiry, delivery location).
 - Trading contracts on an exchange.
 - Require trader to post collateral in a margin account.
 - Mark to market to reduce counterparty risk.
 - Net positions daily for determining proper margin.
 - Make well-capitalized *clearinghouse* all contracts' *counterparty*.
(aka *central counterparty*, CCP)
- First futures in 1730 in Osaka at Dōjima Rice Market.
- Re-emerged in 1848 at Chicago Board of Trade.
- Term “futures” coined in 1865 at CBOT.

Futures: Range of Factors

- Futures are traded on a number of underliers:
 - *Equity Indices*: Domestic/foreign, large-/small-cap.
 - *Interest Rates*: LIBOR/short-term, sovereign bonds.
 - *Foreign Exchange*: Cross-rates with “Big 5,” others.
 - *Ags*: Grains, oilseeds/crush, fibers, livestock, softs.
 - *Metals*: Base, minor, strategic (U_3O_8), precious, steel.
 - *Energy*: Crude+products, LPGs+natgas, coal, power, ethanol.
 - *Other*: Plastics, emissions, shipping, real estate, volatility, storage.
- *N.B.* LME contracts are often called futures; technically forwards.
- This allows easy and cheap trading of many risk factors.
- Thus futures may be used to hedge or invest.

Hedging with Futures

- Hedge usually does not replace supply chain or investments.
- Rather: put hedge on; P&L roughly compensates for price changes.
- Exit contract before actual purchase/sale via supply chain.
- e.g. buy copper futures; exit before delivery of copper wire.
- Can exit a contract in a few ways:
 - Usually trade out of (“offset”) position before expiry.
 - Rare delivery: 1% (metals, energy) to 0.25% (grains, oilseeds).
 - Can exchange contracts for physical goods (“EFP”) before expiry.
- To keep hedge, must *roll* (transition) to new contract at expiry.

Benefits of Futures: Market-wide

- Futures lower cash market volatility.
 - What about onions...?
- Futures increase cash market liquidity, depth.
- Futures have also increased trade, allocative efficiency.
- That lowers prices; consumer benefit exceeds producer burden.
- Futures also allow investors to further diversify portfolios

Benefits of Futures: Specific

- Can use 10Y Treasury note futures to hedge, e.g. \$10,000 DV01.
 - Modified duration $D^* = 8.7$ for bond futures trading at 90;
 - Contract multiplier is \$1000;
 - Futures DV01: $8.7 \times 0.0001 \times \$1000 \times 90 = \$78.3$.
 - Sell $\$10k / \$78.3 \approx 128$ futures contracts to hedge.
- *Index arbitrage*: trade futures vs index members, syncs prices.
 - Trade all 500 S&P 500 stocks, most/all of Topix/Russell 2000.
 - Hence why high-speed Chicago–NYC, London–Frankfurt links.
- We can also use the futures to synthetically short stocks:
 - Go long 1 contract; do an EFP.* Now have stocks long.
 - Go short 1 contract. Now flat: stocks net out futures.
 - Selling stock long gives us a synthetic short position.

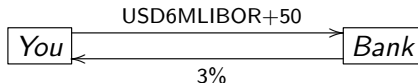
* EFP turns futures long into portfolio of index stocks.

Swaps

- A *swap* is an agreement to exchange cashflows.
- Thus a swap is very similar to a futures contract.
- Futures: Post collateral (ceding interest), get MTM flows.
- Swap: Pay financing flows, receive desired cashflows.
- For swaps, cashflows based on some *notional* amount.
 - The notional amount *never* changes hands.
- Cashflows referred to as *legs*: pay leg, receive leg.
 - Usually, one leg is a floating rate = *financing leg*.
 - Occasionally *resets*: net payment; change floating-rate.
- First swap: 1981, IBM $\rightleftharpoons$ World Bank (CHF $\rightleftharpoons$ USD rates).

Interest Rate Swaps

- Interest rate swaps are the most “plain vanilla” of swaps.
- “Fixed-for-float:” swap fixed- for floating-rate (or vice versa).
- Example: pay USD 6M LIBOR + 50, receive 3% fixed.



- Swap on \$100 MM notional; resets semi-annually.
- 50 bp floating-rate offset is a credit spread.
- Most floating rates reset every 6M, based on LIBOR.
- Fixed rate set by market to equalize NPVs of expected cashflows.
- For same maturity, swap DV01 < bond: no principal repaid.

Other Swaps

- Other swaps: equity index or baskets; variance; assets; credit.
- Instead of fixed rate, might swap equity-related returns.
 - Pay USD 3M LIBOR + 35 bp; receive S&P 500 index return; or
 - Pay USD 1M LIBOR + 20 bp; receive returns on portfolio P .
 - Often reset quarterly (index) or monthly (baskets).
 - Basket swaps: like transactional prime brokerage agreements.
- Credit default and asset swaps let you hedge default risk.
- Basket and asset swaps may also substitute for repo transactions.
- Worry: will counterparty be able to pay cashflow owed?

Swaps vs. Futures

- How do swaps differ from futures? A few key ways:
 - Not standardized;
 - Traded OTC, mostly out of London (for now?);
 - No margin is posted (although some swaps use margin);
 - If not centrally-cleared, incur counterparty risk;
 - Not designed for ease of exit.
- Difficulty of exit means swaps often accrete:
 - Historically: enter into another swap to “exit” a trade.
 - Thus outstanding notionals are nearly meaningless.
 - In bankruptcy, this may allow selective contract enforcement
- To alleviate that, some services now offer *compression*:
 - Net out swaps which cancel risk factors; leave only net exposure.

Central Clearing

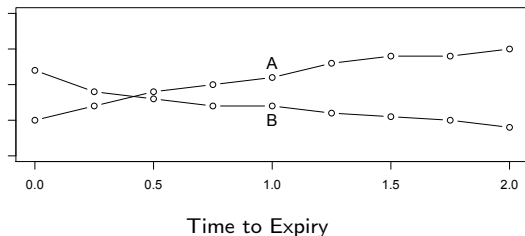
- Recent regulations encourage central clearing of common derivatives.
 - What is the difference from futures then? Not much.
- Central clearing has handled many crises smoothly:
 - 1987 stock market crash;
 - 1991 Persian Gulf War;
 - Refco bankruptcy (failure of #1 CME futures broker);
 - Amaranth bankruptcy (large commodities hedge fund failure);
 - Bear Stearns near-bankruptcy (\$700 bn of exposure at CME alone);
 - Lehman Brothers bankruptcy (\$1.3 tn exposure at CME alone!).

Futures Versus Forwards

- I claim $F_{t,T} = S_t e^{r_t(T-t)}$; w/div. yield d : $F_{t,T} = S_t e^{(r_t-d)(T-t)}$.
 - Can do likewise for convenience yield with commodity futures.
- This implies futures equals forward price $F_{t,T} = f_{t,T}$.
- What if changes in S_t are correlated with changes in r_t ?
 - e.g. Rate cuts correlated with changes in oil prices? bond prices??
 - Then mark-to-market in-/out-flows have asymmetric effect
- Let $\rho_{S,r} = \text{Corr}(dS_t/S_t, dr_t/r_t)$.
 - $\rho_{S,r} > 0 \Rightarrow F_{t,T} > f_{t,T}$ (futures above forwards).
 - $\rho_{S,r} < 0 \Rightarrow F_{t,T} < f_{t,T}$ (futures below forwards).
- If $\rho_{S,r} > 0$, longs get MTM money when rates high.

Futures Curves

- *Near contract*: contract nearest to expiry; usu. most active.
- *Next contract*: contract next-closest to expiry.
- *Futures/forward curve*: plot of contract prices vs. maturities.
 - Indicates relative supply and demand, including cost of carry.



- If futures curve increases (A), market is in *contango*.
- If futures curve decreases (B), market is in *backwardation*.
- Contracts drift along curve, converge to cash ($T = 0$) price.
- Basis = Cash price - Futures price.

Futures Curves: Beautiful Theories

- Keynes (1930) claims producers hedge (sell) aggressively.
 - Thus futures underestimate later spot prices.
 - Futures curve is normally decreasing (backwardated).
- Hotelling (1931): resource exhaustion forecastable.
 - Arbitrage argument: Equilibrium price grows with risk-free rate.
 - However, unique resources may be worth more (e.g. catalysts).
- Beautiful theories. . . yet again, slain by ugly facts:
 - The data just do not agree with these hypotheses.

Futures Curve Models

- Research suggests commodity prices mean-revert, suggests:

$$r_{F,t,T} = \alpha + \beta(\bar{X} - X_t)e^{-\gamma(T-t)} + \epsilon_{t,T}, \quad (1)$$

$$\epsilon_{t,T} \stackrel{\perp}{\sim} (0, \sigma_t e^{-\gamma_2(T-t)} / \text{inventory}), \quad (2)$$

$$X_t = \frac{\sum_{i=1}^n \phi_i X_{t-i}}{\sum_{i=1}^n \phi_i}. \quad (3)$$

- But if cash prices revert, they are stationary; can work w/prices.
- Can then modify first (dynamics) equation to be:

$$F_{t,T} = \alpha + \beta(\bar{X}_t - X_t)e^{-\gamma_1(T-t)} + \beta_2 X_t e^{r_{t,T} + \kappa c_t - d_{t,T} - v_{t,T}} + \epsilon_{t,T}. \quad (4)$$

w/cost-of-carry coefficient κ , convenience yield $v_{t,T}$.

- Then must decide how long averaging window n should be.

The Road Ahead

We covered Futures and Swaps; on to Options Basics next!

- Risk Alleviation: Options (Basics, Valuation), Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

Options Basics

Chapter 20

Introduction

- Here we will discuss options basics. In
- particular, we will discuss:
 - Option Markets;
 - Option Basics and Payoffs;
 - Put-Call Parity;
 - Embedded and Exotic Options;

Options

- An *option* gives holder right to buy/sell *underlying* asset.
 - In general, right is to buy/sell at *strike price*.
 - Does not oblige holder to take action, however.
 - Holder *exercises* right if worthwhile.
- Options reference price, yield, or other underlier “price.”
 - Comparing this price to the strike determines option value.
- The price to buy such an option is called the *premium*.

Puts, Calls, and Moneyness

- Seller (*writer*) of an option cannot choose to exercise.
 - Must provide benefit if exercised; paid *premium* in exchange.
- *Call* options allow buying underlier at strike price.
- *Put* options allow selling underlier at strike price.
- *Moneyness* refers to the value of immediate exercise:

Type	$S_t < K$	$S_t = K$	$S_t > K$
Put	In the money (ITM)	At the money (ATM)	Out of the money (OOM)
Call	Out of the money (OOM)	At the money (ATM)	In the money (ITM)

- Options exercised on *expiry date* only are *European*.
 - *American* options may be exercised any time until expiry.
 - Most stock, interest-rate, and futures options are *American*.
 - Most index and many FX options are *European*.

Options Markets

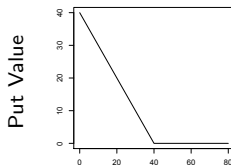
- Explicitly-traded options more recent than forwards: 1400s in Italy.
- Organized options exchanges are even more recent.
 - First: Chicago Board Options Exchange, 1973 (CBOT offshoot).
- Two other boosts to options trading in 1973:
 - Black and Scholes discover closed-form option pricing formula;
 - Advent of handheld calculator to allow computation of prices.
- Many competing US exchanges for stock and index options.
- Globally: stock option use growing; index options most common.
- Unlike futures, US options exchanges use independent clearinghouse.
 - Clearinghouse separated from exchanges; facilitates competition.

Option Payoffs

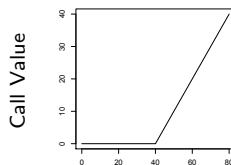
- What is the payoff at maturity for a put or call?
- For strike K , underlier price S_t , expiry at T :
 - Call expiry value: $C_T = (S_T - K)^+$.
 - Put expiry value: $P_T = (K - S_T)^+$.
 - Remember: We only exercise if payoff is positive.
- Payout graphs are typical “hockey sticks.”
- Payoff diagrams may be combined to create basic strategies.
- Helpful way to visualize options/combos.
- Also holds for early exercise of American options.

Options Strategies: Eliminating Upside/Downside

- We can see put and call diagrams to confirm priors:

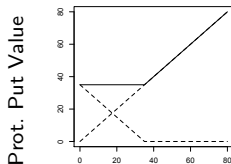


Underlier Price S_T

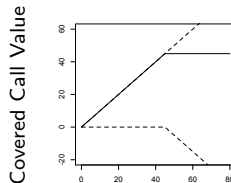


Underlier Price S_T

- *Protective put*: underlier + put eliminates downside.
- *Covered call*: underlier - call eliminates upside, yields income.



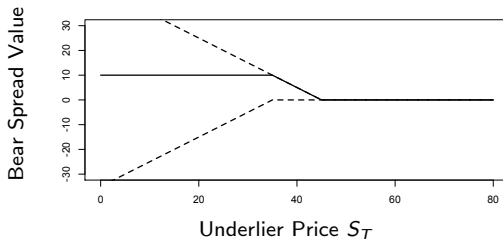
Underlier Price S_T



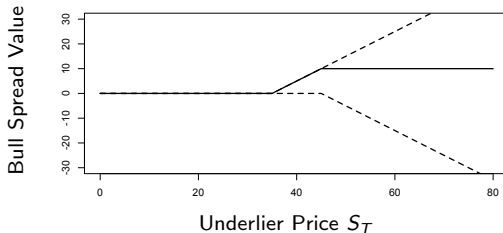
Underlier Price S_T

Options Strategies: Spreads

- *Bear spread*: long put, short lower-strike put; bet on downside.

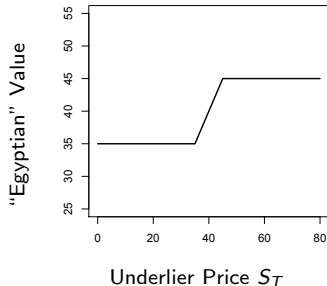
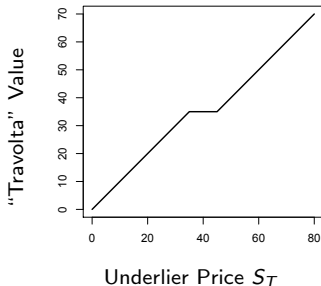


- *Bull spread*: long call, short higher-strike call; bet on upside.



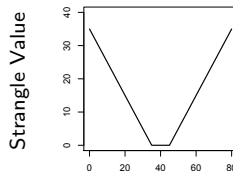
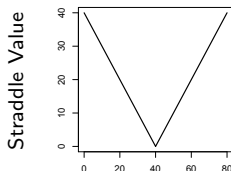
Options Strategies: Collars

- *Collars* can model protection offered in merger terms.
 - Described by behavior in a range of acquirer stock prices.
- *Fixed-price collar*: offers fixed price unless buyer stock moves.
 - If buyer's stock drops/spikes, acquirer pays less/more, aka "Travolta."
- *Fixed-ratio collar*: offers price with floor+ceiling, aka "Egyptian."

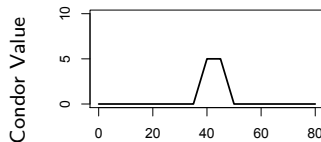
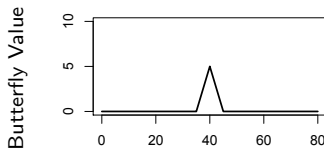


Options Strategies: Trading Volatility

- We can also use options to express a view on volatility.
- *Straddle*: same-strike call+put; *Strangle*: similar, but $K_{\text{put}} < K_{\text{call}}$.



- *Butterfly*: long call, short 2 mid-strike calls, long high-strike call.
- *Condor*: like butterfly, but mid-strike calls have different strikes.



Underlier Price S_T

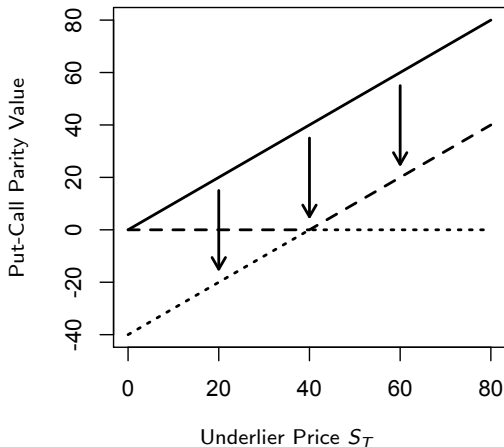
Underlier Price S_T

Time Value

- Even without pricing an option, we know they have value.
- Also know: a call struck at \$30 has value if stock is at \$29.99.
- More volatile stock $\implies$ call has more potential value.
- *Time value*: difference between option price and terminal payoff.
 - Sort of like taking a smoothing/kernel average of nearby prices.
- Some strategies are all about time value, e.g. *calendar spread*:
 - Buy longer-dated option, short shorter-dated option.
 - Effect is like a skewed butterfly spread.

Put-Call Parity: Payoff Diagram

- Long call (—), short put (···): like shifting underlying by $PV(K)$.
- We call this relationship *put-call parity*.



Put-Call Parity

- Thus the five prices we use are related:
 - Prices of calls and puts expiring at T ($C_{t,T}, P_{t,T}$);
 - Prices of underlier and strike (S_t, K); and,
 - Price of money — interest rate $t \rightarrow T$ (r).
- Arbitrage argument (with all typical assumptions):
 - At t : buy underlier @ S_t , put @ $P_{t,T}$; sell call @ $C_{t,T}$.
 - At T : get $S_T + (K - S_T)^+ - (S_T - K)^+ = K$.
 - No arbitrage price $\$K$: $Ke^{-r(T-t)}$ or $\frac{K}{(1+r)^{(T-t)}}$.
 - Should also account for omitted dividend yield d .
- Put-call parity: $C_{K,t,T} - P_{t,T} = S_t e^{-d(T-t)} - Ke^{-r(T-t)}$.

Embedded Options

- Many contracts have embedded options, option-like behavior.
- *Callable bonds* are short a call at some yield*.
- *Convertible bonds* (CBs) allow converting bond to stock.
 - Specified conversion ratio implies # shares, strike price.
 - Stock is issued out of company treasury; dilution results.
- Bonds with *detachable warrants* are very much like CBs.
- *Collateralized loans*: like risk-free bond less collateral put.
- *Real options*: option-like choices in business.

* Options on yields appear to be “mirror imaged”.

Exotic Options

- American/European puts/calls often called *vanilla options*.
- In contrast, *exotic options* involve different payoffs.
 - Payoffs may involve extra optionality or price history.
 - Payoffs may be on oddly-behaved underliers like spreads*.
- Exotic options include:
 - *Asian options*: based on average underlier price;
 - *Swing options*: allow deviations from constant-delivery rate;
 - *Exchange options*: allow switching underlying assets;
 - *Spread options*: may hedge marginal production cost;
 - *Quanto options*: hedge FX and underlier risk;
 - *Barrier options*: knock-in, knock-out; and,
 - *Lookback options*: payoff based on recent past best/worst price.

* Spreads behave oddly because they can assume negative values.

Comparative Statics

- Vanilla options respond in certain ways to key variables:

Variable	Call value	Put value
$S_t \uparrow$	$\uparrow$	$\downarrow$
$K \uparrow$	$\downarrow$	$\uparrow$
$\sigma \uparrow$	$\uparrow$	$\uparrow$
$T \uparrow$	$\uparrow$	$\uparrow$
$r_f \uparrow$	$\uparrow$	$\downarrow$
$d \uparrow$	$\downarrow$	$\uparrow$

Early Exercise

- Does European vs. American option difference matter?
 - Sometimes, but in most circumstances not.
- Would not exercise OOM options early: negative value.
- Would not exercise ATM options early: gives up option value.
- Without dividends, we would not exercise a call option early.
- Immediately pre-dividend, might exercise ITM call early.
- For an American put option, early exercise may be optimal:
 - If $S_t = \$0.01$, $K = \$30$: why lose time value of profit?
 - Thus below some S^* , $P_{t,T}^{\text{Am}} = K - S_t > P_{t,T}^{\text{Eu}} > Ke^{-r(T-t)} - S_t$.

The Road Ahead

We covered option basics; on to option valuation next!

- Risk Alleviation: Option Valuation, Credit, Structured Products; and,
- All Together Now: Active Portfolios, Investment Firms, Crises.

Option Valuation

Chapter 21

Introduction

- Here we will discuss option valuation.
- In particular, we will discuss:
 - Binomial Tree Option Pricing;
 - The Black-Scholes-Merton Option Model;
 - Divergences from Black-Scholes;
 - Recovering the Risk-Neutral Density; and,
 - Real Options.

Option Valuation History

- While options seem esoteric, pricing them was a recurring question.
- Early traders knew of put-call parity.
- Bachelier (1900) priced options for arithmetic Brownian motion.
 - Option pricing used Brownian motion before Einstein did!
- Bronzin (1908) developed similar formulas as Bachelier.
- Others tried, but the issue was the underlier expected return.
- Black and Scholes realized there was a replication argument.
 - That meant discounting could be done at the risk-free rate.
- Advent of handheld calculators made pricing feasible as well.

Simulation

- Easiest approach (from programmer perspective): *simulation*.
 - Use *pseudorandom numbers* to create *random variates*.
 - Use these to simulate final payoff (*Monte Carlo approach*).
- This approach lets us estimate the option value.
- Sometimes, value is based on unusual/extreme events.
- We can then use *importance sampling* for better estimate.
 - Create many rare/extreme scenarios;
 - Those yield better clarity on behavior in those scenarios.
 - Downweight those results to rare/extreme event likelihood.

Simulation: Random Process Choice

Biggest question: what random process is most accurate?

① *Arithmetic Brownian motion*: $dS_t = \mu dt + \sigma_A dW_t$,

$$S_t - S_0 \sim N(\mu t, \sigma_A^2 t). \quad (1)$$

② *Geometric Brownian motion*: $dS_t = \mu S_t dt + \sigma_G S_t dW_t$,

$$\log(S_t/S_0) \sim N((\mu - \sigma_G^2/2)t, \sigma_G^2 t). \quad (2)$$

③ *Ornstein-Uhlenbeck process*: $dS_t = \gamma(S_t - \bar{S})dt + \sigma_O dW_t$,

$$S_t \sim N\left(\bar{S} + \frac{S_0 - \bar{S}}{e^{\gamma t}}, \sigma_O^2 \frac{1 - e^{-2\gamma t}}{2\gamma}\right). \quad (3)$$

④ *Brownian bridge* for known B_T : $dB_t = \frac{\bar{B}_t - B_t}{1-t/T} dt + \sigma_B dW_t$,

$$B_t - B_0 \sim N\left(\bar{B}_t, \sigma_B^2 \frac{t(T-t)}{T^2}\right), \quad \text{for } \bar{B}_t = B_0 + \frac{(B_T - B_0)t}{T}. \quad (4)$$

Simulation: Easy Computation

- With a process model, simulation is very easy.
- For example, can easily simulate call value:

① $\{S_{iT}\}_{i \in \{1 \dots m\}} \leftarrow$ generate m random variates.

② $\bar{V}_T = \frac{1}{m} \sum_{i=1}^m (S_{iT} - K)^+$

③ $\hat{V} = \bar{V}_T e^{-r_f(T-t)}.$

- Another intuitive way to price options: *binomial tree*.
 - At discrete times, underlier moves up or down.
 - Up-then-down may (or may not) be same as down-then-up.
 - If same, may refer to tree as a *mesh* or *lattice*.
- At tree end, find option worth for different outcomes.
- Also find underlier — bond portfolio yielding option payout.
- Portfolio value must equal option value (by LOOP).
- Work backwards in time doing this until at current time.

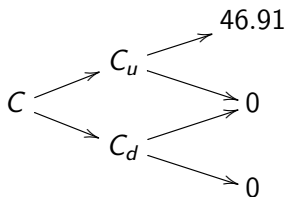
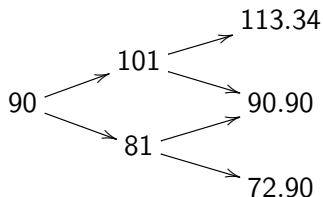
Binomial Tree: One-Step Example

- Assume the following stock, option payoffs in one year:



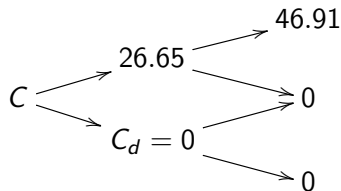
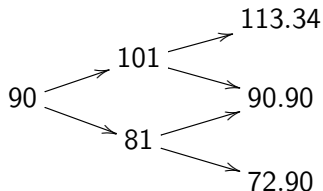
- If $r = 3\%$, what is C ? Note: $PV(\$81) = \78.64 .
- Buy 1 share of stock for \$90, sell \$78.64 of bonds.
- In one year, sell stock and buy back bond:
 - If stock @ \$101, $P\&L = \$101 - \$81 = \$20$.
 - If stock @ \$81, $P\&L = \$81 - \$81 = \$0$.
- Cost of stock – bond portfolio: $\$90 - \$78.64 = \$11.36$.
- To get worth of option, scale portfolio by $21/20$.
- Thus option is worth $\$11.36 \times \frac{21}{20} = \11.93 .

Binomial Tree: Two-Step Example



- Extend to 2 years: If $r = 3\%$, what is C , C_u , and C_d ?
- $PV(\$72.90, \$81, \$90.90) = (\$70.78, \$78.64, \$88.25)$.
- At $t = 1$, up state: Buy stock for \$101, sell \$88.25 of bonds.
- In one year, sell stock and buy back bond:
 - If stock @ \$113.34, $P\&L = \$113.34 - \$90.90 = \$22.44$.
 - If stock @ \$90.90, $P\&L = \$90.90 - \$90.90 = \$0$.
- Cost of stock – bond portfolio: $\$101 - \$88.25 = \$12.75$.
- To get worth of C_u , scale portfolio by $46.91/22.44$.
- Thus C_u is worth $\$12.75 \times \frac{46.91}{22.44} = \26.65 .

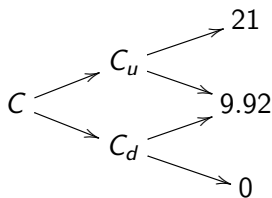
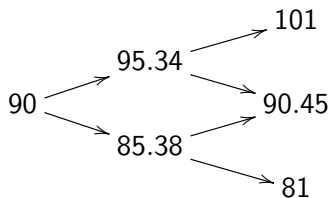
Binomial Tree: Two-Step Example Continued



- Obviously, C_d worth \$0.
- If $r = 3\%$, what is C ?
- Notice this is just a scaled version of One-Step Example.
 - Example I call paid off \$21 or \$0; worth \$11.93.
- Thus $C = \frac{26.65}{21} \times \$11.93 = \$15.14$.

Binomial Tree: Interpolated One-Step Example

- Suppose we interpolate the one-step example for more accuracy.
- Let up, down multipliers be u, d . $90u^2 = 101$, $90d^2 = 81$.
- One-step example had strike of \$81, multiplier of 1.05.

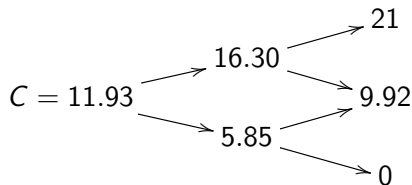
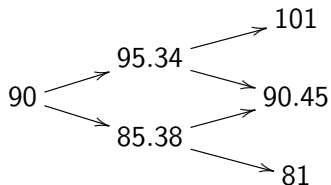


- Finding C_u, C_d is tricky. Solve simultaneous equations.

$$C_u = HS_u + FV(B), \quad C_d = HS_d + FV(B); \quad (5)$$

$$\implies H = \frac{C_u - C_d}{Su - Sd}, \quad B = \frac{uC_d - dC_u}{u - d} PV(1). \quad (6)$$

Binomial Tree: Interpolated One-Step Example Results



- With C_u, C_d we can work backward.
- Results of solving equations relate to replicating portfolio.
- H = *hedge ratio* = how much underlier to hold.
- B = bond holdings (financing).
- Option price = cost of replicating portfolio: $C = HS + B$.

Binomial Tree: Generalizing

- We can generalize what we just did to any binomial tree.
 - Recursively handle sub-trees; work backwards.
- Discrete dividends, path-dependency complicate matters.
 - Typically, we assume a dividend yield δ .
 - Path-dependencies? Price sub-trees or just simulate.
- What are sensible u, d (for geometric Brownian motion)?
 - Cox-Ross-Rubinstein: $u = e^{\sigma\sqrt{\Delta t}}, d = 1/u$.
 - Jarrow-Rudd: $u = e^{(r_f - \delta - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}}, d = e^{(r_f - \delta - \sigma^2/2)\Delta t - \sigma\sqrt{\Delta t}}$.
 - Could even match skewness with Tian (1993) approach.
- All are asymptotically equivalent; but, some converge faster.

Drift and Diffusion

- Assume underlier follows geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (7)$$

- So $\log(S_T) - \log(S_t) \sim N(\mu(T-t), \sigma^2(T-t))$? No.
 - Geometric Brownian motion means correct drift by $-\sigma^2/2$.
- Then $\log(S_T) \sim N(\log(S_t) + (\mu - \sigma^2/2)(T-t), \sigma^2(T-t))$.
- Subtract $\log(K)$ from each side:

$$\log\left(\frac{S_T}{K}\right) \sim N\left(\log\left(\frac{S_t}{K}\right) + \underbrace{\left(\mu - \frac{\sigma^2}{2}\right)}_{=r_f}(T-t), \sigma^2(T-t)\right). \quad (8)$$

- This give distribution of return beyond strike K .

The Risk-Neutral World

- If drift $\mu = r_f$, preceding is *risk-neutral density* ($\mathbb{Q}$ -measure).
- *Physical probabilities*: actual probabilities of events, $\mathbb{P}$ -measure.
- However, $\mathbb{P}$ -measure requires pricing kernel $k > r_f$.
- Example of difference between $\mathbb{P}$ and $\mathbb{Q}$ worlds:
 - Consider 1Y bond w/payoff \$1000 w.p. 0.9 and \$400 w.p. 0.1.
 - If 1Y risk-free rate is 5%, PV(expected cashflows) = \$895.24.
 - Implies YTM = 11.7%; but, market would demand risk premium.
 - If market demands 15%, price = $\$940/1.15 = \817.39 .
 - $\mathbb{Q}$ -measure is then given by:

$$P_{\mathbb{Q}}(\text{no default}) = \left\{ q : \frac{1000q + 400(1 - q)}{1.05} = 817.39 \right\}. \quad (9)$$

$\implies q = 0.76$ and $P(\text{default}) = 0.24 > 0.1$.

- Risk-neutral probabilities assign higher likelihood to bad outcomes.

Black-Scholes-Merton (1973) Model

- Then in the risk-neutral $\mathbb{Q}$ world:

$$P_Q(S_T > K) = \Phi\left(\frac{\ln(\frac{S_t}{K}) + (r_f - \frac{\sigma^2}{2})(T - t)}{\sigma\sqrt{T - t}}\right) = \Phi(d_2); \quad (10)$$

$$\begin{aligned} E_Q(S_T | S_T > K) &= S_t \Phi\left(\frac{\ln(\frac{S_t}{K}) + (r_f + \frac{\sigma^2}{2})(T - t)}{\sigma\sqrt{T - t}}\right) \\ &= S_t \Phi(d_1). \end{aligned} \quad (11)$$

- Thus the Black and Scholes (1973) and Merton (1973) model:

$$C_{t,T} = \underbrace{S_t \Phi(d_1)}_{E_Q(S_T | S_T > K)} - \underbrace{Ke^{-r_f(T-t)} \Phi(d_2)}_{PV(K) \cdot P_Q(S_T > K)}. \quad (12)$$

- Put price follows: $P_{t,T} = Ke^{-r_f(T-t)} \Phi(-d_2) - S_t \Phi(-d_1)$.
- Sort of a complicated DCF valuation.

Option Models for Other Underliers

- Can modify B-S-M for stocks with dividends, commodities.
- Merton (1973) model: dividend yield (like convenience yield).
 - Stock with dividends: dividend yield δ .
 - Commodities: $\delta = y - c = \text{convenience yield} - \text{storage cost}$.

$$C_{t,T} = S_t e^{-\delta(T-t)} \Phi(d_1) - K e^{-r_f(T-t)} \Phi(d_2) \quad (13)$$

$$d_1 = \frac{\ln(S_t/K) + (r_f - \delta + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

- Black (1976) model: Options on futures maturing at $T_1 > T$.

$$C_{t,T} = e^{-r_f(T-t)} [F_{t,T_1} \Phi(d_1) - K \Phi(d_2)] \quad (14)$$

$$d_1 = \frac{\ln(F_{t,T_1}/K) + \frac{\sigma^2}{2}(T-t)}{\sigma\sqrt{T-t}} \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Sensitivities and Risk Factors

- We might trade options because we think they are mispriced.
- Or, we might be a market maker carrying options inventory.
- In these cases, hedging risk factors is critical.
- Analyze option price sensitivity to factors $\Rightarrow$ exposures.
- Use derivatives of option price V wrt factor (*Greeks*).
 - $\Delta = \frac{\partial V}{\partial S}$; often $\Phi(d_1)$ (underlier exposure);
 - $\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}$ (indicates frequency of adjusting Δ hedge);
 - vega $= \frac{\partial V}{\partial \sigma}$ (exposure to 1% σ change; and, no: *not* a Greek letter);
 - $\rho = \frac{\partial V}{\partial r_f}$ (related to DV01);
 - $\psi = \frac{\partial V}{\partial \delta}$ (dividend yield sensitivity);
 - $\theta = \frac{\partial V}{\partial t}$ (time decay; unhedgeable).

Black-Scholes-Merton: Complications

- These results work reasonably well for European options.
- And, we can use these results for many index options.
- However, prices are rarely log-normal; kurtosis usually > 3 .
 - Important for commodities; B-S value should be lower bound.
- American options: must consider value of “early exercise.”
 - Typically, must price American options on a lattice (tree).
- Replication may fail if there are price jumps.
- Also need to worry about *pin risk*:
 - Delta hedging changes price distribution near expiry.
 - Price may be attracted to/repelled from strike prices.
- Replication can also fail during large market moves/data delays.

Portfolio Insurance

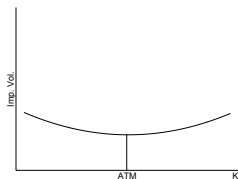
- We can dynamically replicate options with stock, bond.
- If we have a custom portfolio, probably no options listed on it.
- Can trade portfolio and bonds to synthesize an option.
- Create protective put for portfolio, aka *portfolio insurance*.
- Sheer brilliance! Or is it?
- Problem: dynamic replication has certain assumptions.
 - Know bond, stock prices, volatility; ability to transact quickly.
- In a market crash, these assumptions all break.
 - Prices jump; prices/fills delayed; some trading halted; volatility spikes.
- Post-1987 crash, portfolio insurance became less popular.

Implied Volatility

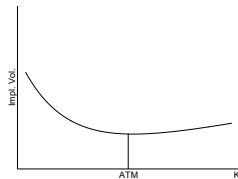
- What if we turn pricing upside-down?
 - Find volatilities which would imply market prices.
- Why would we do this?
 - Volatility is the one parameter whose value we never see.
 - But we see prices, so we examine *implied volatilities*.
- If we plot implied volatilities versus K , often see a curve.
 - *Volatility curve* partly due to using normal distribution.
 - True distribution has fatter tails; B-S-M is lower bound.
 - Also caused by leverage effects and risk aversion.

Volatility Curves

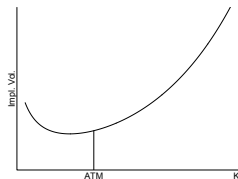
- Volatility curves adjust B-S-M for invalid assumptions.
- Curves may exhibit a *smile* or *smirk*.



Smile for stocks
pre-1987 crash



Smirk for stocks
post-1987 crash



Smirk for
commodities

- Note leverage, reverse leverage, risk aversion effects.
- Often fit volatility curve, then use it for pricing and inference.
- Commodities curve less stable; varies with supply elasticity.

Beyond the Volatility Curve

- Can create a dynamic volatility model which implies curve.

- Heston (1993) *stochastic volatility* model:

$$dS_t = rS_t dt + \sqrt{\Sigma_t} S_t dW_t^S \quad (15)$$

$$d\Sigma_t = \underbrace{a(b - \Sigma_t)dt}_{\text{OU mean reversion}} + \gamma \sqrt{\Sigma_t} dW_t^\Sigma \quad (16)$$

$$dW_t^\Sigma \cdot dW_t^S = \rho \quad (\rho > 0 \Rightarrow \text{leverage effect}). \quad (17)$$

- Can also plot implied volatility vs. K, T to get *volatility surface*.
 - Surface profile typically flattens as $T \uparrow$.
 - If we think surface is roughly stationary, can try to fit it.
 - May want fit to have smooth first, second derivatives.

Extracting Risk-Neutral Densities

- Second derivative of option prices wrt K yields underlier density.
- Create a butterfly w/strikes at $K - h, K, K + h$.
- Triangle area: $\frac{1}{2} \cdot 2h \cdot h = h^2$. Normalize: hold $1/h^2$ butterflies.
- If we write out the butterfly portfolio value, $V_{B,h}$:

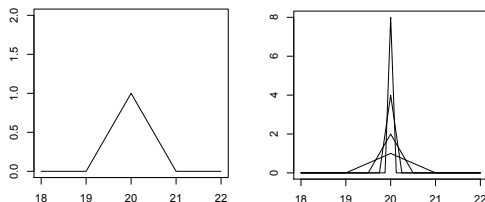
$$V_{B,h} = \frac{C_{K-h,t,T} - 2C_{K,t,T} + C_{K+h,t,T}}{h^2} \quad (18)$$

$$= \frac{P_{K-h,t,T} - 2P_{K,t,T} + P_{K+h,t,T}}{h^2}, \quad (19)$$

- Centered difference formula: $V_{B,h} \approx \frac{\partial^2}{\partial K^2}$ of $C_{K,t,T}$ or $P_{K,t,T}$.
- Take limit to get second derivative:

$$\frac{\partial^2 C_{K,t,T}}{\partial K^2} = \frac{\partial^2 P_{K,t,T}}{\partial K^2} = \lim_{h \downarrow 0} V_{B,h}. \quad (20)$$

Extracting Risk-Neutral Densities: Convoluting Butterflies



- Take limit as $h \downarrow 0$; get Dirac delta function δ .
- Finding the butterfly price = finding its expected value:

$$V_{B,h} = e^{-r_f(T-t)} \int_0^\infty f(S_T) \delta(S_T - K) dS_T = e^{-r_f(T-t)} f(K), \quad (21)$$

where f = risk-neutral density of S_T .

- Thus finding $\frac{\partial^2 C_{K,t,T}}{\partial K^2} = \frac{\partial^2 P_{K,t,T}}{\partial K^2}$ for all K gives us $f e^{-r_f(T-t)}$.

Extracting Risk-Neutral Densities: Improvements

- This can be tough if we lack option prices for all strikes.
- To fix that, we use vol curve to find all option prices.
- Then use those prices to extract risk-neutral density.
- Great — if you are a $\mathbb{Q}$ -quant (arbitrage pricers).
- What about $\mathbb{P}$ -quants (alpha believers)?
- Can we get physical density $\mathbb{P}$ from risk-neutral $\mathbb{Q}$?
- Not easily. Often must invert with an uncertain quantity.

Real Options

- *Real options* are option-like decisions we can make.
- May be embedded in contracts; typically, give possibility of:
 - Starting, stopping, canceling, or abandoning;
 - Increasing, decreasing, lengthening, or shortening;
 - Switching assets.
- Examples of real options:
 - Reactivate/lease a gold mine?
 - Store inventory for busy season?
 - Start and run “peaker” plant to create electricity?
 - Increase factory capacity to create more high-margin product?
 - Purchase more fuel in later months at agreed price?

Real Options: Valuation

- Valuing real options is usually done in one of many ways:
 - Comparative analysis;
 - Discounted cash flow (DCF) analysis;
 - Monte Carlo simulation analysis;
 - Using option models for similar options; or,
 - Stochastic optimization.
- We'll examine two of these approaches.

DCF Analysis of Real Options

- DCF analysis is common method for valuing decision/project:
 - Estimate/predict future cashflows; and,
 - Compute net present value (NPV) of those cashflows.
- The DCF approach neglects two key questions:
 - What rate to use for DCF analysis? (CAPM-based? WACC?)
 - What horizon do we analyze to?
- This does not even begin to consider term-structure of rates.
- Analysis/investment horizon is equally thorny:
 - What if horizon is random or affected by option exercise?
- DCF analysis also neglects current market prices, volatility.

Analyzing Real Options Using Option Models

- Can also use Black-Scholes-Merton and related models.
 - This is called a *real options approach**
- A real options approach requires regularity conditions:
 - 1 Option must have starting, ending dates $t_0 < T \leq \infty$.
 - 2 Risk factors $S_1, S_2, \dots$ must be clearly identified.
 - 3 $S_1, S_2, \dots$ must be continuously traded (for hedging).
 - 4 Must have stochastic process models for $S_1, S_2, \dots$.
 - 5 Must know exact option form (e.g. exercise type, payoff).
 - 6 Option cannot violate market completeness, measurability.
- Assumptions 3 and 4 are most critical.
- Assumption 5 rarely holds; try various models and guess/average.

* Real options and a real options approach are different. Confusing, eh?

Example: Value of Gold Mine Lease

- Consider leasing a gold mine for ten years.
 - Mine can produce 10,000 oz./year; extraction costs \$1000/oz.
 - Cash gold @ \$1300/oz; futures $\geq 1Y$ @ \$1200/oz; vol. = 30%.
 - WACC = 8%; risk-free $r_f = 3\%$; storage $c = y$ conv. yld.
- DCF: Discount E(cashflows) (\$300/oz, then \$200/oz.) at WACC.
- Real options approach: Annuity of annual/quarterly options.*
- Simulate real option values:
 - Geometric Brownian motion (GBM) or Ornstein-Uhlenbeck (OU)?

Method	DCF	GBM real opt.	OU real opt.	Sim GBM	Sim OU
Annually	\$14.3 mn	\$37.7 mn	\$23.9 mn	\$41.7 mn	\$20.8 mn
Quarterly	\$15.1 mn	\$33.3 mn	\$20.3 mn	\$60.5 mn	\$21.9 mn

- Production constraints: use tree or stochastic optimization.
- If gold $\leq$ \$1000/oz., DCF valuation deeply flawed.

* I interpolated between cash and futures for 3M–9M.